Course Notes

Abstract Algebra III

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Chapter 1

Lecture 1: Modules

Message: QQ, BB. HW 30%, Mid 20%, Final 50%

1.1 Basic Definitions and Examples

Definition 1.1.1. *Let R be a ring with 1, and let M be an abelian group. We say that M is a left R-module if there is a map*

$$R \times M \to M, \quad (r, m) \mapsto rm,$$

such that

$$r(m+n) = rm + rn,$$
$$(r+s)m = rm + sm,$$
$$(rs)m = r(sm),$$
$$1m = m$$

for all $r, s \in R$ and $m, n \in M$.
*A **right R-module** is defined similarly, using a map $(m, r) \mapsto mr$ and the axiom $m(rs) = (mr)s$. If R is commutative, left R-modules and right R-modules coincide. In general, a right R-module may be viewed as a left R^{op}-module.*

For each $r \in R$, the map $\varphi_r \colon M \to M$ given by $\varphi_r(m) = rm$ is a homomorphism of abelian groups. Let

$$\mathrm{End}(M) = \{\, \varphi \colon M \to M \mid \varphi \text{ is a homomorphism of abelian groups} \,\}.$$

Then $\mathrm{End}(M)$ is a ring under composition.

Theorem 1.1.2. *Let M be an abelian group. Then M is an R-module if and only if there is a ring homomorphism*

$$\rho \colon R \to \mathrm{End}(M).$$

Proof. ($\Rightarrow$) Suppose M is an R-module. Define $\rho\colon R \to \mathrm{End}(M)$ by

$$\rho(r)(m) = rm \qquad (r \in R,\ m \in M).$$

For each $r \in R$, the map $\rho(r)\colon M \to M$ is a homomorphism of abelian groups since

$$\rho(r)(m_1 + m_2) = r(m_1 + m_2) = rm_1 + rm_2.$$

Moreover, ρ is a ring homomorphism:

- $\rho(r+s)(m) = (r+s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r+s) = \rho(r) + \rho(s)$;
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$;
- $\rho(1)(m) = 1m = m$, so $\rho(1) = \mathrm{id}_M$.

($\Leftarrow$) Suppose $\rho\colon R \to \mathrm{End}(M)$ is a ring homomorphism. Define scalar multiplication by

$$(r, m) \mapsto \rho(r)(m).$$

Then

$$\begin{aligned} r(m_1 + m_2) &= \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2), \\ (r+s)m &= \rho(r+s)(m) = \rho(r)(m) + \rho(s)(m), \\ (rs)m &= \rho(rs)(m) = \rho(r)(\rho(s)(m)) = r(sm), \\ 1m &= \rho(1)(m) = \mathrm{id}_M(m) = m. \end{aligned}$$

Hence M is an R-module. ∎

Example 1.1.3. *(1) A vector space over a field is a module over that field.*
(2) An abelian group is a $\mathbb{Z}$-module.
(3) Any ideal of a ring R is naturally an R-module.
(4) Let M be an abelian group. Then M is an $\mathrm{End}(M)$-module via

$$\mathrm{End}(M) \times M \to M, \quad (\varphi, m) \mapsto \varphi(m).$$

(5) Let V be a vector space over a field K, and let $A\colon V \to V$ be a linear transformation. Then V is a $K[x]$-module via

$$K[x] \times V \to V, \quad (f(x), v) \mapsto f(A)v.$$

(6) If $|M| = 1$, then M is called the zero module.
(7) Let R be a ring with 1 and let $A \in M_n(R)$. Then the solution set of $AX = 0$, where $X \in R^n$, is a right R-module.

Definition 1.1.4 (R-submodule)**.** *Let R be a ring with 1, and let M be a left R-module. A subset $N \subseteq M$ is called an* R-submodule *of M if*

1. $0 \in N$;

2. $n_1, n_2 \in N$ *implies* $n_1 + n_2 \in N$;

3. $n \in N$ *implies* $-n \in N$;

4. $r \in R$ *and* $n \in N$ *imply* $rn \in N$.

Equivalently, N *is a subgroup of the abelian group* $(M, +)$ *that is closed under the* R*-action.*

Example 1.1.5. *(1) If* V *is a vector space over a field* K*, then every subspace of* V *is a* K*-submodule of* V*.*
(2) If G *is an abelian group, then every subgroup of* G *is a* $\mathbb{Z}$*-submodule of* G*.*
(3) The left regular module ${}_RR$ *has precisely the left ideals of* R *as its submodules.*
(4) Let G *be an abelian group, viewed as an* $\mathrm{End}(G)$*-module. Then the* $\mathrm{End}(G)$*-submodules of* G *are exactly the fully invariant subgroups of* G*.*

Remark 1.1.6. *A natural question is the following: are submodules of finitely generated modules again finitely generated? In general the answer is no. This property holds for all submodules if and only if the ring is Noetherian.*

Example 1.1.7 (Submodule of a finitely generated module that is not finitely generated)**.** *Let* k *be a field, and consider the polynomial ring*

$$R = k[x_1, x_2, x_3, \dots]$$

in infinitely many variables.

- *The ring* R *is a cyclic left* R*-module, generated by* 1*.*
- *Let* $I = \langle x_1, x_2, x_3, \dots \rangle$*. Then* I *is a submodule of the left* R*-module* R*.*
- *The submodule* I *is not finitely generated. Indeed, suppose*

$$I = \langle f_1, \dots, f_n \rangle$$

for some polynomials $f_1, \dots, f_n \in R$*. Let* N *be the largest index of any variable appearing in any of the* f_i*. Then every element of* $\langle f_1, \dots, f_n \rangle$ *lies in the subring* $k[x_1, \dots, x_N]$*, but* $x_{N+1} \in I$ *does not. This is a contradiction.*

Thus I *is a submodule of the finitely generated* R*-module* R *which is not finitely generated.*

Definition 1.1.8. *Let* M *be an* R*-module, and let* $(M_i)_{i \in I}$ *be a family of* R*-submodules of* M*. Then*

$$\bigcap_{i \in I} M_i$$

is an R-submodule of M. Moreover,

$$\sum_{i \in I} M_i = \left\{ \sum_{i \in I}^{\text{finite}} m_i \;\middle|\; m_i \in M_i \right\}$$

is also an R-submodule of M.

Definition 1.1.9. *Let M be an R-module, and let $S \subseteq M$. The submodule generated by S is*

$$\langle S \rangle = \left\{ \sum_{j=1}^{n} a_j s_j \;\middle|\; n \geq 0,\ a_j \in R,\ s_j \in S \right\}.$$

Equivalently,

$$\langle S \rangle = \bigcap_{\substack{N \leq M \\ S \subseteq N}} N.$$

If $M = \langle S \rangle$, then we say that M is generated by S. In particular, if $|S| = 1$, then M is called cyclic*; if $|S| < \infty$, then M is called* finitely generated.

Definition 1.1.10. *Let R be a ring, and let M be a left R-module. We say that M is* Noetherian *if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilizes; that is, there exists $n \geq 1$ such that

$$M_n = M_{n+1} = M_{n+2} = \cdots.$$

Equivalently, M is Noetherian if every submodule of M is finitely generated.

1.2 Quotient Modules and Homomorphism Theorems

Definition 1.2.1. *Let R be a ring, let M be a left R-module, and let N be a submodule of M. The* quotient module *(or* factor module*) of M by N is the quotient abelian group*

$$M/N = \{\, x + N \mid x \in M \,\},$$

with scalar multiplication defined by

$$r(x + N) = rx + N \qquad (r \in R,\ x \in M).$$

This is well defined, and with this operation M/N becomes a left R-module.

Example 1.2.2. *(1) If I is a left ideal of a ring R, then R/I is a quotient R-module of the left regular module ${}_RR$.*
(2) For $n \in \mathbb{Z}$, the subgroup $n\mathbb{Z}$ is a submodule of the $\mathbb{Z}$-module $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is a cyclic quotient module.

Definition 1.2.3. *Let R be a ring, and let M and N be left R-modules. A group homomorphism*

$$f\colon M \to N$$

*is called an R-*module homomorphism *(or R-*linear map*) if*

$$f(rx) = rf(x)$$

for all $r \in R$ and $x \in M$.

Definition 1.2.4. *Let $f\colon M \to N$ be an R-module homomorphism. The* kernel *of f is the set*

$$\ker f = \{\, x \in M \mid f(x) = 0 \,\}.$$

The kernel is a submodule of M. The image *of f is the set*

$$\operatorname{Im} f = \{\, f(x) \mid x \in M \,\}.$$

The image is a submodule of N.
If $\ker f = 0$, then f is injective. If $\operatorname{Im} f = N$, then f is surjective. A bijective R-module homomorphism is called an isomorphism*.*

Example 1.2.5. *Let R be a commutative ring, and let M and N be R-modules. Then $\operatorname{Hom}_R(M, N)$ is naturally an R-module.*

Theorem 1.2.6. *If $f\colon M \to N$ is an R-module homomorphism, then there is a canonical R-module isomorphism*

$$M/\ker f \to \operatorname{Im} f.$$

Theorem 1.2.7. *Let N be an R-submodule of M, and let $\pi_N\colon M \to M/N$ be the quotient map. Then π_N induces a natural bijection*

$$\{\, R\text{-submodules of } M \text{ containing } N \,\} \to \{\, R\text{-submodules of } M/N \,\}.$$

Theorem 1.2.8. *If $N \subseteq H$ are two R-submodules of M, then*

$$M/H \cong (M/N)/(H/N).$$

Theorem 1.2.9. *If N and H are R-submodules of M, then*

$$\frac{H+N}{N} \cong \frac{H}{H\cap N}.$$

Proposition 1.2.10. *Let M be an R-module, let N be a submodule of M, let M' be an R-module, and let $\varphi\colon M \to M'$ be a morphism such that $N \subseteq \ker\varphi$. Then there exists a unique morphism*

$$\varphi_N \colon M/N \to M'$$

such that the following diagram commutes. Moreover,

$$\ker\varphi_N = (\ker\varphi)/N.$$

$$\begin{tikzcd} M \arrow[r, "\varphi"] \arrow[d, "\pi_N"'] & M' \\ M/N \arrow[ur, dashed, "\varphi_N"'] & \end{tikzcd}$$

Remark 1.2.11. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- *S is an R-module,*
- *$\theta : M \to S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,*

then there exists a unique isomorphism $\psi : M/N \to S$ such that the following diagram commutes.

$$\begin{tikzcd} M \arrow[r, two heads, "\theta"] \arrow[d, two heads, "\pi_N"'] & S \\ M/N \arrow[ur, dashed, "\exists!\,\psi"'] & \end{tikzcd}$$

Proof. Since (S, θ) satisfies the same universal property as $(M/N, \pi_N)$, we may apply that property to the canonical projection

$$\pi_N : M \to M/N.$$

Because $N \subseteq \ker(\pi_N)$, there exists a unique morphism

$$u : S \to M/N$$

such that

$$u \circ \theta = \pi_N.$$

On the other hand, since $(M/N, \pi_N)$ has the universal property of the quotient by N, and since $N \subseteq \ker(\theta)$, there exists a unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

We now show that ψ and u are inverse to each other.

First, compute

$$(u \circ \psi) \circ \pi_N = u \circ (\psi \circ \pi_N) = u \circ \theta = \pi_N.$$

But the identity morphism $\mathrm{id}_{M/N} : M/N \to M/N$ also satisfies

$$\mathrm{id}_{M/N} \circ \pi_N = \pi_N.$$

By the uniqueness part of the universal property of $(M/N, \pi_N)$, it follows that

$$u \circ \psi = \mathrm{id}_{M/N}.$$

Similarly,

$$(\psi \circ u) \circ \theta = \psi \circ (u \circ \theta) = \psi \circ \pi_N = \theta.$$

But the identity morphism $\mathrm{id}_S : S \to S$ also satisfies

$$\mathrm{id}_S \circ \theta = \theta.$$

By the uniqueness part of the universal property of (S, θ), we obtain

$$\psi \circ u = \mathrm{id}_S.$$

Therefore ψ is an isomorphism, with inverse u.

Finally, the uniqueness of ψ follows directly from the universal property of $(M/N, \pi_N)$: indeed,ψ is the unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

Hence there exists a unique isomorphism

$$\psi : M/N \to S$$

for which the diagram commutes. □

Proposition 1.2.12. *There is a unique isomorphism* $\bar{\varphi} : \operatorname{coim} \varphi \to \operatorname{Im} \varphi$ *such that the following diagram commutes:*

$$\begin{tikzcd}
M \arrow[r, "\varphi"] \arrow[d, "\pi_{\ker(\varphi)}"'] & M' \\
\mathrm{coim}(\varphi) \arrow[r, "\overline{\varphi}"] & \mathrm{Im}(\varphi) \arrow[u, hook]
\end{tikzcd}$$

Recall:

$$\begin{aligned}
&\varphi : M \to M', \quad R\text{-morphism} \\
&\mathrm{coker}(\varphi) = M'/\,\mathrm{Im}(\varphi) \\
&\mathrm{coim}(\varphi) = M/\ker\varphi
\end{aligned}$$

Definition 1.2.13. *Let M be an R-module and I an ideal of R. Denote by IM the submodule of M generated by all elements am where $a \in I, m \in M$.*

$$IM = \left\{ \sum_{finite} a_i m_i \mid a_i \in I,\ m_i \in M \right\}$$

Proposition 1.2.14. *Let M be an R-module. Then $R \to \mathrm{End}(M)$ is a ring homomorphism.*

1. *$R \to \mathrm{End}(M)$ factors through the natural surjection $R \to R/I$ if and only if $IM = 0$:*

$$\begin{tikzcd}
R \arrow[rr, "\rho"] \arrow[dr, two heads, "\pi_I"'] & & \mathrm{End}(M) \\
& R/I \arrow[ur, dashed, "\exists\,\rho_I"'] &
\end{tikzcd}$$

   Where ρ factors through π_I means $\exists \rho_I : R/I \to \mathrm{End}(M)$ such that the diagram above commutes.

2. *The module M/IM is naturally endowed with a structure of R/I-module.*

3. *Let $\varphi : M \to M'$ be an R-homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I-modules*

$$M/IM \to M'/IM'.$$

4. *If $M = M_1 \oplus \cdots \oplus M_n$, then $M/IM = M_1/IM_1 \oplus \cdots \oplus M_n/IM_n$.*

5. *$R^n/IR^n \cong (R/IR)^n$.*

Proof. **(1)** Suppose first that ρ factors through π_I. Then there exists a ring homomorphism

$$\rho_I : R/I \to \mathrm{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

If $a \in I$, then $\pi_I(a) = 0$, so

$$\rho(a) = \rho_I(0) = 0.$$

Therefore $am = 0$ for every $m \in M$. Since IM is generated by elements of the form am with $a \in I$ and $m \in M$, it follows that

$$IM = 0.$$

Conversely, assume that $IM = 0$. Then for every $a \in I$ and every $m \in M$ we have

$$\rho(a)(m) = am = 0.$$

Thus $\rho(a) = 0$, so $I \subseteq \ker \rho$. Hence ρ factors uniquely through the quotient ring R/I: there exists a unique ring homomorphism

$$\rho_I : R/I \to \mathrm{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

(2) Define an action of R/I on M/IM by

$$(r + I) \cdot (m + IM) := rm + IM.$$

We must show that this is well-defined.

If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

If $m + IM = m' + IM$, then $m - m' \in IM$. Since IM is a submodule of M, we have

$$r(m - m') \in IM,$$

so

$$rm + IM = rm' + IM.$$

Thus the action is well-defined. The module axioms follow immediately from the module axioms on M. Therefore M/IM is naturally an R/I-module.

(3) Let $\varphi : M \to M'$ be an R-module homomorphism. Take any element of IM, say

$$x = \sum_{j=1}^{t} a_j m_j \qquad (a_j \in I,\ m_j \in M).$$

Then

$$\varphi(x) = \sum_{j=1}^{t} \varphi(a_j m_j) = \sum_{j=1}^{t} a_j \varphi(m_j).$$

Since each $a_j \in I$ and each $\varphi(m_j) \in M'$, it follows that

$$a_j \varphi(m_j) \in IM'.$$

Hence $\varphi(x) \in IM'$, and therefore

$$\varphi(IM) \subseteq IM'.$$

Thus we may define

$$\overline{\varphi} : M/IM \to M'/IM', \qquad \overline{\varphi}(m + IM) = \varphi(m) + IM'.$$

This is well-defined: if $m + IM = m' + IM$, then $m - m' \in IM$, so

$$\varphi(m) - \varphi(m') = \varphi(m - m') \in IM',$$

hence

$$\varphi(m) + IM' = \varphi(m') + IM'.$$

Finally,$\overline{\varphi}$ is R/I-linear, since

$$\overline{\varphi}\big((r+I)\cdot(m+IM)\big) = \overline{\varphi}(rm+IM) = \varphi(rm)+IM' = r\varphi(m)+IM' = (r+I)\cdot\overline{\varphi}(m+IM).$$

(4) Assume that

$$M = M_1 \oplus \cdots \oplus M_n.$$

We first show that

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

If $x \in IM$, then x is a finite sum of elements of the form am with $a \in I$ and $m \in M$. Writing

$$m = m_1 + \cdots + m_n \qquad (m_i \in M_i),$$

we get

$$am = am_1 + \cdots + am_n,$$

and each $am_i \in IM_i$. Hence $x \in IM_1 + \cdots + IM_n$, so

$$IM \subseteq IM_1 + \cdots + IM_n.$$

The reverse inclusion is immediate because each $M_i \subseteq M$, hence each $IM_i \subseteq IM$. Therefore

$$IM = IM_1 + \cdots + IM_n.$$

To see that the sum is direct, suppose

$$x_1 + \cdots + x_n = 0 \qquad \text{with } x_i \in IM_i \subseteq M_i.$$

Since

$$M = M_1 \oplus \cdots \oplus M_n$$

is a direct sum, we must have $x_i = 0$ for all i. Thus

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

Now define

$$\Psi : M_1/IM_1 \oplus \cdots \oplus M_n/IM_n \to M/IM$$

by

$$\Psi(m_1 + IM_1, \ldots, m_n + IM_n) = (m_1 + \cdots + m_n) + IM.$$

This map is well-defined and R/I-linear. It is surjective because every element of M can be written as $m_1 + \cdots + m_n$. It is injective because

$$(m_1 + \cdots + m_n) + IM = IM$$

means that

$$m_1 + \cdots + m_n \in IM = IM_1 \oplus \cdots \oplus IM_n,$$

so each $m_i \in IM_i$. Hence

$$m_i + IM_i = IM_i$$

for all i, and the kernel is trivial. Therefore Ψ is an isomorphism:

$$M/IM \cong M_1/IM_1 \oplus \cdots \oplus M_n/IM_n.$$

(5) Since

$$R^n = R \oplus \cdots \oplus R$$

(n copies), part (4) yields

$$R^n/IR^n \cong R/IR \oplus \cdots \oplus R/IR.$$

Therefore

$$R^n/IR^n \cong (R/IR)^n.$$

This completes the proof. $\square$

1.3 Direct Sums and Free Modules

Definition 1.3.1. *Let $(M_i)_{i\in I}$ be a family of R-modules. The* direct product *of $(M_i)_{i\in I}$ is the module*

$$\prod_{i\in I} M_i = \left\{ \varphi\colon I \to \bigcup_{i\in I} M_i \;\middle|\; \varphi(i) \in M_i \text{ for all } i \in I \right\},$$

with componentwise addition and scalar multiplication:

$$R \times \prod_{i\in I} M_i \to \prod_{i\in I} M_i, \quad (r,\varphi) \mapsto (r\varphi(i))_{i\in I}$$

The direct sum *of $(M_i)_{i\in I}$ is the submodule*

$$\bigoplus_{i\in I} M_i = \left\{ \varphi \in \prod_{i\in I} M_i \;\middle|\; \#\{\, i \in I \mid \varphi(i) \neq 0 \,\} < \infty \right\}.$$

These two modules agree when $|I| < \infty$. In the category of R-modules, the direct product is the categorical product and the direct sum is the categorical coproduct.

Theorem 1.3.2. *Suppose that $M_1, \dots, M_n$ are R-submodules of M such that $M = M_1 + \dots + M_n$. Let $I = [n]$. Then the following conditions are equivalent:*

1. *The map $\varphi\colon \bigoplus_{i\in I} M_i \to M$ given by $\varphi((m_i)_{i\in I}) = \sum_{i\in I} m_i$ is an isomorphism.*
2. *Any element of M can be uniquely written as a sum of elements of M_i.*
3. *The zero element can be uniquely written as a sum of elements of the M_i.*
4. *For each $i \in I$, one has*
$$M_i \cap \sum_{j\neq i} M_j = 0.$$

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

(1) $\Rightarrow$ (2)**.** If φ is an isomorphism, then for every $m \in M$ there exists a unique

$$(m_1, \dots, m_n) \in \bigoplus_{i=1}^{n} M_i$$

such that

$$m = \varphi(m_1, \dots, m_n) = m_1 + \dots + m_n.$$

Hence every element of M can be written uniquely as a sum of elements of the M_i.
(2) $\Rightarrow$ (3)**.** Apply (2) to the element $0 \in M$.

(3) ⇒ (4). Fix $i \in I$, and let

$$x \in M_i \cap \sum_{j \neq i} M_j.$$

Then $x \in M_i$, and there exist $m_j \in M_j$ for $j \neq i$ such that

$$x = \sum_{j \neq i} m_j.$$

Therefore

$$0 = x - \sum_{j \neq i} m_j$$

is a decomposition of 0 as a sum of elements from the submodules $M_1, \dots, M_n$. By the uniqueness in (3), this decomposition must be the trivial one. In particular,

$$x = 0.$$

Hence

$$M_i \cap \sum_{j \neq i} M_j = 0.$$

(4) ⇒ (1). Since

$$M = M_1 + \cdots + M_n,$$

the map φ is surjective. It remains to show that φ is injective.

Suppose that

$$\varphi(m_1, \dots, m_n) = 0,$$

that is,

$$m_1 + \cdots + m_n = 0 \qquad \text{with } m_i \in M_i.$$

For each $i \in I$, we have

$$m_i = -\sum_{j \neq i} m_j \in \sum_{j \neq i} M_j.$$

Since also $m_i \in M_i$, it follows that

$$m_i \in M_i \cap \sum_{j \neq i} M_j = 0.$$

Thus $m_i = 0$ for all i, so

$$\ker \varphi = 0.$$

Therefore φ is injective, and hence an isomorphism.

This proves that the four conditions are equivalent. □

Definition 1.3.3 (free module)**.** *Let M be an R-module.*

1. $x_1, \ldots, x_n \in M$ *are linearly independent if*

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0 \to a_1 = a_2 = \cdots = a_n = 0.$$

2. M *is called free if it is generated by a linearly independent subset. This subset is called a basis of* M*.*

3. *If* M *is an* R*-module with a basis* $x_1, \ldots, x_n$*, then* $M = \langle x_1 \rangle \oplus \cdots \oplus \langle x_n \rangle$*.*

Example 1.3.4. $R \oplus \cdots \oplus R = R^n$.

Theorem 1.3.5. *Let* R *be a commutative ring with* 1*. Let* M *be a finitely generated free* R*-module. Then any basis of* M *has the same cardinality. The number of elements in a basis is called the rank of the free module.*

Proof. Let $B = \{x_i\}_{i\in I}$ be a basis of M. We first show that any basis is finite. Since M is finitely generated, choose generators $y_1, \ldots, y_n$ of M. Each y_k is a finite R–linear combination of elements of B, so there exists a finite subset $I_0 \subset I$ such that

$$y_1, \ldots, y_n \in \sum_{i\in I_0} Rx_i.$$

Hence

$$M = \sum_{k=1}^{n} Ry_k \subseteq \sum_{i\in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i\in I_0} Rx_i$. If $I \setminus I_0 \neq \varnothing$, pick $j \in I \setminus I_0$. Then

$$x_j \in M = \sum_{i\in I_0} Rx_i,$$

contradicting that B is linearly independent. Thus $I = I_0$ is finite.

Now let $x_1, \ldots, x_r$ and $y_1, \ldots, y_s$ be two bases of M. Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F-vector space. Write $\overline{m}$ for the class of $m \in M$ in M/JM, and $\overline{a}$ for the class of $a \in R$ in F. We claim that $\overline{x_1}, \ldots, \overline{x_r}$ form an F–basis of M/JM.

Spanning. For any $m \in M$, write $m = \sum_{i=1}^{r} a_ix_i$. Then

$$\overline{m} = \sum_{i=1}^{r} \overline{a_i}\,\overline{x_i},$$

so $\overline{x_1}, \ldots, \overline{x_r}$ span M/JM.

Linear independence. Suppose

$$\sum_{i=1}^{r} \overline{a_i}\,\overline{x_i} = 0 \quad \text{in } M/JM.$$

Then $\sum_{i=1}^{r} a_i x_i \in JM$, so there exist $a'_1, \ldots, a'_r \in J$ such that

$$\sum_{i=1}^{r} a_i x_i = \sum_{i=1}^{r} a'_i x_i.$$

Hence

$$\sum_{i=1}^{r} (a_i - a'_i)\, x_i = 0.$$

By linear independence of $x_1, \ldots, x_r$, we get $a_i - a'_i = 0$ for all i, so $a_i \in J$, thus $\overline{a_i} = 0$ for all i. Therefore $\overline{x_1}, \ldots, \overline{x_r}$ are F-linearly independent, and form an F–basis of M/JM.

By the same argument,$\overline{y_1}, \ldots, \overline{y_s}$ also form an F–basis of M/JM. Hence

$$r = \dim_F(M/JM) = s.$$

Therefore any two bases of M have the same cardinality. $\square$

Chapter 2

Lecture 2: Tensor Product

2.1 Annihilator, Torsion and Projectivity of Free Modules

Definition 2.1.1. *Let M be an R-module, and let $m \in M$. The cyclic submodule generated by m is*

$$Rm = \langle m \rangle = \{\, rm \mid r \in R \,\}.$$

The annihilator *of m in R is*

$$\mathrm{Ann}_R(m) = \{\, r \in R \mid rm = 0 \,\}.$$

It is a left ideal of R.

Proposition 2.1.2. *For every $m \in M$, there is an isomorphism of left R-modules*

$$R/\,\mathrm{Ann}_R(m) \cong Rm.$$

In particular, every cyclic module is isomorphic to a quotient of R.

Definition 2.1.3. *An element $m \in M$ is called*

- ***torsion-free*** *if* $\mathrm{Ann}_R(m) = 0$*;*
- ***torsion*** *if* $\mathrm{Ann}_R(m) \neq 0$.

A module is called torsion-free *if every nonzero element is torsion-free.*

Definition 2.1.4. *The* torsion submodule *of M is*

$$\mathrm{Tor}(M) = \{\, m \in M \mid \mathrm{Ann}_R(m) \neq 0 \,\}.$$

Lemma 2.1.5. *If R is an integral domain, then the torsion part of M is a submodule of M.*

Proof. We verify the submodule criteria.

Non-empty. Since $r \cdot 0_M = 0_M$ for every $r \in R$ and $R \neq 0$, we have $\mathrm{Ann}_R(0_M) = R \neq 0$, so $0_M \in \mathrm{Tor}(M)$.

Closed under addition. Let $m, n \in \mathrm{Tor}(M)$. Then there exist $\lambda, \mu \in R \setminus \{0\}$ such that $\lambda m = 0$ and $\mu n = 0$. Hence

$$(\lambda\mu)(m+n) = \mu(\lambda m) + \lambda(\mu n) = 0.$$

Since R is an integral domain and $\lambda, \mu \neq 0$, we have $\lambda\mu \neq 0$, so $\mathrm{Ann}_R(m+n) \neq 0$ and $m + n \in \mathrm{Tor}(M)$.

Closed under scalar multiplication. Let $m \in \mathrm{Tor}(M)$ and $r \in R$. Choose $\lambda \in R \setminus \{0\}$ with $\lambda m = 0$. Then

$$\lambda(rm) = r(\lambda m) = 0,$$

so $\mathrm{Ann}_R(rm) \neq 0$ and $rm \in \mathrm{Tor}(M)$.

Therefore $\mathrm{Tor}(M)$ is a submodule of M. $\square$

Example 2.1.6. *(1) The element $0 \in M$ is torsion.*
(2) Let R be a commutative ring with 1. Then R is an integral domain if and only if the left regular module ${}_RR$ is torsion-free. In this case $\mathrm{Tor}({}_RR) = 0$ and $\mathrm{Tor}(R^n) = 0$ for all $n \geq 1$.
(3) If I is a nonzero ideal of R, then $\mathrm{Tor}(R/I) = R/I$ when R/I is regarded as an R-module. If we regard R/I as an R/I-module, then it is torsion-free if and only if I is a prime ideal.
(4) More generally, let I be any nonzero ideal of R, and let M be an R-module. Then $\mathrm{Tor}(M/IM) = M/IM$ when M/IM is regarded as an R-module.
(5) Even when I is prime, M/IM need not be torsion-free as an R/I-module. For $\overline{m} \in M/IM$, one has $\mathrm{Ann}_{R/I}(\overline{m}) \neq 0$ if and only if there exists $r \in R \setminus I$ with $rm \in IM$. Thus primeness of I does not rule out nonzero operators in R/I annihilating nonzero classes in M/IM.
For instance, let $R = k[x,y]$, let $I = (y)$, and let $M = R/(x) \cong k[y]$. Then $IM = (y)$ and $M/IM \cong k$, while $x \cdot M = 0$ but $x \notin I$; hence $x + I \in R/I$ is nonzero and annihilates all of M/IM. If instead I is maximal, then R/I is a field, nonzero scalars act invertibly, and M/IM is always torsion-free over R/I. Item (3) is the case $M = R$.

Remark 2.1.7. *Let R be a commutative ring with 1, let M be an R-module, and fix $r \in R$. Then the map*

$$f\colon M \to M, \qquad f(m) = rm,$$

is an R-module homomorphism. If moreover M is torsion-free and $r \neq 0$, then f is a monomorphism.

Lemma 2.1.8. *If R is an integral domain and M is an R-module, then*

1. $\operatorname{Tor}(\operatorname{Tor}(M)) = \operatorname{Tor}(M)$.
2. $\operatorname{Tor}(M/\operatorname{Tor}(M)) = 0$.

Proof.

1. Since $\operatorname{Tor}(M)$ is a submodule of M, its torsion submodule is

$$\operatorname{Tor}(\operatorname{Tor}(M)) = \{ m \in \operatorname{Tor}(M) \mid \operatorname{Ann}_R(m) \neq 0 \}.$$

Every $m \in \operatorname{Tor}(M)$ already satisfies $\operatorname{Ann}_R(m) \neq 0$, so $\operatorname{Tor}(\operatorname{Tor}(M)) \subseteq \operatorname{Tor}(M)$. Conversely,$\operatorname{Tor}(M) \subseteq \operatorname{Tor}(\operatorname{Tor}(M))$ because the annihilator of m in R does not depend on the ambient module: for any $m \in \operatorname{Tor}(M)$, its annihilator computed in the submodule $\operatorname{Tor}(M)$ is the same as $\operatorname{Ann}_R(m) \neq 0$, so $m \in \operatorname{Tor}(\operatorname{Tor}(M))$. Hence $\operatorname{Tor}(\operatorname{Tor}(M)) = \operatorname{Tor}(M)$.

2. Let $\pi \colon M \to M/\operatorname{Tor}(M)$ be the canonical projection. Write $\overline{m} = \pi(m) = m + \operatorname{Tor}(M)$. Suppose $\overline{m} \in \operatorname{Tor}(M/\operatorname{Tor}(M))$, i.e. there exists $\lambda \in R \setminus \{0\}$ such that

$$\lambda \overline{m} = \overline{0} \quad \text{in } M/\operatorname{Tor}(M).$$

This means $\lambda m \in \operatorname{Tor}(M)$, so there exists $\mu \in R \setminus \{0\}$ with $\mu(\lambda m) = 0$, i.e.

$$(\mu\lambda)\, m = 0.$$

Since R is an integral domain and $\mu, \lambda \neq 0$, we have $\mu\lambda \neq 0$, hence $m \in \operatorname{Tor}(M)$, so $\overline{m} = \overline{0}$. Therefore $\operatorname{Tor}(M/\operatorname{Tor}(M)) = 0$.

$\square$

Definition 2.1.9. *Let J be a subscript set, denote $R^J = \prod_{j\in J} R . R^{(J)} = \bigoplus_{j \in J} R$. Let M be an R-module. Let $x_j \in M$ where $j \in J$ be a family of elements of M. Then the submodule generated by $(x_j)_{j\in J}$ is denoted by $\sum_{j \in J} Rx_j$.*

Corollary 2.1.10. *We have a morphism of R-modules $\phi : R^{(J)} \to M$ such that $\phi(r_j)_{j\in J} = \sum_{j\in J} r_j x_j$ for all $(r_j)_{j \in J} \in R^{(J)}$.*

- *ϕ is injective if and only if $\{x_j\}_{j\in J}$ are linearly independent.*
- *ϕ is surjective if and only if $\{x_j\}_{j\in J}$ generate M.*
- *ϕ is an isomorphism if and only if M is free with basis $\{x_j\}_{j\in J}$.*

Lemma 2.1.11. *Let L be a free R-module with basis $\{x_j\}_{j\in J}$. For any R-module M, the map*

$$\Phi : \mathrm{Hom}_R(L, M) \to M^J, \qquad f \mapsto (f(x_j))_{j\in J}$$

is a bijection.

Proof. We prove that Φ is injective and surjective.

Injectivity. Let $f, g \in \mathrm{Hom}_R(L, M)$ and suppose that

$$\Phi(f) = \Phi(g).$$

Then for every $j \in J$ we have

$$f(x_j) = g(x_j).$$

Since $\{x_j\}_{j\in J}$ is a basis of L, every element $x \in L$ can be written uniquely as a finite sum

$$x = \sum_{j\in J} a_j x_j$$

with $a_j \in R$ and all but finitely many a_j equal to 0. Hence

$$f(x) = f\left(\sum_{j\in J} a_j x_j\right) = \sum_{j\in J} a_j f(x_j) = \sum_{j\in J} a_j g(x_j) = g\left(\sum_{j\in J} a_j x_j\right) = g(x).$$

Thus $f = g$, so Φ is injective.

Surjectivity. Let $(m_j)_{j\in J} \in M^J$ be given. We define a map

$$f : L \to M$$

by declaring that for

$$x = \sum_{j\in J} a_j x_j \qquad (a_j \in R,\ \text{all but finitely many } a_j = 0),$$

we set

$$f(x) := \sum_{j\in J} a_j m_j.$$

This is well defined because each element of L has a unique expression as a finite linear combination of the basis elements.

Now let

$$x = \sum_{j\in J} a_j x_j, \qquad y = \sum_{j\in J} b_j x_j, \qquad r \in R.$$

Then

$$f(x + y) = f\left(\sum_{j\in J} (a_j + b_j) x_j\right) = \sum_{j\in J} (a_j + b_j) m_j = \sum_{j\in J} a_j m_j + \sum_{j\in J} b_j m_j = f(x) + f(y),$$

and

$$f(rx) = f\left(\sum_{j\in J}(ra_j)x_j\right) = \sum_{j\in J}(ra_j)m_j = r\sum_{j\in J}a_jm_j = rf(x).$$

So f is an R-module homomorphism. Moreover, for every $j \in J$,

$$f(x_j) = m_j.$$

Therefore

$$\Phi(f) = (m_j)_{j\in J}.$$

Thus Φ is surjective.

Since Φ is both injective and surjective, it is a bijection. □

Proposition 2.1.12. *Let L be a free R-module.*

1. *Let $\pi : X \to Y$ be an epimorphism of R-modules. For any R-module homomorphism $\varphi : L \to Y$ there exists a morphism $\tilde{\varphi} : L \to X$ such that the following diagram commutes:*

$$\begin{tikzcd} & L \arrow[dl, "\tilde{\varphi}"'] \arrow[d, "\varphi"] & \\ X \arrow[r, "\pi"'] & Y \arrow[r] & 0 \end{tikzcd}$$

2. *In particular, for any R-module M, any epimorphism $\pi : M \to L$ has a section, that is, there exists a morphism $\sigma : L \to M$ such that $\pi \circ \sigma = \mathrm{id}_L$. In that case σ is injective,$\mathrm{Im}(\sigma) \cong L$, and*

$$M = \ker\pi \oplus \mathrm{Im}(\sigma).$$

$$\begin{tikzcd} 0 \arrow[r] & \ker\pi \arrow[r] & M \arrow[r, two heads, shift left, "\pi"] & L \arrow[l, dashed, shift left, "\sigma"] \arrow[r] & 0 \end{tikzcd}$$

Proof. 1. Since L is free, let $\{x_j\}_{j\in J}$ be a basis of L.

For each $j \in J$, since $\pi : X \to Y$ is surjective, there exists an element

$$\tilde{x}_j \in X$$

such that

$$\pi(\tilde{x}_j) = \varphi(x_j).$$

Because L is free with basis $\{x_j\}_{j\in J}$, there exists a unique R-module homomorphism

$$\tilde{\varphi} : L \to X$$

satisfying

$$\tilde{\varphi}(x_j) = \tilde{x}_j \qquad \text{for all } j \in J.$$

We now check that $\pi \circ \tilde{\varphi} = \varphi$. For each basis element x_j,

$$(\pi \circ \tilde{\varphi})(x_j) = \pi(\tilde{\varphi}(x_j)) = \pi(\tilde{x}_j) = \varphi(x_j).$$

Hence $\pi \circ \tilde{\varphi}$ and φ agree on a basis of L, so they are equal as R-module homomorphisms. Therefore the diagram commutes.

2. Apply part (1) with

$$X = M, \qquad Y = L, \qquad \varphi = \mathrm{id}_L\,.$$

Since $\pi : M \to L$ is an epimorphism, there exists an R-module homomorphism

$$\sigma : L \to M$$

such that

$$\pi \circ \sigma = \mathrm{id}_L\,.$$

Thus σ is a section of π.

We first show that σ is injective. If $\sigma(\ell) = 0$ for some $\ell \in L$, then

$$\ell = (\pi \circ \sigma)(\ell) = \pi(0) = 0.$$

Hence $\ker \sigma = 0$, so σ is injective.

Next, since σ is injective, it induces an isomorphism

$$L \xrightarrow{\sim} \mathrm{Im}(\sigma), \qquad \ell \mapsto \sigma(\ell).$$

Thus $\mathrm{Im}(\sigma) \cong L$.

It remains to prove that

$$M = \ker \pi \oplus \mathrm{Im}(\sigma).$$

First, let $m \in M$. Then

$$m = \big(m - \sigma(\pi(m))\big) + \sigma(\pi(m)).$$

Now

$$\pi\big(m - \sigma(\pi(m))\big) = \pi(m) - (\pi \circ \sigma)(\pi(m)) = \pi(m) - \pi(m) = 0,$$

so

$$m - \sigma(\pi(m)) \in \ker \pi,$$

while clearly

$$\sigma(\pi(m)) \in \mathrm{Im}(\sigma).$$

Hence

$$M = \ker \pi + \mathrm{Im}(\sigma).$$

Finally, suppose

$$z \in \ker \pi \cap \operatorname{Im}(\sigma).$$

Then $z = \sigma(\ell)$ for some $\ell \in L$, and also $\pi(z) = 0$. Therefore

$$0 = \pi(z) = \pi(\sigma(\ell)) = (\pi \circ \sigma)(\ell) = \ell.$$

So $z = \sigma(\ell) = 0$. Thus

$$\ker \pi \cap \operatorname{Im}(\sigma) = 0.$$

Therefore

$$M = \ker \pi \oplus \operatorname{Im}(\sigma).$$

□

Exercise 2.1.13. *Let L be a free R-module. Let P be a direct summand of L. Show that whenever*

$$\pi : X \to Y$$

is an epimorphism and

$$\varphi : P \to Y$$

is a morphism, there exists a morphism

$$\tilde{\varphi} : P \to X$$

such that

$$\varphi = \pi \circ \tilde{\varphi}.$$

Proof. Since P is a direct summand of L, there exist R-module homomorphisms

$$i : P \to L \qquad \text{and} \qquad p : L \to P$$

such that

$$p \circ i = \mathrm{id}_P .$$

(Here i is the inclusion and p is the projection.)

Consider the composite

$$\psi := \varphi \circ p : L \to Y.$$

Since L is free and $\pi : X \to Y$ is an epimorphism, by the lifting property of free modules there exists an R-module homomorphism

$$\widetilde{\psi} : L \to X$$

such that

$$\pi \circ \widetilde{\psi} = \psi.$$

Now define

$$\widetilde{\varphi} := \widetilde{\psi} \circ i : P \to X.$$

Then

$$\pi \circ \widetilde{\varphi} = \pi \circ \widetilde{\psi} \circ i = \psi \circ i = \varphi \circ p \circ i = \varphi \circ \mathrm{id}_P = \varphi.$$

Therefore there exists a morphism $\widetilde{\varphi} : P \to X$ such that

$$\varphi = \pi \circ \widetilde{\varphi}.$$

□

2.2 Tensor product

Let R be a commutative ring, and let M and N be R-modules. Roughly speaking, the tensor product $M \otimes_R N$ linearizes bilinear maps out of $M \times N$.

Definition 2.2.1. *Let R be a commutative ring, and let M, N, and K be R-modules. A map*

$$f\colon M \times N \to K$$

is called bilinear *if for all $r \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$, one has*

- $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$*;*
- $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$*;*
- $f(rm, n) = rf(m, n)$*;*
- $f(m, rn) = rf(m, n)$*.*

Example 2.2.2. *If $f\colon M \times N \to K$ is bilinear, then*

$$f(x, 0) = 0 = f(0, y)$$

for all $x \in M$ and $y \in N$. Moreover,

$$f(-x, y) = -f(x, y), \qquad f(x, -y) = -f(x, y)$$

for all $x \in M$ and $y \in N$.

Definition 2.2.3. *The* tensor product *of M and N is a pair $(M \otimes_R N, t_{M,N})$, where $M \otimes_R N$ is an R-module and*

$$t_{M,N}\colon M \times N \to M \otimes_R N$$

is a bilinear map, usually written $t_{M,N}(m, n) = m \otimes n$, such that the following universal property holds:
whenever X is an R-module and $f\colon M \times N \to X$ is bilinear, there exists a unique

R-module homomorphism

$$\tilde{f}\colon M \otimes_R N \to X$$

such that the following diagram commutes:

$$\begin{tikzcd}
M \times N \arrow[r, "f"] \arrow[d, "t_{M,N}"'] & X \\
M \otimes_R N \arrow[ur, dashed, "\tilde{f}"'] &
\end{tikzcd}$$

Theorem 2.2.4. *The tensor product of M and N exists and is unique up to isomorphism.*

Proof. We first prove uniqueness.

Suppose (T_1, t_1) and (T_2, t_2) are two tensor products of M and N. By the universal property, there exist unique R-module homomorphisms

$$\varphi\colon T_1 \to T_2, \qquad \psi\colon T_2 \to T_1$$

such that

$$t_2 = \varphi \circ t_1, \qquad t_1 = \psi \circ t_2.$$

Hence

$$\varphi \circ \psi \circ t_2 = \varphi \circ t_1 = t_2, \qquad \psi \circ \varphi \circ t_1 = \psi \circ t_2 = t_1.$$

By uniqueness in the universal property, it follows that

$$\varphi \circ \psi = \mathrm{id}_{T_2}, \qquad \psi \circ \varphi = \mathrm{id}_{T_1}.$$

Therefore φ and ψ are inverse isomorphisms, so $T_1 \cong T_2$.

Now we prove existence.

Let F be the free R-module with basis $M \times N$, say

$$F = \bigoplus_{(m,n) \in M \times N} R e_{m,n}.$$

Let B be the submodule of F generated by the following elements:

- $e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}$;
- $e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}$;
- $e_{rm,n} - r e_{m,n}$;
- $e_{m,rn} - r e_{m,n}$,

where $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$.

Define

$$M \otimes_R N = F/B, \qquad t_{M,N}(m,n) = e_{m,n} + B.$$

Then $t_{M,N}\colon M \times N \to M \otimes_R N$ is a well-defined bilinear map.

It remains to verify the universal property.

Let X be an R-module, and let $f\colon M \times N \to X$ be bilinear. Since F is free on the set $M \times N$, there exists a unique R-module homomorphism

$$\hat{f}\colon F \to X$$

such that $\hat{f}(e_{m,n}) = f(m,n)$ for all $(m,n) \in M \times N$.

We claim that $B \subseteq \ker \hat{f}$. Indeed, since f is bilinear,

$$\begin{aligned}
\hat{f}(e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}) &= f(m_1+m_2,n) - f(m_1,n) - f(m_2,n) = 0,\\
\hat{f}(e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}) &= f(m,n_1+n_2) - f(m,n_1) - f(m,n_2) = 0,\\
\hat{f}(e_{rm,n} - re_{m,n}) &= f(rm,n) - rf(m,n) = 0,\\
\hat{f}(e_{m,rn} - re_{m,n}) &= f(m,rn) - rf(m,n) = 0.
\end{aligned}$$

Since B is generated by these elements and $\ker \hat{f}$ is a submodule, we obtain $B \subseteq \ker \hat{f}$.

By the universal property of quotient modules, $\hat{f}$ induces a unique R-module homomorphism

$$\tilde{f}\colon M \otimes_R N = F/B \to X$$

such that $\tilde{f} \circ \pi = \hat{f}$, where $\pi\colon F \to F/B$ is the canonical projection. In particular,

$$\tilde{f}(m \otimes n) = \tilde{f}(e_{m,n} + B) = \hat{f}(e_{m,n}) = f(m,n),$$

so $\tilde{f} \circ t_{M,N} = f$.

For uniqueness, suppose $g\colon M \otimes_R N \to X$ is another R-module homomorphism such that $g \circ t_{M,N} = f$. Then

$$g(m \otimes n) = f(m,n) = \tilde{f}(m \otimes n)$$

for all $m \in M$ and $n \in N$. Since the elementary tensors generate $M \otimes_R N$ as an R-module, it follows that $g = \tilde{f}$.

Therefore $(M \otimes_R N, t_{M,N})$ is a tensor product of M and N. $\square$

Definition 2.2.5. *The set of elements of the form $m \otimes n$ for $m \in M$ and $n \in N$ is called the set of elementary tensors. This set generates $M \otimes_R N$ as an R-module.*

Example 2.2.6. *Fix $m \in M$. The map $N \to M \otimes_R N, n \mapsto m \otimes n$ is a morphism of R-modules. Its image $m \otimes_R N = \{m \otimes n \mid n \in N\}$ is a submodule of $M \otimes_R N$ which is isomorphic to a quotient $N/\operatorname{Ann}_R(m)N$.*

Proposition 2.2.7. *We have the following isomorphisms:*

1. *$(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$ where $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$.*

2. $M_1 \otimes_R M_2 \cong M_2 \otimes_R M_1$ *where* $m_1 \otimes m_2 \mapsto m_2 \otimes m_1$.

3. $M_1 \otimes_R (M_2 \oplus M_3) \cong (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R M_3)$ *where* $m_1 \otimes (m_2 \oplus m_3) \mapsto (m_1 \otimes m_2) + (m_1 \otimes m_3)$. *And more generally,*

$$M_1 \otimes_R (\bigoplus_{i=1}^{n} M_i) \cong \bigoplus_{i=1}^{n} (M_1 \otimes_R M_i)$$

where $m_1 \otimes (m_2 \oplus \cdots \oplus m_n) \mapsto (m_1 \otimes m_2) + \cdots + (m_1 \otimes m_n)$.

4. $R \otimes_R M \cong M$ *where* $r \otimes m \mapsto rm$. *And more generally,* $R^{(I)} \otimes_R M \cong M^{(I)}$ *where* $(r_i)_{i\in I} \otimes m \mapsto (r_i m)_{i\in I}$.

Proof. We prove (1):$(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$.

Step 1. We construct $\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$.

Fix $m_3 \in M_3$. For each m_3, define $f_{m_3} : M_1 \times M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $f_{m_3}(m_1, m_2) = m_1 \otimes (m_2 \otimes m_3)$. This is bilinear in (m_1, m_2), so by the universal property of $M_1 \otimes_R M_2$, there exists a unique R-module homomorphism $\tilde{f}_{m_3} : M_1 \otimes_R M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ such that $\tilde{f}_{m_3}(m_1 \otimes m_2) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd}
M_1 \times M_2 \arrow[r, "f_{m_3}"] \arrow[d, "t_{M_1,M_2}"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\
M_1 \otimes_R M_2 \arrow[ur, dashed, "\tilde{f}_{m_3}"'] &
\end{tikzcd}$$

Now define $g : (M_1 \otimes_R M_2) \times M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $g(\alpha, m_3) = \tilde{f}_{m_3}(\alpha)$. We verify that g is bilinear. Linearity in the first argument follows because each $\tilde{f}_{m_3}$ is an R-module homomorphism. For the second argument, on elementary tensors:

$$\begin{aligned}
g(m_1 \otimes m_2, m_3 + m_3') &= m_1 \otimes (m_2 \otimes (m_3 + m_3')) \\
&= m_1 \otimes (m_2 \otimes m_3 + m_2 \otimes m_3') \\
&= m_1 \otimes (m_2 \otimes m_3) + m_1 \otimes (m_2 \otimes m_3') \\
&= g(m_1 \otimes m_2, m_3) + g(m_1 \otimes m_2, m_3'), \\
g(m_1 \otimes m_2, rm_3) &= m_1 \otimes (m_2 \otimes rm_3) \\
&= m_1 \otimes (r(m_2 \otimes m_3)) \\
&= r(m_1 \otimes (m_2 \otimes m_3)) \\
&= r\, g(m_1 \otimes m_2, m_3).
\end{aligned}$$

Since elementary tensors generate $M_1 \otimes_R M_2$, bilinearity holds on all of $(M_1 \otimes_R M_2) \times M_3$.

By the universal property of $(M_1 \otimes_R M_2) \otimes_R M_3$, there exists a unique R-module homomorphism

$$\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$$

such that $\varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd}
(M_1 \otimes_R M_2) \times M_3 \arrow[r, "g"] \arrow[d, "t"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\
(M_1 \otimes_R M_2) \otimes_R M_3 \arrow[ur, dashed, "\varphi"'] &
\end{tikzcd}$$

Step 2. By an entirely symmetric argument (fixing m_1 first, then extending), we obtain an R-module homomorphism

$$\psi : M_1 \otimes_R (M_2 \otimes_R M_3) \to (M_1 \otimes_R M_2) \otimes_R M_3$$

such that $\psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3$.

Step 3. We show φ and ψ are inverses. On elementary tensors:

$$(\psi \circ \varphi)((m_1 \otimes m_2) \otimes m_3) = \psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3,$$

$$(\varphi \circ \psi)(m_1 \otimes (m_2 \otimes m_3)) = \varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3).$$

Since elementary tensors generate both modules,$\psi \circ \varphi = \mathrm{id}$ and $\varphi \circ \psi = \mathrm{id}$. The overall situation is summarized by the following commutative diagram:

$$\begin{tikzcd}
& M_1 \times M_2 \times M_3 \arrow[dl] \arrow[dr] & \\
(M_1 \otimes_R M_2) \otimes_R M_3 \arrow[rr, shift left, "\varphi"] & & M_1 \otimes_R (M_2 \otimes_R M_3) \arrow[ll, shift left, "\psi"]
\end{tikzcd}$$

where the diagonal arrows send (m_1, m_2, m_3) to $(m_1 \otimes m_2) \otimes m_3$ and $m_1 \otimes (m_2 \otimes m_3)$ respectively. Hence φ is an isomorphism with $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$. □

Theorem 2.2.8. *Let M, N, P be R-modules. We have a canonical isomorphism:*

$$\phi_{M,N,P} : \mathrm{Hom}_R(M \otimes_R N, P) \to \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$$

such that $\phi_{M,N,P}(f)(m)(n) = f(m \otimes n)$. Furthermore, if R is commutative, then $\phi_{M,N,P}$ is an R-module isomorphism. If R is not commutative, then $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism.

Proof. Let $f \in \mathrm{Hom}_R(M \otimes_R N, P)$. For each $m \in M$, define $\phi_{M,N,P}(f)(m) : N \to P$ by

$$\phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

We verify that $\phi_{M,N,P}(f)(m) \in \mathrm{Hom}_R(N, P)$: for $n_1, n_2 \in N$ and $r \in R$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m)(n_1 + n_2) &= f(m \otimes (n_1 + n_2)) = f(m \otimes n_1 + m \otimes n_2) \\
&= f(m \otimes n_1) + f(m \otimes n_2), \\
\phi_{M,N,P}(f)(m)(rn) &= f(m \otimes rn) = f(r(m \otimes n)) \\
&= rf(m \otimes n) = r\, \phi_{M,N,P}(f)(m)(n).
\end{aligned}$$

Next, we verify that $\phi_{M,N,P}(f) \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$: for $m_1, m_2 \in M$,$r \in R$, and any $n \in N$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m_1 + m_2)(n) &= f((m_1 + m_2) \otimes n) \\
&= f(m_1 \otimes n) + f(m_2 \otimes n) \\
\phi_{M,N,P}(f)(rm)(n) &= f(rm \otimes n) \\
&= f(r(m \otimes n)) \\
&= rf(m \otimes n) \\
&= (r\,\phi_{M,N,P}(f)(m))(n).
\end{aligned}$$

So

$$\phi_{M,N,P}(f)(m_1 + m_2) = \phi_{M,N,P}(f)(m_1) + \phi_{M,N,P}(f)(m_2)$$

and $\phi_{M,N,P}(f)(rm) = r\,\phi_{M,N,P}(f)(m)$, confirming R-linearity.

Let $g \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$. Define $\hat{g} : M \times N \to P$ by $\hat{g}(m, n) = g(m)(n)$. We check that $\hat{g}$ is R-bilinear:

$$\begin{aligned}
\hat{g}(m_1 + m_2, n) &= g(m_1 + m_2)(n) = g(m_1)(n) + g(m_2)(n) = \hat{g}(m_1, n) + \hat{g}(m_2, n), \\
\hat{g}(m, n_1 + n_2) &= g(m)(n_1 + n_2) = g(m)(n_1) + g(m)(n_2) = \hat{g}(m, n_1) + \hat{g}(m, n_2), \\
\hat{g}(rm, n) &= g(rm)(n) = (rg(m))(n) = rg(m)(n) = r\,\hat{g}(m, n), \\
\hat{g}(m, rn) &= g(m)(rn) = rg(m)(n) = r\,\hat{g}(m, n).
\end{aligned}$$

By the universal property of $M \otimes_R N$, there exists a unique R-module homomorphism $\psi_{M,N,P}(g) : M \otimes_R N \to P$ such that $\psi_{M,N,P}(g)(m \otimes n) = g(m)(n)$:

$$\begin{tikzcd}
M \times N \arrow[r, "\hat{g}"] \arrow[d, "t_{M,N}"'] & P \\
M \otimes_R N \arrow[ur, dashed, "\psi_{M,N,P}(g)"'] &
\end{tikzcd}$$

For any $f \in \mathrm{Hom}_R(M \otimes_R N, P)$:

$$\psi_{M,N,P}(\phi_{M,N,P}(f))(m \otimes n) = \phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

Since elementary tensors generate $M \otimes_R N$, we have $\psi_{M,N,P}(\phi_{M,N,P}(f)) = f$.

For any $g \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$, for all $m \in M$ and $n \in N$:

$$\phi_{M,N,P}(\psi_{M,N,P}(g))(m)(n) = \psi_{M,N,P}(g)(m \otimes n) = g(m)(n).$$

So $\phi_{M,N,P}(\psi_{M,N,P}(g))(m) = g(m)$ for all m, hence $\phi_{M,N,P}(\psi_{M,N,P}(g)) = g$.

Therefore $\phi_{M,N,P}$ is a bijection of sets (in fact, of abelian groups).

It remains to show that $\phi_{M,N,P}$ respects the appropriate module structure.

Both $\mathrm{Hom}_R(M \otimes_R N, P)$ and $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$ are abelian groups under pointwise addition. We verify $\phi_{M,N,P}$ is a group homomorphism: for $f_1, f_2 \in \mathrm{Hom}_R(M \otimes_R N, P)$,

$$\begin{aligned}\phi_{M,N,P}(f_1 + f_2)(m)(n) &= (f_1 + f_2)(m \otimes n) \\ &= f_1(m \otimes n) + f_2(m \otimes n) \\ &= \phi_{M,N,P}(f_1)(m)(n) + \phi_{M,N,P}(f_2)(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(f_1 + f_2) = \phi_{M,N,P}(f_1) + \phi_{M,N,P}(f_2)$. Hence $\phi_{M,N,P}$ is a $\mathbb{Z}$-module isomorphism.

If R is commutative,$\mathrm{Hom}_R(M \otimes_R N, P)$ carries an R-module structure via $(r \cdot f)(x) = rf(x)$, and similarly for $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$. Then

$$\begin{aligned}\phi_{M,N,P}(rf)(m)(n) &= (rf)(m \otimes n) \\ &= rf(m \otimes n) \\ &= r\,\phi_{M,N,P}(f)(m)(n) \\ &= (r\,\phi_{M,N,P}(f))(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(rf) = r\,\phi_{M,N,P}(f)$, and $\phi_{M,N,P}$ is an R-module isomorphism.

If R is not commutative,$\mathrm{Hom}_R(M \otimes_R N, P)$ does not naturally carry an R-module structure (scalar multiplication by r need not commute with R-linearity), so $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism. $\square$

Definition 2.2.9. *Let $f : M \to M'$ and $g : N \to N'$ be R-module homomorphisms. The tensor product of f and g is the R-module homomorphism:*

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

defined by $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$.

Definition 2.2.10. *Let $M_i, i \in I$ be a family of R-modules. Use the following notation:*

$$L^n(M_1, \ldots, M_n; N) = \{f : M_1 \times \cdots \times M_n \to N \mid f \text{ is } n\text{-linear }\}$$

In particular,

$$L^2(M_1, M_2; N) \cong L(M_1 \otimes M_2, N) \cong \mathrm{Hom}_R(M_1 \otimes M_2, N)$$

or equivalently,

$$L^2(M_1 \otimes M_2, N) \cong L(M_1, L(M_2, N))$$

Proposition 2.2.11. *Let M, N be two finitely generated free R-modules. Denote*

$$M^\vee = \mathrm{Hom}_R(M, R)$$

Then we have a canonical isomorphism:

$$M^\vee \otimes N \cong \mathrm{Hom}_R(M, N)$$

Proof. Since M is a finitely generated free R-module, we have $M \cong R^m$ for some $m \geq 1$. We construct an explicit isomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N).$$

Define a bilinear map $\beta : M^\vee \times N \to \mathrm{Hom}_R(M, N)$ by

$$\beta(f, n)(m) = f(m)\, n, \qquad f \in M^\vee,\ n \in N,\ m \in M.$$

We check R-bilinearity. For $f_1, f_2 \in M^\vee$,$n, n_1, n_2 \in N$,$m \in M$, and $r \in R$:

$$\begin{aligned}
\beta(f_1 + f_2, n)(m) &= (f_1 + f_2)(m)\, n = f_1(m)\, n + f_2(m)\, n \\
&= \beta(f_1, n)(m) + \beta(f_2, n)(m), \\
\beta(f, n_1 + n_2)(m) &= f(m)(n_1 + n_2) = f(m)\, n_1 + f(m)\, n_2 \\
&= \beta(f, n_1)(m) + \beta(f, n_2)(m), \\
\beta(rf, n)(m) &= (rf)(m)\, n = r\, f(m)\, n = r\, \beta(f, n)(m), \\
\beta(f, rn)(m) &= f(m)\, (rn) = r\, f(m)\, n = r\, \beta(f, n)(m).
\end{aligned}$$

By the universal property of the tensor product, there exists a unique R-module homomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N), \qquad \Phi(f \otimes n)(m) = f(m)\, n.$$

Now let $\{e_1, \ldots, e_m\}$ be a basis of $M \cong R^m$, and let $\{e_1^*, \ldots, e_m^*\}$ be the dual basis; that is,

$$e_i^* \in M^\vee, \qquad e_i^*(e_j) = \delta_{ij}.$$

Define

$$\Psi : \mathrm{Hom}_R(M, N) \to M^\vee \otimes_R N, \qquad \Psi(g) = \sum_{i=1}^m e_i^* \otimes g(e_i).$$

We verify Ψ is an R-module homomorphism: for $g_1, g_2 \in \mathrm{Hom}_R(M, N)$ and $r \in R$,

$$\begin{aligned}
\Psi(g_1 + g_2) &= \sum_{i=1}^m e_i^* \otimes (g_1 + g_2)(e_i) = \sum_{i=1}^m e_i^* \otimes (g_1(e_i) + g_2(e_i)) \\
&= \sum_{i=1}^m e_i^* \otimes g_1(e_i) + \sum_{i=1}^m e_i^* \otimes g_2(e_i) = \Psi(g_1) + \Psi(g_2), \\
\Psi(rg) &= \sum_{i=1}^m e_i^* \otimes (rg)(e_i) = \sum_{i=1}^m e_i^* \otimes r\, g(e_i) \\
&= r \sum_{i=1}^m e_i^* \otimes g(e_i) = r\, \Psi(g).
\end{aligned}$$

We show Φ and Ψ are mutual inverses. For any $g \in \mathrm{Hom}_R(M,N)$ and $m = \sum_j r_j e_j \in M$:

$$\begin{aligned}
\Phi(\Psi(g))(m) &= \Phi\Big(\sum_{i=1}^{m} e_i^* \otimes g(e_i)\Big)\Big(\sum_j r_j e_j\Big) \\
&= \sum_{i=1}^{m} e_i^*\Big(\sum_j r_j e_j\Big) g(e_i) = \sum_{i=1}^{m} r_i\, g(e_i) \\
&= g\Big(\sum_{i=1}^{m} r_i e_i\Big) = g(m).
\end{aligned}$$

So $\Phi \circ \Psi = \mathrm{id}$.

For any elementary tensor $f \otimes n \in M^\vee \otimes_R N$:

$$\begin{aligned}
\Psi(\Phi(f \otimes n)) &= \sum_{i=1}^{m} e_i^* \otimes \Phi(f \otimes n)(e_i) = \sum_{i=1}^{m} e_i^* \otimes f(e_i)\, n \\
&= \sum_{i=1}^{m} f(e_i)\,(e_i^* \otimes n) = \Big(\sum_{i=1}^{m} f(e_i)\, e_i^*\Big) \otimes n = f \otimes n,
\end{aligned}$$

where the last equality uses $f = \sum_{i=1}^{m} f(e_i)\, e_i^*$(expansion of f in the dual basis). Since elementary tensors generate $M^\vee \otimes_R N$ and $\Psi \circ \Phi$ is R-linear, we have $\Psi \circ \Phi = \mathrm{id}$.

Therefore Φ is an R-module isomorphism, giving $M^\vee \otimes_R N \cong \mathrm{Hom}_R(M,N)$. $\square$

Proposition 2.2.12. *Assume that R is a subring of S, and M is an R-module. Then the bilinear map:*

$$S \times (S \otimes_R M) \to S \otimes_R M$$

defined by

$$(s, s' \otimes m) \mapsto ss' \otimes m$$

defines a structure of S-module on the R-module $S \otimes_R M$.
More generally, let $f : R \to T$ be a ring homomorphism. And view T as an R-module via $R \times T \to T, (\lambda, \mu) \mapsto f(\lambda)\mu$. Then the bilinear map:

$$T \times (T \otimes_R M) \to T \otimes_R M$$

defined by

$$(t, t' \otimes m) \mapsto tt' \otimes m$$

defines a structure of T-module on the R-module $T \otimes_R M$.

Proof. It suffices to prove the more general statement in (2), since (1) is the special case where $T = S$ and $f : R \hookrightarrow S$ is the inclusion map.

For each fixed $t \in T$, define a map

$$\beta_t : T \times M \longrightarrow T \otimes_R M, \qquad (t', m) \longmapsto tt' \otimes m.$$

This map is R-bilinear. Hence, by the universal property of the tensor product, there exists a unique R-linear map

$$\mu_t : T \otimes_R M \longrightarrow T \otimes_R M$$

such that

$$\mu_t(t' \otimes m) = tt' \otimes m \qquad \text{for all } t' \in T,\ m \in M.$$

Now define

$$t \cdot x := \mu_t(x), \qquad t \in T,\ x \in T \otimes_R M.$$

We claim that this gives a left T-module structure on $T \otimes_R M$.

Since the elementary tensors $t' \otimes m$ generate $T \otimes_R M$ as an abelian group, it is enough to verify the module axioms on elementary tensors.

Let $t_1, t_2, t' \in T$ and $m \in M$. Then

$$(t_1 + t_2) \cdot (t' \otimes m) = (t_1 + t_2)t' \otimes m = t_1t' \otimes m + t_2t' \otimes m = t_1 \cdot (t' \otimes m) + t_2 \cdot (t' \otimes m).$$

Also,

$$(t_1t_2) \cdot (t' \otimes m) = t_1t_2t' \otimes m = t_1 \cdot (t_2t' \otimes m) = t_1 \cdot \big(t_2 \cdot (t' \otimes m)\big).$$

Moreover,

$$1_T \cdot (t' \otimes m) = 1_Tt' \otimes m = t' \otimes m.$$

Finally, for any $x, y \in T \otimes_R M$,

$$t \cdot (x + y) = \mu_t(x + y) = \mu_t(x) + \mu_t(y) = t \cdot x + t \cdot y,$$

because μ_t is R-linear, hence additive.

Therefore the operation

$$T \times (T \otimes_R M) \longrightarrow T \otimes_R M, \qquad (t,\, t' \otimes m) \longmapsto tt' \otimes m$$

defines a left T-module structure on $T \otimes_R M$.

This proves (2), and hence also (1). □

Proposition 2.2.13. *Let I be an ideal of R. Let M be an R-module. Then the morphism of R-modules:*

$$M \to (R/I) \otimes_R M$$

defined by

$$m \mapsto 1_{R/I} \otimes m$$

induces an isomorphism of R-modules:

$$M/(IM) \cong (R/I) \otimes_R M$$

as R/I-modules.

Proof. Define
$$\varphi : M \longrightarrow (R/I) \otimes_R M, \qquad m \longmapsto 1_{R/I} \otimes m.$$
This is an R-module homomorphism. For any $a \in I$ and $m \in M$, we have
$$\varphi(am) = 1_{R/I} \otimes am = (1_{R/I} \cdot a) \otimes m = (a + I) \otimes m = 0,$$
since $a + I = 0$ in R/I. Hence $\varphi(IM) = 0$, so $IM \subseteq \ker(\varphi)$.

Therefore, by the universal property of the quotient,φ induces an R-module homomorphism
$$\overline{\varphi} : M/IM \longrightarrow (R/I) \otimes_R M, \qquad m + IM \longmapsto 1_{R/I} \otimes m.$$

Now define
$$\beta : (R/I) \times M \longrightarrow M/IM, \qquad (r + I, m) \longmapsto rm + IM.$$
We first check that β is well defined. If $r + I = r' + I$, then $r - r' \in I$, so
$$(r - r')m \in IM,$$
hence
$$rm + IM = r'm + IM.$$
Also,β is R-balanced, since for any $a \in R$,
$$\beta((r + I)a, m) = \beta(ra + I, m) = ram + IM = \beta(r + I, am).$$
Thus, by the universal property of the tensor product, there exists a unique homomorphism
$$\psi : (R/I) \otimes_R M \longrightarrow M/IM$$
such that
$$\psi((r + I) \otimes m) = rm + IM \qquad \text{for all } r \in R,\ m \in M.$$

We now show that $\overline{\varphi}$ and ψ are inverse to each other.

For any $m \in M$,
$$(\psi \circ \overline{\varphi})(m + IM) = \psi(1_{R/I} \otimes m) = 1m + IM = m + IM.$$
So
$$\psi \circ \overline{\varphi} = \mathrm{id}_{M/IM}.$$

For any elementary tensor $(r + I) \otimes m \in (R/I) \otimes_R M$,
$$(\overline{\varphi} \circ \psi)\big((r + I) \otimes m\big) = \overline{\varphi}(rm + IM) = 1_{R/I} \otimes rm = (r + I) \otimes m.$$

Since elementary tensors generate $(R/I) \otimes_R M$, it follows that

$$\overline{\varphi} \circ \psi = \mathrm{id}_{(R/I)\otimes_R M}.$$

Hence $\overline{\varphi}$ is an isomorphism:

$$M/IM \cong (R/I) \otimes_R M.$$

Finally, both modules carry natural R/I-module structures, and the maps $\overline{\varphi}$ and ψ are R/I-linear. Therefore this is an isomorphism of R/I-modules. $\square$

2.3 Balanced product

Definition 2.3.1. *Let R be a ring (not necessarily commutative). Let M_R be a right R-module and ${}_RN$ be a left R-module. The balanced product of M and N is defined as a pair (P, f) where P is an abelian group and $f : M \times N \to P$ is a bilinear map satisfies:*

- $f(rm, n) = f(m, rn)$ *for all* $r \in R$*,*$m \in M$*,*$n \in N$.
- $f(m + m', n) = f(m, n) + f(m', n)$ *for all* $m, m' \in M$*,*$n \in N$.
- $f(m, n + n') = f(m, n) + f(m, n')$ *for all* $m \in M$*,*$n, n' \in N$.

Definition 2.3.2. *A tensor product of M_R and ${}_RN$ is a balanced product $(M \otimes_R N, t_{M,N})$ such that if (P, f) is any balanced product of M_R and ${}_RN$, then there exists a unique abelian group homomorphism $\tilde{f} : M \otimes_R N \to P$ such that the following diagram commutes:*

$$\begin{tikzcd} M \times N \arrow[r, "t_{M,N}"] \arrow[d, "f"'] & M \otimes_R N \arrow[dl, dashed, "\tilde{f}"'] \\ P & \end{tikzcd}$$

Chapter 3

Lecture 3: Categories, Bimodules

3.1 Categories

Definition 3.1.1. *A category $\mathcal{C}$ consists of:*

- *A class of objects* $\mathrm{ob}(\mathcal{C})$.
- *A set of morphisms* $\mathrm{Hom}_{\mathcal{C}}(A, B)$ *for each pair of objects* $A, B \in \mathcal{C}$.
- *A composition map* $\mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$ *such that* $(\phi, \psi) \mapsto \phi \circ \psi$ *for each triple of objects* $A, B, C \in \mathcal{C}$.

It is assumed that objects and morphisms satisfy the following conditions:

- If $(A, B) \neq (C, D)$, then
$$\mathrm{Hom}(A, B) \cap \mathrm{Hom}(C, D) = \varnothing.$$
- If
$$f \in \mathrm{Hom}(A, B), \qquad g \in \mathrm{Hom}(B, C), \qquad h \in \mathrm{Hom}(C, D),$$
then
$$(h \circ g) \circ f = h \circ (g \circ f).$$
- (Unit) For every object A we have an element
$$1_A \in \mathrm{Hom}(A, A)$$
such that
$$f \circ 1_A = f \qquad \text{for any } f \in \mathrm{Hom}(A, B),$$
and
$$1_A \circ g = g \qquad \text{for any } g \in \mathrm{Hom}(B, A).$$

Example 3.1.2. *The following are examples of categories.*

1. *The category of sets: the objects are sets, and the morphisms are maps. It is denoted by* **Set**.

2. *The category of groups: the objects are groups, and the morphisms are group homomorphisms. It is denoted by* **Grp**.

3. *The category of abelian groups: the objects are abelian groups, and the morphisms are group homomorphisms. It is denoted by* **Ab**.

4. *The category of R-modules: the objects are R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-***Mod***(left R-modules, for right R-modules, we use R-***Mod***).*

5. *The category of finitely generated R-modules: the objects are finitely generated R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-***mod**.

6. *The category of topological spaces: the objects are topological spaces, and the morphisms are continuous maps. It is denoted by* **Top**.

Definition 3.1.3 (Subcategory)**.** *A category $\mathcal{D}$ is called a* subcategory *of a category $\mathcal{C}$ if*

1. *the objects of $\mathcal{D}$ form a subclass of the objects of $\mathcal{C}$;*

2. *for any objects A, B of $\mathcal{D}$, we have*

   - $\operatorname{Hom}_{\mathcal{D}}(A, B) \subseteq \operatorname{Hom}_{\mathcal{C}}(A, B)$;
   - *the identity morphisms and the composition of morphisms in $\mathcal{D}$ are the same as those in $\mathcal{C}$.*

The subcategory $\mathcal{D}$ of $\mathcal{C}$ is called full *if*

$$\operatorname{Hom}_{\mathcal{D}}(A, B) = \operatorname{Hom}_{\mathcal{C}}(A, B)$$

for any objects A, B of $\mathcal{D}$.

Example 3.1.4. *The category* **Ab** *is a full subcategory of* **Grp**.
*The category R-***mod** *is a full subcategory of R-***Mod**.
The category **Grp** *is not a full subcategory of* **Set**.

Definition 3.1.5. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A* covariant functor

$$F : \mathcal{C} \to \mathcal{D}$$

consists of the following data:

- *for every object A of $\mathcal{C}$, an object $F(A)$ of $\mathcal{D}$;*
- *for each pair of objects A, B of $\mathcal{C}$, a map*

$$F : \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B)), \qquad f \mapsto F(f).$$

These data are required to satisfy the following conditions:

1. *If $g \circ f$ is defined in $\mathcal{C}$, then*

$$F(g) \circ F(f) = F(g \circ f).$$

2. *For every object A of $\mathcal{C}$,*

$$F(1_A) = 1_{F(A)}.$$

Proposition 3.1.6. *Functors preserve diagram:*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g"'] & B \arrow[d, "h"] \\ & C \end{tikzcd} \qquad \xrightarrow{F} \qquad \begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[dr, "F(g)"'] & F(B) \arrow[d, "F(h)"] \\ & F(AC \end{tikzcd}$$

Let M and N be R-modules. Then

$$\mathrm{Hom}_R(M, N) = \{\, f : M \to N \mid f \text{ is an } R\text{-module homomorphism} \,\}.$$

This is an abelian group.

Moreover,

$$\mathrm{End}_R(M) = \mathrm{Hom}_R(M, M),$$

and $\mathrm{End}_R(M)$ is a ring.

Example 3.1.7. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab}$$

is a functor.

More precisely, it acts as follows:

$$N \longmapsto \mathrm{Hom}_R(M, N), \qquad f : N \to N' \longmapsto \mathrm{Hom}_R(M, f).$$

Given an R-module homomorphism

$$f : N \to N',$$

the induced map

$$\mathrm{Hom}_R(M, f) : \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(M, N')$$

is defined by

$$\mathrm{Hom}_R(M, f)(h) = f \circ h \qquad (h \in \mathrm{Hom}_R(M, N)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd} & M \arrow[dl, "h"'] \arrow[dr, "f\circ h"] & \\ N \arrow[rr, "f"'] & & N' \end{tikzcd}$$

Hence

$$\mathrm{Hom}_R(M, -)$$

is a covariant functor.

Definition 3.1.8. *Let $\mathcal{C}$ be a category. The* opposite category *(or* dual category*) $\mathcal{C}^{\mathrm{op}}$ is defined as follows:*

- $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$*;*
- *for any objects A, B,*

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A).$$

The composition in $\mathcal{C}^{\mathrm{op}}$ is defined by reversing the order of composition in $\mathcal{C}$: if

$$f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B), \qquad g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B, C),$$

then

$$g \circ f \text{ in } \mathcal{C}^{\mathrm{op}}$$

is defined to be

$$f \circ g \text{ in } \mathcal{C}.$$

We call $\mathcal{C}^{\mathrm{op}}$ the dual category of $\mathcal{C}$.

This is illustrated by the following diagrams:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g\circ f"'] & B \arrow[d, "g"] \\ & C \end{tikzcd} \qquad \text{in } \mathcal{C}^{\mathrm{op}},$$

while the corresponding picture in $\mathcal{C}$ is

$$\begin{tikzcd} A & B \arrow[l, "f"'] \\ & C \arrow[ul, "f\circ g"] \arrow[u, "g"'] \end{tikzcd} \qquad \text{in } \mathcal{C}.$$

Definition 3.1.9. *A* contravariant functor *from a category $\mathcal{C}$ to a category $\mathcal{D}$ is a covariant functor*

$$\mathcal{C}^{\mathrm{op}} \to \mathcal{D}.$$

Example 3.1.10. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab}$$

is a contravariant functor.

More precisely, it is defined on objects by

$$N \longmapsto \mathrm{Hom}_R(N, M),$$

and on morphisms by sending an R-module homomorphism

$$f : N \to N'$$

to the induced group homomorphism

$$\mathrm{Hom}_R(f, M) : \mathrm{Hom}_R(N', M) \to \mathrm{Hom}_R(N, M),$$

given by

$$\mathrm{Hom}_R(f, M)(h) = h \circ f \qquad (h \in \mathrm{Hom}_R(N', M)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd} N \arrow[d, "f"'] \arrow[dr, "h\circ f"] & \\ N' \arrow[r, "h"'] & M \end{tikzcd}$$

3.2 Exact Sequences

Definition 3.2.1. *Let $\{M_i\}_{i\in\mathbb{Z}}$ be a family of R-modules, and let*

$$\varphi_i : M_i \to M_{i+1}$$

be R-module homomorphisms for each $i \in \mathbb{Z}$. Then

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is called a sequence *of R-modules.*

Definition 3.2.2. *The sequence*

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is said to be exact at M_i *if*

$$\mathrm{Im}(\varphi_{i-1}) = \ker(\varphi_i).$$

It is called an exact sequence *if it is exact at every term.*

A short exact sequence is a sequence of the form

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0.$$

It is exact if and only if

$$\ker(\iota) = 0, \qquad \mathrm{Im}(\pi) = M'', \qquad \mathrm{Im}(\iota) = \ker(\pi).$$

Equivalently,ι is injective,π is surjective, and

$$\mathrm{Im}(\iota) = \ker(\pi).$$

Hence, for every short exact sequence

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0,$$

we have

$$M/M' \cong M''.$$

Definition 3.2.3. *Let R, S be rings, and let*

$$F : R\text{-Mod} \to S\text{-Mod}$$

be a covariant functor.
We say that F is left exact *if for any exact sequence*

$$0 \to N \xrightarrow{\varphi} M \xrightarrow{\psi} P,$$

the sequence

$$0 \to F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P)$$

is exact.
We say that F is right exact *if for any exact sequence*

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} P \to 0,$$

the sequence

$$F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \to 0$$

is exact.

Definition 3.2.4. *Let*

$$G : R\text{-Mod} \to S\text{-Mod}$$

be a contravariant functor. We say that G is left exact *(respectively,* right exact*) if it is left exact (respectively, right exact) when regarded as a covariant functor*

$$G : (R\text{-Mod})^{\mathrm{op}} \to S\text{-Mod}.$$

A functor is called exact *if it is both left exact and right exact.*

Let M be an R-module. Then:

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is left exact,}$$

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is left exact,}$$

and

$$M \otimes_R - : R\text{-Mod} \to \mathrm{Ab} \quad \text{is right exact.}$$

Definition 3.2.5. *A functor $F : \mathcal{C} \to \mathcal{D}$ is called* exact *if it is both left exact and right exact.*

Definition 3.2.6. *Let M be an R-module.*

1. *M is called* projective *if the functor*

$$\mathrm{Hom}_R(M, -)$$

   is exact.

2. *M is called* injective *if the functor*

$$\mathrm{Hom}_R(-, M)$$

   is exact.

3. *M is called* flat *if the functor*

$$- \otimes_R M$$

is exact.

If R is noncommutative, then in order for

$$M \otimes_R -$$

to be well-defined,M must be a right R-module.

3.3 Bimodules

Definition 3.3.1. *Let R and S be rings. An abelian group M is called a* left-R-right-S-bimodule *if M is both a left R-module and a right S-module such that the two scalar multiplications satisfy*

$$r(xs) = (rx)s \qquad \textit{for all } r \in R,\ s \in S,\ x \in M.$$

Notation.

$$_RM \qquad \text{left } R\text{-module,}$$

$$M_S \qquad \text{right } S\text{-module,}$$

$$_RM_S \qquad \text{left } R\text{-right } S\text{-bimodule.}$$

Let R, S be rings, and let $_RU_S$ be a bimodule.
For each left R-module $_RM$, there are two naturally induced S-module structures:

$$\mathrm{Hom}_R(_RU_S, {_RM}), \qquad \text{a left } S\text{-module,}$$

$$\mathrm{Hom}_R(_RM, {_RU_S}), \qquad \text{a right } S\text{-module.}$$

More precisely, for each $s \in S$:
if

$$f \in \mathrm{Hom}_R(_RU_S, {_RM}),$$

define

$$s \cdot f : U \to M, \qquad u \mapsto f(us).$$

Similarly, if

$$f \in \mathrm{Hom}_R(_RM, {_RU_S}),$$

define

$$f \cdot s : M \to U, \qquad m \mapsto f(m)s.$$

Thus we obtain functors

$$\mathrm{Hom}_R({}_RU_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod},$$

given on objects by

$${}_RM \longmapsto \mathrm{Hom}_R({}_RU_S, {}_RM),$$

which is covariant, and

$$\mathrm{Hom}_R(-, {}_RU_S) : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Mod}\text{-}S,$$

given on objects by

$${}_RM \longmapsto \mathrm{Hom}_R({}_RM, {}_RU_S),$$

which is contravariant.

On morphisms, for

$$f : M \to M',$$

the induced map

$$\mathrm{Hom}_R({}_RU_S, {}_RM) \longrightarrow \mathrm{Hom}_R({}_RU_S, {}_RM')$$

is given by

$$h \longmapsto f \circ h.$$

the induced map

$$\mathrm{Hom}_R({}_RM', {}_RU_S) \longrightarrow \mathrm{Hom}_R({}_RM, {}_RU_S)$$

is given by

$$h \longmapsto h \circ f.$$

Theorem 3.3.2. *The functor*

$$\mathrm{Hom}_R({}_RU_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod}$$

is left exact.

Proof. Let

$$0 \longrightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$$

be an exact sequence in R-**Mod**. Consider the induced sequence

$$0 \longrightarrow \mathrm{Hom}_R({}_RU_S, N') \xrightarrow{\mathrm{Hom}_R(U,f)} \mathrm{Hom}_R({}_RU_S, N) \xrightarrow{\mathrm{Hom}_R(U,g)} \mathrm{Hom}_R({}_RU_S, N'').$$

We show that this sequence is exact.

First,$\mathrm{Hom}_R(U, f)$ is injective. Let

$$\varphi \in \mathrm{Hom}_R(U, N')$$

and suppose that

$$\mathrm{Hom}_R(U,f)(\varphi) = f \circ \varphi = 0.$$

Then for every $u \in U$,

$$f(\varphi(u)) = 0.$$

Since f is injective, it follows that $\varphi(u) = 0$ for all $u \in U$. Hence $\varphi = 0$, so $\mathrm{Hom}_R(U,f)$ is injective.

Next, we show that

$$\mathrm{Im}\big(\mathrm{Hom}_R(U,f)\big) \subseteq \ker\big(\mathrm{Hom}_R(U,g)\big).$$

Indeed, if

$$\varphi \in \mathrm{Hom}_R(U,N'),$$

then

$$\mathrm{Hom}_R(U,g)\big(\mathrm{Hom}_R(U,f)(\varphi)\big) = g \circ f \circ \varphi = 0,$$

because $g \circ f = 0$. Therefore

$$\mathrm{Im}\big(\mathrm{Hom}_R(U,f)\big) \subseteq \ker\big(\mathrm{Hom}_R(U,g)\big).$$

Finally, let

$$\psi \in \ker\big(\mathrm{Hom}_R(U,g)\big).$$

Then $\psi \in \mathrm{Hom}_R(U,N)$ and

$$g \circ \psi = 0.$$

Thus for every $u \in U$,

$$g(\psi(u)) = 0,$$

so $\psi(u) \in \ker g = \mathrm{Im}\, f$ by exactness. Hence for each $u \in U$ there exists some $y \in N'$ such that

$$f(y) = \psi(u).$$

Since f is injective, such y is unique. Define

$$\varphi : U \to N', \qquad u \mapsto y,$$

where y is the unique element satisfying $f(y) = \psi(u)$.

Then

$$f \circ \varphi = \psi.$$

It remains to check that φ is R-linear. For $u, v \in U$,

$$f(\varphi(u+v)) = \psi(u+v) = \psi(u) + \psi(v) = f(\varphi(u)) + f(\varphi(v)) = f(\varphi(u) + \varphi(v)).$$

Since f is injective,

$$\varphi(u+v) = \varphi(u) + \varphi(v).$$

Similarly, for $r \in R$ and $u \in U$,

$$f(\varphi(ru)) = \psi(ru) = r\psi(u) = rf(\varphi(u)) = f(r\varphi(u)),$$

so again by injectivity of f,

$$\varphi(ru) = r\varphi(u).$$

Thus $\varphi \in \mathrm{Hom}_R(U, N')$, and

$$\mathrm{Hom}_R(U, f)(\varphi) = f \circ \varphi = \psi.$$

Therefore

$$\ker\big(\mathrm{Hom}_R(U, g)\big) \subseteq \mathrm{Im}\big(\mathrm{Hom}_R(U, f)\big).$$

Combining the two inclusions, we get

$$\mathrm{Im}\big(\mathrm{Hom}_R(U, f)\big) = \ker\big(\mathrm{Hom}_R(U, g)\big).$$

Hence

$$0 \longrightarrow \mathrm{Hom}_R({}_RU_S, N') \xrightarrow{\mathrm{Hom}_R(U,f)} \mathrm{Hom}_R({}_RU_S, N) \xrightarrow{\mathrm{Hom}_R(U,g)} \mathrm{Hom}_R({}_RU_S, N'')$$

is exact, and so the functor $\mathrm{Hom}_R({}_RU_S, -)$ is left exact. □

Theorem 3.3.3. *Let ${}_RU_S$ be an R-S-bimodule. Then the contravariant functor*

$$\mathrm{Hom}_R(-, U) : R\text{-Mod} \longrightarrow \text{Mod-}S$$

is left exact.

Proof. Since $\mathrm{Hom}_R(-, U)$ is contravariant, to say that it is left exact means the following: for every short exact sequence of left R-modules

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0,$$

the induced sequence of right S-modules

$$0 \longrightarrow \mathrm{Hom}_R(C, U) \xrightarrow{\mathrm{Hom}_R(g,U)} \mathrm{Hom}_R(B, U) \xrightarrow{\mathrm{Hom}_R(f,U)} \mathrm{Hom}_R(A, U)$$

is exact.

We verify this.

First,$\mathrm{Hom}_R(g, U)$ is injective. Indeed, let $\varphi \in \mathrm{Hom}_R(C, U)$ and suppose

$$\mathrm{Hom}_R(g, U)(\varphi) = \varphi \circ g = 0.$$

Since g is surjective, it follows that $\varphi = 0$. Hence $\mathrm{Hom}_R(g, U)$ is injective.

Next, we show

$$\operatorname{Im}\big(\operatorname{Hom}_R(g,U)\big) \subseteq \ker\big(\operatorname{Hom}_R(f,U)\big).$$

Let $\varphi \in \operatorname{Hom}_R(C,U)$. Then

$$\operatorname{Hom}_R(f,U)\big(\operatorname{Hom}_R(g,U)(\varphi)\big) = (\varphi \circ g) \circ f = \varphi \circ (g \circ f) = 0,$$

because $g \circ f = 0$. Therefore

$$\operatorname{Im}\big(\operatorname{Hom}_R(g,U)\big) \subseteq \ker\big(\operatorname{Hom}_R(f,U)\big).$$

Finally, we prove the reverse inclusion. Let $\psi \in \operatorname{Hom}_R(B,U)$ satisfy

$$\operatorname{Hom}_R(f,U)(\psi) = \psi \circ f = 0.$$

Then ψ vanishes on $\operatorname{Im}(f) = \ker(g)$. Hence ψ factors through the quotient

$$B/\ker(g).$$

Since g is surjective and $\ker(g) = \operatorname{Im}(f)$, we have

$$B/\ker(g) \cong C.$$

Therefore there exists a unique R-module homomorphism $\overline{\psi} : C \to U$ such that

$$\psi = \overline{\psi} \circ g.$$

Thus

$$\psi \in \operatorname{Im}\big(\operatorname{Hom}_R(g,U)\big),$$

so

$$\ker\big(\operatorname{Hom}_R(f,U)\big) \subseteq \operatorname{Im}\big(\operatorname{Hom}_R(g,U)\big).$$

Combining the two inclusions, we obtain

$$\operatorname{Im}\big(\operatorname{Hom}_R(g,U)\big) = \ker\big(\operatorname{Hom}_R(f,U)\big).$$

Hence the sequence

$$0 \longrightarrow \operatorname{Hom}_R(C,U) \xrightarrow{\operatorname{Hom}_R(g,U)} \operatorname{Hom}_R(B,U) \xrightarrow{\operatorname{Hom}_R(f,U)} \operatorname{Hom}_R(A,U)$$

is exact. This proves that $\operatorname{Hom}_R(-,U)$ is left exact. $\square$

Let ${}_SM_R$ be a left-S-right-R-bimodule,${}_RN$ be a left R-module, consider the tensor product $M \otimes_R N$.

For each $s \in S$, the mapping

$$M \times N \longrightarrow M \otimes_R N, \qquad (m,n) \longmapsto (sm) \otimes n$$

is balanced.

So there is a unique abelian group homomorphism

$$u(s) : M \otimes_R N \longrightarrow M \otimes_R N$$

such that

$$u(s)(m \otimes n) = (sm) \otimes n.$$

Only need to check:

$$u : S \longrightarrow \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$$

is a ring homomorphism.

If M_R, ${}_R N_T$, then $M \otimes_R N$ is a right T-module:

$$(m \otimes n)t = m \otimes (nt).$$

Lemma 3.3.4. *Let ${}_S M_R$ and ${}_R N_T$ be bimodules. Then $M \otimes_R N$ carries a natural structure of a left S-right T-bimodule, defined on elementary tensors by*

$$s(m \otimes n) = (sm) \otimes n, \qquad (m \otimes n)t = m \otimes (nt),$$

for all $s \in S$,$t \in T$,$m \in M$, and $n \in N$.

Proof. Fix $s \in S$. Define

$$\beta_s : M \times N \to M \otimes_R N, \qquad \beta_s(m, n) = sm \otimes n.$$

We claim that β_s is R-balanced. Indeed, it is additive in each variable, and for $r \in R$,

$$\beta_s(mr, n) = s(mr) \otimes n = (sm)r \otimes n = sm \otimes rn = \beta_s(m, rn).$$

Hence, by the universal property of the tensor product, there exists a unique group homomorphism

$$\lambda_s : M \otimes_R N \to M \otimes_R N$$

such that

$$\lambda_s(m \otimes n) = sm \otimes n.$$

We write $\lambda_s(x) = sx$.

Similarly, for fixed $t \in T$, define

$$\gamma_t : M \times N \to M \otimes_R N, \qquad \gamma_t(m, n) = m \otimes nt.$$

Again γ_t is R-balanced, since for $r \in R$,

$$\gamma_t(mr, n) = mr \otimes nt = m \otimes r(nt) = m \otimes (rn)t = \gamma_t(m, rn).$$

Thus there exists a unique group homomorphism

$$\rho_t : M \otimes_R N \to M \otimes_R N$$

such that

$$\rho_t(m \otimes n) = m \otimes nt.$$

We write $\rho_t(x) = xt$.

It remains to verify the bimodule axioms. Since elementary tensors generate $M \otimes_R N$, it suffices to check them on $m \otimes n$:

$$(s_1 + s_2)(m \otimes n) = ((s_1 + s_2)m) \otimes n = (s_1m) \otimes n + (s_2m) \otimes n = s_1(m \otimes n) + s_2(m \otimes n),$$

$$(s_1s_2)(m \otimes n) = ((s_1s_2)m) \otimes n = s_1((s_2m) \otimes n) = s_1\big(s_2(m \otimes n)\big),$$

$$1_S(m \otimes n) = (1_Sm) \otimes n = m \otimes n,$$

and similarly

$$(m \otimes n)(t_1 + t_2) = m \otimes n(t_1 + t_2) = (m \otimes nt_1) + (m \otimes nt_2),$$

$$((m \otimes n)t_1)t_2 = (m \otimes nt_1)t_2 = m \otimes (nt_1)t_2 = m \otimes n(t_1t_2) = (m \otimes n)(t_1t_2),$$

$$(m \otimes n)1_T = m \otimes n.$$

Finally,

$$\big(s(m \otimes n)\big)t = ((sm) \otimes n)t = (sm) \otimes nt = s(m \otimes nt) = s\big((m \otimes n)t\big).$$

Therefore $M \otimes_R N$ is a left S-right T-bimodule. □

Lemma 3.3.5. *Let M_R,M'_R be right R-modules, and let ${}_RN$,${}_RN'$ be left R-modules. Suppose*

$$f_1, f_2, f \in \mathrm{Hom}_R(M, M'), \qquad g_1, g_2, g \in \mathrm{Hom}_R(N, N').$$

Then for each pair (f, g) there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n) \qquad (m \in M,\ n \in N).$$

Moreover,

1. *$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g)$;*

2. *$f \otimes (g_1 + g_2) = (f \otimes g_1) + (f \otimes g_2)$;*

3. *$f \otimes 0 = 0$ and $0 \otimes g = 0$;*

4. $1_M \otimes 1_N = 1_{M \otimes_R N}$.

Proof. Define

$$\beta : M \times N \to M' \otimes_R N', \qquad \beta(m, n) = f(m) \otimes g(n).$$

We check that β is R-balanced. It is additive in each variable because f and g are module homomorphisms. Also, for $r \in R$,

$$\beta(mr, n) = f(mr) \otimes g(n) = f(m)r \otimes g(n) = f(m) \otimes rg(n) = f(m) \otimes g(rn) = \beta(m, rn).$$

Hence, by the universal property of $M \otimes_R N$, there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

Now each asserted identity is verified on elementary tensors, and therefore on all of $M \otimes_R N$.

For (1), for all $m \in M$ and $n \in N$,

$$\big(((f_1 + f_2) \otimes g)(m \otimes n)\big) = (f_1 + f_2)(m) \otimes g(n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n),$$

while

$$\big((f_1 \otimes g) + (f_2 \otimes g)\big)(m \otimes n) = (f_1 \otimes g)(m \otimes n) + (f_2 \otimes g)(m \otimes n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n).$$

Hence

$$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g).$$

The proof of (2) is analogous:

$$(f \otimes (g_1 + g_2))(m \otimes n) = f(m) \otimes (g_1 + g_2)(n) = f(m) \otimes g_1(n) + f(m) \otimes g_2(n),$$

which equals

$$((f \otimes g_1) + (f \otimes g_2))(m \otimes n).$$

For (3),

$$(f \otimes 0)(m \otimes n) = f(m) \otimes 0 = 0, \qquad (0 \otimes g)(m \otimes n) = 0 \otimes g(n) = 0,$$

so both maps are the zero homomorphism.

For (4),

$$(1_M \otimes 1_N)(m \otimes n) = 1_M(m) \otimes 1_N(n) = m \otimes n,$$

hence $1_M \otimes 1_N$ is the identity on $M \otimes_R N$. $\square$

Lemma 3.3.6. *Let M_R,M'_R,M''_R be right R-modules, and let ${}_RN$,${}_RN'$,${}_RN''$ be left R-modules. Let*

$$f : M \to M', \qquad f' : M' \to M'', \qquad g : N \to N', \qquad g' : N' \to N''$$

be R-module homomorphisms. Then

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g).$$

Proof. Again it suffices to check the equality on elementary tensors. For $m \in M$ and $n \in N$,

$$\big((f' \otimes g') \circ (f \otimes g)\big)(m \otimes n) = (f' \otimes g')\big(f(m) \otimes g(n)\big) = f'(f(m)) \otimes g'(g(n)),$$

while

$$\big((f' \circ f) \otimes (g' \circ g)\big)(m \otimes n) = (f' \circ f)(m) \otimes (g' \circ g)(n) = f'(f(m)) \otimes g'(g(n)).$$

Thus both homomorphisms agree on the generators $m \otimes n$ of $M \otimes_R N$, hence they are equal. ☐

Definition 3.3.7. *Let ${}_SU_R$ be a bimodule. Define*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

by

$$M \longmapsto U \otimes_R M, \qquad f \longmapsto \mathrm{id}_U \otimes f.$$

Similarly, Let ${}_RU_S$ be a bimodule.

$$- \otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

by

$$N \longmapsto N \otimes_R U, \qquad g \longmapsto g \otimes \mathrm{id}_U .$$

Proposition 3.3.8. *Let ${}_SU_R$ be a bimodule. Then*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

is a covariant functor.
Similarly, if ${}_RU_S$ is a bimodule, then

$$- \otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

is a covariant functor.

Proof. We prove the first statement. For each left R-module M, the tensor product

$$U \otimes_R M$$

is naturally a left S-module via

$$s(u \otimes m) = (su) \otimes m \qquad (s \in S,\ u \in U,\ m \in M).$$

For each R-module homomorphism $f : M \to N$, define

$$U \otimes_R f : U \otimes_R M \to U \otimes_R N$$

by

$$(U \otimes_R f)(u \otimes m) = u \otimes f(m).$$

Equivalently,

$$U \otimes_R f = \mathrm{id}_U \otimes f.$$

This is well-defined and is an S-module homomorphism.

It remains to verify the functorial properties. For the identity map $\mathrm{id}_M : M \to M$, we have

$$U \otimes_R \mathrm{id}_M = \mathrm{id}_U \otimes \mathrm{id}_M = \mathrm{id}_{U \otimes_R M}.$$

If

$$M \xrightarrow{f} N \xrightarrow{g} P$$

are R-module homomorphisms, then for every $u \otimes m \in U \otimes_R M$,

$$(U \otimes_R (g \circ f))(u \otimes m) = u \otimes (g \circ f)(m) = g(u \otimes f(m)) = ((U \otimes_R g) \circ (U \otimes_R f))(u \otimes m).$$

Hence

$$U \otimes_R (g \circ f) = (U \otimes_R g) \circ (U \otimes_R f).$$

Therefore $U \otimes_R -$ is a covariant functor from R-Mod to S-Mod.

The proof of the second statement is entirely similar. If ${}_R U_S$ is a bimodule and N is a right R-module, then

$$N \otimes_R U$$

is naturally a right S-module via

$$(n \otimes u)s = n \otimes (us).$$

For a homomorphism $g : N \to N'$ in Mod-R, define

$$g \otimes_R U = g \otimes \mathrm{id}_U : N \otimes_R U \to N' \otimes_R U$$

by

$$(g \otimes_R U)(n \otimes u) = g(n) \otimes u.$$

One checks immediately that identities and compositions are preserved. Thus

$$- \otimes_R U : \text{Mod-}R \to \text{Mod-}S$$

is also a covariant functor. $\square$

Theorem 3.3.9. *Let ${}_S U_R$ be a bimodule. Then the functor*

$$U \otimes_R - \;:\; R\text{-}Mod \longrightarrow S\text{-}Mod$$

is right exact. In other words, if

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is an exact sequence of left R-modules, then

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact.

Proof. Since

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is exact, we have

$$\ker g = \operatorname{Im} f \qquad \text{and} \qquad N'' \cong N/\operatorname{Im} f.$$

Let

$$\pi : N \longrightarrow N/\operatorname{Im} f$$

be the canonical projection. Then there exists an isomorphism

$$\alpha : N/\operatorname{Im} f \longrightarrow N''$$

such that

$$g = \alpha \circ \pi.$$

Now consider the map

$$\Phi : U \otimes_R N \longrightarrow U \otimes_R (N/\operatorname{Im} f), \qquad u \otimes n \longmapsto u \otimes \pi(n).$$

By the universal property of tensor products,Φ is a well-defined S-module homomorphism.

We claim that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

First, since $\pi \circ f = 0$, functoriality gives

$$\Phi \circ (\mathrm{id}_U \otimes f) = \mathrm{id}_U \otimes (\pi \circ f) = 0.$$

Hence

$$\operatorname{Im}(\mathrm{id}_U \otimes f) \subseteq \ker \Phi.$$

To prove the reverse inclusion, define

$$\overline{\Phi} : (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f) \longrightarrow U \otimes_R (N/\operatorname{Im} f)$$

by

$$\overline{\Phi}\big((u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f)\big) = u \otimes \pi(n).$$

This is well defined because Φ vanishes on $\operatorname{Im}(\mathrm{id}_U \otimes f)$.

Next define

$$\Psi : U \times (N/\operatorname{Im} f) \longrightarrow (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f)$$

by

$$\Psi(u, \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f),$$

where $\overline{n} = n + \operatorname{Im} f$.

We check that Ψ is well defined. If $\overline{n} = \overline{n'}$, then

$$n - n' \in \operatorname{Im} f,$$

so there exists $x \in N'$ such that

$$n - n' = f(x).$$

Hence

$$(u \otimes n) - (u \otimes n') = u \otimes (n - n') = u \otimes f(x) = (\mathrm{id}_U \otimes f)(u \otimes x) \in \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Thus

$$(u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f) = (u \otimes n') + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

So Ψ is well defined. It is clearly R-balanced, hence induces an S-module homomorphism

$$\overline{\Psi} : U \otimes_R (N/\operatorname{Im} f) \longrightarrow (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f)$$

such that

$$\overline{\Psi}(u \otimes \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

It is immediate that $\overline{\Phi}$ and $\overline{\Psi}$ are inverse to each other. Therefore

$$(U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f) \cong U \otimes_R (N/\operatorname{Im} f).$$

It follows that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Finally, since $\alpha : N/\operatorname{Im} f \to N''$ is an isomorphism, the induced map

$$\mathrm{id}_U \otimes \alpha : U \otimes_R (N/\operatorname{Im} f) \longrightarrow U \otimes_R N''$$

is an isomorphism. Moreover,

$$\mathrm{id}_U \otimes g = (\mathrm{id}_U \otimes \alpha) \circ \Phi.$$

Hence

$$\ker(\mathrm{id}_U \otimes g) = \ker \Phi = \mathrm{Im}(\mathrm{id}_U \otimes f),$$

and $\mathrm{id}_U \otimes g$ is surjective because Φ is surjective and $\mathrm{id}_U \otimes \alpha$ is an isomorphism.

Therefore

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact. □

Theorem 3.3.10. *Let ${}_R U_S$ be a left-R-right-S-bimodule. Then the functor*

$$- \otimes_R U \ : \ Mod - R \longrightarrow Mod - S$$

is right exact.

Chapter 4

Lecture 4: Projectives, Injectives

4.1 Projective, Injective and Flat Modules

Definition 4.1.1. *Let R be a ring.*

- *A left R-module P is called* projective *if the functor*

$$\mathrm{Hom}_R(P, -) : R - \mathrm{Mod} \longrightarrow \mathbf{Ab}$$

  is exact.

- *A left R-module I is called* injective *if the contravariant functor*

$$\mathrm{Hom}_R(-, I) : R - \mathrm{Mod} \longrightarrow \mathbf{Ab}$$

  is exact.

- *A left R-module F is called* flat *if the functor*

$$- \otimes_R F : \text{Mod-}R \longrightarrow \mathbf{Ab}$$

  is exact.

Theorem 4.1.2. *Let ${}_R P$ be a left R-module. The following statements are equivalent.*

1. *Whenever we are given a commutative diagram*

$$\begin{tikzcd} & P \arrow[dl, dashed, "\widetilde{\gamma}"'] \arrow[d, "\gamma"] & \\ M \arrow[r, two heads, "g"'] & N \arrow[r] & 0 \end{tikzcd}$$

   with g an epimorphism, there exists an R-homomorphism

$$\widetilde{\gamma} : P \to M$$

   such that $g \circ \widetilde{\gamma} = \gamma$.

2. *For every epimorphism $f : M \to N$ of left R-modules, the induced map*

$$\mathrm{Hom}_R(P, f) : \mathrm{Hom}_R(P, M) \longrightarrow \mathrm{Hom}_R(P, N)$$

is surjective.

3. *For every ring S and every right S-module structure on P making ${}_R P_S$ an R-S-bimodule, the functor*

$$\mathrm{Hom}_R(P, -) : R - \mathrm{Mod} \longrightarrow S - \mathrm{Mod}$$

is exact.

4. *For every short exact sequence*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

of left R-modules, the sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact.

Proof. We first recall two standard facts.

1. For every left R-module P, the functor

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to \mathbf{Ab}$$

is left exact.

2. If ${}_R P_S$ is an R-S-bimodule and M is a left R-module, then $\mathrm{Hom}_R(P, M)$ is naturally a left S-module via

$$(s \cdot \varphi)(p) := \varphi(ps) \qquad (s \in S,\ \varphi \in \mathrm{Hom}_R(P, M),\ p \in P).$$

Moreover, for every R-homomorphism $h : M \to N$, the induced map

$$\mathrm{Hom}_R(P, h) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$$

is S-linear.

(1) $\Longleftrightarrow$(2). Let $f : M \to N$ be an epimorphism. For $\gamma \in \mathrm{Hom}_R(P, N)$, saying that there exists $\widetilde{\gamma} : P \to M$ such that

$$f \circ \widetilde{\gamma} = \gamma$$

is exactly saying that γ lies in the image of

$$\mathrm{Hom}_R(P, f) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N).$$

Hence (1) and (2) are equivalent.

(2) $\Longrightarrow$(4). Let

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be a short exact sequence of left R-modules. Since $\mathrm{Hom}_R(P, -)$ is left exact, the sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'')$$

is exact. Since v is an epimorphism, (2) implies that $\mathrm{Hom}_R(P, v)$ is surjective. Therefore

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact, proving (4).

(4) $\Longrightarrow$(2). Let $f : M \to N$ be an epimorphism. Apply (4) to the short exact sequence

$$0 \longrightarrow \ker(f) \longrightarrow M \xrightarrow{f} N \longrightarrow 0.$$

Then

$$\mathrm{Hom}_R(P, f) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$$

is surjective. Thus (2) holds.

(3) $\Longrightarrow$(4). Take $S = \mathbb{Z}$, with the canonical right $\mathbb{Z}$-module structure on P. Then ${}_RP_{\mathbb{Z}}$ is an R-$\mathbb{Z}$-bimodule, and $\mathbb{Z}$-Mod $= \mathbf{Ab}$. Hence (3) gives that

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to \mathbf{Ab}$$

is exact, which is exactly statement (4).

(4) $\Longrightarrow$(3). Fix a ring S and a right S-module structure on P making ${}_RP_S$ an R-S-bimodule. Let

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be a short exact sequence of left R-modules. By (4), the induced sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact as a sequence of abelian groups. By the remark above, each term carries a natural left S-module structure and each map is S-linear. Since exactness in S-Mod is equivalent to exactness of the underlying abelian groups, the above sequence is exact in S-Mod. Therefore

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to S\text{-Mod}$$

is exact. Hence (3) holds.

Therefore (1), (2), (3), and (4) are equivalent. $\square$

Proposition 4.1.3. *Free modules are projective.*

Proof. Let $F = \bigoplus_{i\in I} Re_i$ be a free left R-module with basis $\{e_i\}_{i\in I}$. Let $g : M \to N$ be an epimorphism and let $\gamma : F \to N$ be an R-homomorphism. For each $i \in I$, choose $m_i \in M$ such that

$$g(m_i) = \gamma(e_i).$$

Since F is free, there exists a unique R-homomorphism

$$\widetilde{\gamma} : F \to M$$

such that $\widetilde{\gamma}(e_i) = m_i$ for all $i \in I$. Then

$$g\widetilde{\gamma}(e_i) = g(m_i) = \gamma(e_i)$$

for every basis element, so $g\widetilde{\gamma} = \gamma$. Hence F is projective. $\square$

Theorem 4.1.4. *Let ${}_RP$ be a left R-module. The following statements are equivalent.*

1. *P is projective.*
2. *Every epimorphism*
$$f : M \longrightarrow P \longrightarrow 0$$
splits, that is, there exists an R-homomorphism $g : P \to M$ such that
$$f \circ g = \mathrm{id}_P\,.$$
3. *P is isomorphic to a direct summand of a free module.*

Proof. **(1) $\Longrightarrow$(2).** Apply the lifting property to the diagram

$$\begin{tikzcd} & P \arrow[dl, dashed, "g"'] \arrow[d, "\mathrm{id}_P"] & \\ M \arrow[r, two heads, "f"'] & P \arrow[r] & 0 \end{tikzcd}$$

to obtain $f \circ g = \mathrm{id}_P$.

(2) $\Longrightarrow$(3). Every module is a homomorphic image of a free module, so choose an epimorphism

$$\pi : F \to P$$

with F free. By (2),π splits, so there exists $s : P \to F$ such that

$$\pi \circ s = \mathrm{id}_P\,.$$

Therefore

$$F \cong \ker \pi \oplus s(P) \cong \ker \pi \oplus P,$$

and hence P is a direct summand of the free module F.

(3) $\Longrightarrow$(1). Write

$$F \cong P \oplus Q$$

with F free. Since free modules are projective, it suffices to show that a direct summand of a projective module is projective. Let $f : M \to N$ be an epimorphism and

$$\alpha : P \to N$$

an R-homomorphism. Define

$$\widehat{\alpha} : P \oplus Q \to N, \qquad \widehat{\alpha}(p, q) = \alpha(p).$$

Because F is projective, there exists

$$\widehat{\beta} : P \oplus Q \to M$$

such that $f \circ \widehat{\beta} = \widehat{\alpha}$. Let

$$\iota : P \to P \oplus Q, \qquad p \mapsto (p, 0).$$

Then

$$f \circ (\widehat{\beta} \circ \iota) = \widehat{\alpha} \circ \iota = \alpha.$$

Hence P is projective. □

Corollary 4.1.5. *A left R-module P is finitely generated and projective if and only if for some R-module P' and some integer $n \geq 0$ there is an R-isomorphism*

$$P \oplus P' \cong R^n.$$

Proof. If $P \oplus P' \cong R^n$, then P is a direct summand of a finite free module, hence finitely generated and projective.

Conversely, assume that P is finitely generated and projective. By the theorem above, there exists a free module

$$F = \bigoplus_{i \in I} Re_i$$

and an R-module Q such that

$$F \cong P \oplus Q.$$

Let $p : F \to P$ and $s : P \to F$ be the projection and inclusion corresponding to this direct sum decomposition. Choose generators $x_1, \ldots, x_m$ of P. Each $s(x_j)$ is a finite linear combination of the basis elements $\{e_i\}_{i \in I}$, so there exists a finite subset $J \subseteq I$ such that

$$s(x_1), \ldots, s(x_m) \in F_J := \bigoplus_{j \in J} Re_j.$$

Since the x_j generate P, we have $s(P) \subseteq F_J$. Hence the restriction

$$p|_{F_J} : F_J \to P$$

is surjective, and the map $s : P \to F_J$ is a section of $p|_{F_J}$. Therefore

$$F_J \cong P \oplus \ker(p|_{F_J}).$$

Since $F_J \cong R^{|J|}$, the result follows. $\square$

4.2 Schanuel Lemma

Theorem 4.2.1 (Schanuel Lemma). *Let*

$$0 \longrightarrow K \longrightarrow P \xrightarrow{\lambda} M \longrightarrow 0$$

and

$$0 \longrightarrow K' \longrightarrow P' \xrightarrow{\lambda'} M \longrightarrow 0$$

be short exact sequences of left R-modules, with P and P' projective. Then

$$K' \oplus P \cong K \oplus P'.$$

Proof. Define

$$Y := P \times_M P' = \{(p, p') \in P \oplus P' \mid \lambda(p) = \lambda'(p')\}.$$

Let

$$\mu : Y \to P, \qquad \mu(p, p') = p,$$

and

$$\mu' : Y \to P', \qquad \mu'(p, p') = p'.$$

Then Y fits into the pullback diagram

$$\begin{tikzcd} Y \arrow[r, "\mu'"] \arrow[d, "\mu"'] & P' \arrow[d, "\lambda'"] \\ P \arrow[r, "\lambda"'] & M \end{tikzcd}$$

that is, the square commutes and Y is the fiber product of λ and λ'.

We claim that there are short exact sequences

$$0 \longrightarrow K' \xrightarrow{i'} Y \xrightarrow{\mu} P \longrightarrow 0, \qquad i'(k') = (0, k'),$$

and

$$0 \longrightarrow K \xrightarrow{i} Y \xrightarrow{\mu'} P' \longrightarrow 0, \qquad i(k) = (k, 0).$$

First,μ is surjective. Indeed, let $p \in P$. Since λ' is surjective, there exists $p' \in P'$ such that

$$\lambda'(p') = \lambda(p).$$

Then $(p, p') \in Y$, so $\mu(p, p') = p$.

Next,

$$\ker(\mu) = \{(0, p') \in Y \mid \lambda'(p') = 0\} = \{(0, k') \mid k' \in K'\} = \mathrm{Im}(i').$$

Hence

$$0 \longrightarrow K' \xrightarrow{i'} Y \xrightarrow{\mu} P \longrightarrow 0$$

is exact.

Similarly,μ' is surjective, and

$$\ker(\mu') = \{(p, 0) \in Y \mid \lambda(p) = 0\} = \{(k, 0) \mid k \in K\} = \mathrm{Im}(i).$$

Thus

$$0 \longrightarrow K \xrightarrow{i} Y \xrightarrow{\mu'} P' \longrightarrow 0$$

is exact.

Since P is projective and $\mu : Y \to P$ is an epimorphism, there exists an R-homomorphism $s : P \to Y$ such that

$$\mu \circ s = \mathrm{id}_P.$$

This is expressed by the commutative diagram

$$\begin{tikzcd} & Y \arrow[d, two heads, "\mu"] \\ P \arrow[ur, dashed, "s"] \arrow[r, "\mathrm{id}_P"'] & P \end{tikzcd}$$

so the sequence

$$0 \longrightarrow K' \longrightarrow Y \xrightarrow{\mu} P \longrightarrow 0$$

splits. Therefore,

$$Y \cong K' \oplus P.$$

Likewise, since P' is projective and $\mu' : Y \to P'$ is an epimorphism, there exists $s' : P' \to Y$ such that

$$\mu' \circ s' = \mathrm{id}_{P'},$$

and hence

$$0 \longrightarrow K \longrightarrow Y \xrightarrow{\mu'} P' \longrightarrow 0$$

splits. Therefore,

$$Y \cong K \oplus P'.$$

Combining the two decompositions of Y, we obtain

$$K' \oplus P \cong Y \cong K \oplus P',$$

and hence

$$K' \oplus P \cong K \oplus P'.$$

□

Lemma 4.2.2. *Let M be a left R-module. Then*

$$\mathrm{Hom}_R({}_RR, M) \cong M$$

as left R-modules.

Proof. Define

$$\Phi : \mathrm{Hom}_R({}_RR, M) \to M, \qquad f \mapsto f(1).$$

This map is R-linear. Define also

$$\Psi : M \to \mathrm{Hom}_R({}_RR, M), \qquad m \mapsto (r \mapsto rm).$$

For $f \in \mathrm{Hom}_R({}_RR, M)$ and $r \in R$,

$$\Psi(\Phi(f))(r) = rf(1) = f(r \cdot 1) = f(r),$$

so $\Psi\Phi = \mathrm{id}$. Also

$$\Phi(\Psi(m)) = \Psi(m)(1) = 1m = m,$$

hence Φ and Ψ are inverse isomorphisms. □

Corollary 4.2.3. *For any ring R with 1,*

$$\mathrm{End}_R({}_RR) \cong R^{\mathrm{op}}$$

as rings.

Proof. By the lemma, every $f \in \mathrm{End}_R({}_RR)$ is determined by $f(1)$, and

$$f(r) = rf(1)$$

for all $r \in R$. Thus

$$\Theta : \mathrm{End}_R({}_RR) \to R^{\mathrm{op}}, \qquad f \mapsto f(1)$$

is bijective. Moreover, for $f, g \in \mathrm{End}_R({}_RR)$,

$$(f \circ g)(1) = f(g(1)) = g(1)f(1),$$

which is multiplication in the opposite ring. Therefore Θ is a ring isomorphism. □

4.3 Projective Morphisms

Lemma 4.3.1. *Let R be a commutative ring with 1, and let X,Y,M be R-modules. Then there is an R-module homomorphism*

$$\Theta_M^{X,Y} : \mathrm{Hom}_R(X, M) \otimes_R \mathrm{Hom}_R(M, Y) \longrightarrow \mathrm{Hom}_R(X, Y), \qquad \varphi \otimes \psi \longmapsto \psi \circ \varphi.$$

Its image is precisely the set of all morphisms $f : X \to Y$ which factor through some finite direct power M^n of M.

Proof. First define

$$b : \mathrm{Hom}_R(X, M) \times \mathrm{Hom}_R(M, Y) \longrightarrow \mathrm{Hom}_R(X, Y), \qquad b(\varphi, \psi) = \psi \circ \varphi.$$

Since R is commutative, composition is R-balanced in the sense that

$$b(r\varphi, \psi) = \psi \circ (r\varphi) = r(\psi \circ \varphi) = b(\varphi, r\psi)$$

for all $r \in R$,$\varphi \in \mathrm{Hom}_R(X, M)$, and $\psi \in \mathrm{Hom}_R(M, Y)$; moreover b is additive in each variable. Hence b induces a unique R-linear map

$$\Theta_M^{X,Y} : \mathrm{Hom}_R(X, M) \otimes_R \mathrm{Hom}_R(M, Y) \to \mathrm{Hom}_R(X, Y)$$

such that

$$\Theta_M^{X,Y}(\varphi \otimes \psi) = \psi \circ \varphi.$$

Now let

$$f = \Theta_M^{X,Y}\Big(\sum_{i=1}^{n} \varphi_i \otimes \psi_i\Big) = \sum_{i=1}^{n} \psi_i \circ \varphi_i.$$

Define morphisms

$$\alpha : X \to M^n, \qquad x \longmapsto (\varphi_1(x), \dots, \varphi_n(x)),$$

and

$$\beta : M^n \to Y, \qquad (m_1, \dots, m_n) \longmapsto \sum_{i=1}^{n} \psi_i(m_i).$$

Then for every $x \in X$,

$$(\beta \circ \alpha)(x) = \beta\big(\varphi_1(x), \dots, \varphi_n(x)\big) = \sum_{i=1}^{n} \psi_i(\varphi_i(x)) = f(x).$$

Thus $f = \beta \circ \alpha$, so every element in the image of $\Theta_M^{X,Y}$ factors through some finite direct power M^n.

Conversely, suppose that a morphism $f : X \to Y$ factors through M^n, say

$$f : X \xrightarrow{u} M^n \xrightarrow{v} Y.$$

Let $\pi_i : M^n \to M$ be the i th projection and $\iota_i : M \to M^n$ the i th inclusion. Since

$$\mathrm{id}_{M^n} = \sum_{i=1}^{n} \iota_i \circ \pi_i,$$

we obtain

$$f = v \circ u = v \circ \mathrm{id}_{M^n} \circ u = \sum_{i=1}^{n} v \circ \iota_i \circ \pi_i \circ u = \sum_{i=1}^{n} (v \circ \iota_i) \circ (\pi_i \circ u).$$

Therefore

$$f = \Theta_M^{X,Y} \Big(\sum_{i=1}^{n} (\pi_i \circ u) \otimes (v \circ \iota_i) \Big),$$

so f lies in the image of $\Theta_M^{X,Y}$.

Hence the image of $\Theta_M^{X,Y}$ is exactly the set of morphisms $X \to Y$ that factor through some finite direct power M^n. $\square$

Definition 4.3.2. *Let R be a commutative ring with 1. For an R-module X, write*

$$X^* := \mathrm{Hom}_R(X, R).$$

For R-modules X and Y, define

$$\tau_{X,Y} : X^* \otimes_R Y \longrightarrow \mathrm{Hom}_R(X, Y), \qquad \varphi \otimes y \longmapsto \big(x \mapsto \varphi(x) y\big).$$

Definition 4.3.3. *A morphism $f : X \to Y$ is called a* projective morphism *if it factors through a finitely generated projective module.*

Proposition 4.3.4. *Let R be a commutative ring with 1. A morphism $f : X \to Y$ is projective if and only if*

$$f \in \mathrm{Im}(\tau_{X,Y}).$$

Proof. By the previous lemma with $M = R$, the image of $\tau_{X,Y}$ consists exactly of those morphisms that factor through some finite free module R^n.

Since finite free modules are finitely generated projective, every morphism in $\mathrm{Im}(\tau_{X,Y})$ is projective.

Conversely, suppose $f : X \to Y$ factors through a finitely generated projective module P:

$$X \xrightarrow{u} P \xrightarrow{v} Y.$$

Choose Q and n such that

$$P \oplus Q \cong R^n.$$

Let $i : P \to R^n$ and $p : R^n \to P$ be the inclusion and projection corresponding to this decomposition, so that $p \circ i = \mathrm{id}_P$. Then

$$f = v \circ p \circ i \circ u,$$

hence f factors through R^n. Therefore $f \in \mathrm{Im}(\tau_{X,Y})$. $\square$

Definition 4.3.5. *For R-modules X and Y, we write*

$$\mathrm{Hom}_R^{\mathrm{pr}}(X, Y)$$

for the set of projective morphisms from X to Y.

Lemma 4.3.6. *For any R-module M, the set*

$$\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$$

is a two-sided ideal of the ring $\mathrm{End}_R(M)$.

Proof. We first show that $\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ is an additive subgroup of $\mathrm{End}_R(M)$.

Let $f, g \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M)$. Then there exist finitely generated projective R-modules P, Q and morphisms

$$M \xrightarrow{u} P \xrightarrow{v} M, \qquad M \xrightarrow{u'} Q \xrightarrow{v'} M$$

such that

$$f = v \circ u, \qquad g = v' \circ u'.$$

Consider the morphisms

$$\alpha : M \to P \oplus Q, \qquad \alpha(x) = (u(x), u'(x)),$$

and

$$\beta : P \oplus Q \to M, \qquad \beta(p, q) = v(p) + v'(q).$$

Then

$$(f + g)(x) = v(u(x)) + v'(u'(x)) = \beta(\alpha(x))$$

for all $x \in M$, so

$$f + g = \beta \circ \alpha.$$

Since $P \oplus Q$ is again a finitely generated projective R-module, it follows that $f + g \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M)$.

Also, the zero endomorphism factors through the zero module, and if $f = v \circ u$, then

$$-f = (-v) \circ u,$$

so $\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ is an additive subgroup of $\mathrm{End}_R(M)$.

Now let $f \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ and let $a, b \in \mathrm{End}_R(M)$. Choose a factorization

$$M \xrightarrow{u} P \xrightarrow{v} M$$

with P finitely generated projective and $f = v \circ u$. Then

$$a \circ f \circ b = a \circ v \circ u \circ b = (a \circ v) \circ (u \circ b),$$

so $a \circ f \circ b$ still factors through the same finitely generated projective module P. Hence

$$a \circ f \circ b \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M).$$

Therefore $\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ is a two-sided ideal of $\mathrm{End}_R(M)$. □

Theorem 4.3.7. *Let R be a commutative ring with 1, and let P be an R-module. The following statements are equivalent.*

1. *P is a finitely generated projective module.*
2. $$\mathrm{Hom}_R^{\mathrm{pr}}(P, P) = \mathrm{End}_R(P).$$
3. *For every R-module X, the morphism*
   $$\tau_{X,P} : X^* \otimes_R P \longrightarrow \mathrm{Hom}_R(X, P)$$
   is an isomorphism.
4. *For every R-module X, the morphism*
   $$\tau_{P,X} : P^* \otimes_R X \longrightarrow \mathrm{Hom}_R(P, X)$$
   is an isomorphism.
5. *The morphism*
   $$\tau_{P,P} : P^* \otimes_R P \longrightarrow \mathrm{End}_R(P)$$
   is an isomorphism.

Proof. **(1) $\Longrightarrow$(2).** Since P itself is finitely generated projective, the identity morphism

$$\mathrm{id}_P : P \to P$$

is projective. By the previous lemma,$\mathrm{Hom}_R^{\mathrm{pr}}(P, P)$ is a two-sided ideal of $\mathrm{End}_R(P)$ containing id_P, and therefore equals the whole endomorphism ring.

(3) $\Longrightarrow$(5) and **(4) $\Longrightarrow$(5)** are immediate by taking $X = P$.

(2) $\Longrightarrow$(3). By (2),id_P is projective, so there exist $\varphi_1, \ldots, \varphi_n \in P^*$ and $p_1, \ldots, p_n \in P$

such that

$$\mathrm{id}_P = \tau_{P,P}\Big(\sum_{i=1}^{n} \varphi_i \otimes p_i\Big).$$

Equivalently,

$$p = \sum_{i=1}^{n} \varphi_i(p)p_i \qquad \text{for all } p \in P.$$

For any R-module X, define

$$\sigma_X : \mathrm{Hom}_R(X, P) \to X^* \otimes_R P, \qquad \alpha \mapsto \sum_{i=1}^{n} (\varphi_i \circ \alpha) \otimes p_i.$$

Then for $\alpha \in \mathrm{Hom}_R(X, P)$ and $x \in X$,

$$\tau_{X,P}(\sigma_X(\alpha))(x) = \sum_{i=1}^{n} (\varphi_i \circ \alpha)(x)p_i = \sum_{i=1}^{n} \varphi_i(\alpha(x))p_i = \alpha(x),$$

so $\tau_{X,P} \circ \sigma_X = \mathrm{id}$.

Conversely, let

$$\xi = \sum_{j=1}^{m} \psi_j \otimes q_j \in X^* \otimes_R P.$$

Then

$$\begin{aligned}
\sigma_X(\tau_{X,P}(\xi)) &= \sum_{i=1}^{n} \Big(\varphi_i \circ \tau_{X,P}(\xi)\Big) \otimes p_i \\
&= \sum_{i=1}^{n} \left(x \mapsto \varphi_i\Big(\sum_{j=1}^{m} \psi_j(x)q_j\Big) \right) \otimes p_i \\
&= \sum_{i=1}^{n} \sum_{j=1}^{m} \big(\psi_j \varphi_i(q_j)\big) \otimes p_i \\
&= \sum_{j=1}^{m} \psi_j \otimes \Big(\sum_{i=1}^{n} \varphi_i(q_j)p_i\Big) \\
&= \sum_{j=1}^{m} \psi_j \otimes q_j \\
&= \xi.
\end{aligned}$$

Therefore σ_X is the inverse of $\tau_{X,P}$, and $\tau_{X,P}$ is an isomorphism.

(2) $\Longrightarrow$(4). Using the same decomposition of id_P, define for any R-module X

$$\rho_X : \mathrm{Hom}_R(P, X) \to P^* \otimes_R X, \qquad \beta \mapsto \sum_{i=1}^{n} \varphi_i \otimes \beta(p_i).$$

For $\beta \in \mathrm{Hom}_R(P, X)$ and $p \in P$,

$$\tau_{P,X}(\rho_X(\beta))(p) = \sum_{i=1}^{n} \varphi_i(p)\beta(p_i) = \beta\Big(\sum_{i=1}^{n} \varphi_i(p)p_i\Big) = \beta(p),$$

so $\tau_{P,X}\rho_X = \mathrm{id}$. Conversely, for

$$\eta = \sum_{j=1}^{m} \varphi'_j \otimes x_j \in P^* \otimes_R X,$$

we have

$$\begin{aligned}
\rho_X(\tau_{P,X}(\eta)) &= \sum_{i=1}^{n} \varphi_i \otimes \tau_{P,X}(\eta)(p_i) \\
&= \sum_{i=1}^{n} \varphi_i \otimes \Big(\sum_{j=1}^{m} \varphi'_j(p_i)x_j\Big) \\
&= \sum_{j=1}^{m}\Big(\sum_{i=1}^{n} \varphi'_j(p_i)\varphi_i\Big) \otimes x_j \\
&= \sum_{j=1}^{m} \varphi'_j \otimes x_j \\
&= \eta,
\end{aligned}$$

because

$$\varphi'_j(p) = \varphi'_j\Big(\sum_{i=1}^{n} \varphi_i(p)p_i\Big) = \sum_{i=1}^{n} \varphi_i(p)\varphi'_j(p_i)$$

for all $p \in P$. Hence $\tau_{P,X}$ is an isomorphism.

(5) $\Longrightarrow$(1). Since $\tau_{P,P}$ is surjective, we can write

$$\mathrm{id}_P = \tau_{P,P}\Big(\sum_{i=1}^{n} \varphi_i \otimes p_i\Big)$$

for some $\varphi_i \in P^*$ and $p_i \in P$. Define

$$\iota : P \to R^n, \qquad \iota(p) = (\varphi_1(p), \dots, \varphi_n(p)),$$

and

$$\pi : R^n \to P, \qquad \pi(a_1, \dots, a_n) = \sum_{i=1}^{n} a_i p_i.$$

Then

$$(\pi \circ \iota)(p) = \sum_{i=1}^{n} \varphi_i(p)p_i = p$$

for all $p \in P$. Thus $\pi \circ \iota = \mathrm{id}_P$, so P is a direct summand of the finite free module R^n. Therefore P is finitely generated and projective. $\square$

4.4 Projective Covers

Definition 4.4.1. *Let R be a ring with 1. An epimorphism*

$$f : U \to V$$

is called an essential epimorphism *if no proper submodule of U is mapped surjectively onto V by f. Equivalently, whenever*

$$g : W \to U$$

is a morphism such that $f \circ g$ is an epimorphism, then g is an epimorphism.

Proof. Suppose first that no proper submodule of U maps onto V. Let $g : W \to U$ satisfy $f \circ g$ epimorphic. Then

$$f(g(W)) = V.$$

By assumption,$g(W)$ cannot be a proper submodule of U, so $g(W) = U$ and g is surjective.

Conversely, assume that whenever $f \circ g$ is epi, then g is epi. Let $U_0 \subsetneq U$ be a proper submodule and let

$$\iota : U_0 \hookrightarrow U$$

be the inclusion. If $f(U_0) = V$, then $f \circ \iota$ is epi, hence ι is epi, which is impossible. Thus $f(U_0) \neq V$. ☐

Definition 4.4.2. *A* projective cover *of an R-module U is an essential epimorphism*

$$P \to U$$

with P projective.

Proposition 4.4.3. *Let*

$$U \xrightarrow{f} V \xrightarrow{g} W$$

be R-homomorphisms. If any two of f,g, and $g \circ f$ are essential epimorphisms, then so is the third.

Proof. **If f and g are essential, then $g \circ f$ is essential.** Let $U_0 \subsetneq U$ be a proper submodule. Since f is essential,

$$f(U_0) \neq V.$$

As g is essential,$g(f(U_0)) \neq W$. Hence $(g \circ f)(U_0) \neq W$, so $g \circ f$ is essential.

If g and $g \circ f$ are essential, then f is essential. Since $g \circ f$ is epic,f is epic. Let $U_0 \subsetneq U$ be proper. If $f(U_0) = V$, then

$$(g \circ f)(U_0) = W,$$

contradicting that $g \circ f$ is essential. Thus $f(U_0) \neq V$, so f is essential.

If f and $g \circ f$ are essential, then g is essential. Since $g \circ f$ is epic and f is epic,g is epic. Let $V_0 \subsetneq V$ be a proper submodule, and set

$$U_0 := f^{-1}(V_0).$$

Because f is surjective,U_0 is a proper submodule of U. Since $g \circ f$ is essential,

$$(g \circ f)(U_0) \neq W.$$

But

$$(g \circ f)(U_0) = g(V_0),$$

so $g(V_0) \neq W$. Hence g is essential. ☐

4.5 Noetherian and Artinian Modules

Definition 4.5.1. *An R-module M is called* Noetherian *if for every ascending chain of submodules*

$$L_1 \subseteq L_2 \subseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \qquad \text{for all } n \geq n_0.$$

Lemma 4.5.2. *For an R-module M, the following are equivalent.*

1. *M is Noetherian.*
2. *Every submodule of M is finitely generated.*
3. *Every non-empty set of submodules of M has a maximal element.*

Proof. **(1) $\Longrightarrow$(3).** Suppose $\mathcal{S}$ is a non-empty set of submodules of M having no maximal element. Choose $L_1 \in \mathcal{S}$. Since L_1 is not maximal, there exists $L_2 \in \mathcal{S}$ with

$$L_1 \subsetneq L_2.$$

Continuing inductively, we obtain a strictly ascending chain

$$L_1 \subsetneq L_2 \subsetneq L_3 \subsetneq \cdots,$$

contradicting that M is Noetherian.

(3) $\Longrightarrow$(2). Let $N \leq M$. Consider the set

$$\mathcal{T} := \{L \leq N \mid L \text{ is finitely generated}\}.$$

This set is non-empty because $0 \in \mathcal{T}$. By (3),$\mathcal{T}$ has a maximal element, say L. If $L \neq N$, choose $x \in N \setminus L$. Then

$$L + Rx$$

is a finitely generated submodule of N strictly containing L, contradicting the maximality of L. Hence $L = N$, so N is finitely generated.

(2) $\Longrightarrow$(1). Let

$$L_1 \subseteq L_2 \subseteq \cdots$$

be an ascending chain of submodules, and set

$$L := \bigcup_{n \geq 1} L_n.$$

Then L is a submodule of M, so by (2) it is finitely generated, say

$$L = Rx_1 + \cdots + Rx_m.$$

Each generator x_j lies in some L_{n_j}. If $N = \max\{n_1, \ldots, n_m\}$, then

$$x_1, \ldots, x_m \in L_N,$$

so $L = L_N$. Hence

$$L_n = L_N \qquad \text{for all } n \geq N,$$

and the chain stabilizes. $\square$

Proposition 4.5.3. *Let*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be a short exact sequence of R-modules. Then M is Noetherian if and only if both M' and M'' are Noetherian.

Proof. Let $\pi : M \to M''$ be the quotient map, and identify M' with its image in M.

First assume that M is Noetherian. Since M' is a submodule of M, it is Noetherian. Now let $N'' \leq M''$ be any submodule. Then $\pi^{-1}(N'')$ is a submodule of M, hence is finitely generated. Since $\pi(\pi^{-1}(N'')) = N''$, it follows that N'' is finitely generated. Therefore M'' is Noetherian.

Conversely, assume that both M' and M'' are Noetherian. Let $N \leq M$ be any submodule. Then $N \cap M'$ is a submodule of M', so it is finitely generated. Also $\pi(N)$ is a submodule of M'', so it is finitely generated. Choose elements $x_1, \ldots, x_r \in N$ such that

$$\pi(x_1), \ldots, \pi(x_r)$$

generate $\pi(N)$, and choose elements $y_1, \ldots, y_s \in N \cap M'$ which generate $N \cap M'$.

We claim that

$$N = Rx_1 + \cdots + Rx_r + Ry_1 + \cdots + Ry_s.$$

Indeed, let $x \in N$. Since $\pi(x) \in \pi(N)$, there exist $a_1, \ldots, a_r \in R$ such that

$$\pi(x) = \sum_{i=1}^{r} a_i \pi(x_i) = \pi\left(\sum_{i=1}^{r} a_i x_i\right).$$

Hence

$$x - \sum_{i=1}^{r} a_i x_i \in \ker(\pi) \cap N = M' \cap N = N \cap M'.$$

Since $y_1, \ldots, y_s$ generate $N \cap M'$, there exist $b_1, \ldots, b_s \in R$ such that

$$x - \sum_{i=1}^{r} a_i x_i = \sum_{j=1}^{s} b_j y_j.$$

Therefore

$$x = \sum_{i=1}^{r} a_i x_i + \sum_{j=1}^{s} b_j y_j,$$

proving the claim. Thus N is finitely generated.

Since every submodule $N \leq M$ is finitely generated, M is Noetherian. □

Definition 4.5.4. *An R-module M is called* Artinian *if for every descending chain of submodules*

$$L_1 \supseteq L_2 \supseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \qquad \text{for all } n \geq n_0.$$

Lemma 4.5.5. *An R-module M is Artinian if and only if every non-empty set of submodules of M contains a minimal element.*

Proof. The proof is the descending-chain analogue of the Noetherian case. □

Remark 4.5.6. *If an R-module M is Noetherian (respectively Artinian), then we also say that M satisfies the ascending chain condition (respectively descending chain condition) on submodules, abbreviated as ACC (respectively DCC).*

Proposition 4.5.7. *Let*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be a short exact sequence of R-modules. Then M is Artinian if and only if both M' and M'' are Artinian.

Proof. The proof is completely analogous to the Noetherian case, using minimal elements instead of maximal elements. □

Definition 4.5.8. *An R-module M is called* simple *(or* irreducible*) if it has exactly two submodules:*

$$0 \qquad \textit{and} \qquad M.$$

Chapter 5

Lecture 5: Semisimplicity

5.1 Semisimple Modules and the Radical

Definition 5.1.1. *An R-module M is called* semisimple *if it is a direct sum of simple submodules.*

Theorem 5.1.2 (Schur's lemma)**.** *Let S_1 and S_2 be simple R-modules, and let*

$$\varphi : S_1 \to S_2$$

be an R-homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. In particular,$\mathrm{End}_R(S)$ *is a division ring for every simple R-module S.*

Proof. Assume $\varphi \neq 0$. Since $\ker(\varphi)$ is a submodule of the simple module S_1, we have $\ker(\varphi) = 0$ or $\ker(\varphi) = S_1$. The latter is impossible because $\varphi \neq 0$, hence $\ker(\varphi) = 0$ and φ is injective.

Likewise,$\mathrm{Im}(\varphi)$ is a nonzero submodule of the simple module S_2, so $\mathrm{Im}(\varphi) = S_2$. Thus φ is surjective and therefore an isomorphism.

Taking $S_1 = S_2 = S$, every nonzero endomorphism of S is invertible, which means that $\mathrm{End}_R(S)$ is a division ring. □

Lemma 5.1.3. *Let S be a simple left R-module. Then S is a quotient of the regular left module ${}_RR$. More precisely, for every nonzero $x \in S$ there is an isomorphism*

$$S \cong R/\operatorname{Ann}_R(x).$$

In particular,$\operatorname{Ann}_R(x)$ *is a maximal left ideal of R.*

Proof. Fix $0 \neq x \in S$ and define

$$\pi_x : R \to S, \qquad r \mapsto rx.$$

Since S is simple and Rx is a nonzero submodule of S, we have $Rx = S$. Hence π_x is surjective. Its kernel is exactly

$$\ker(\pi_x) = \operatorname{Ann}_R(x) := \{r \in R \mid rx = 0\}.$$

By the first isomorphism theorem,

$$S \cong R/\operatorname{Ann}_R(x).$$

Because S is simple, the quotient $R/\operatorname{Ann}_R(x)$ is simple as a left R-module, so $\operatorname{Ann}_R(x)$ is a maximal left ideal. $\square$

Definition 5.1.4. *Let U be an R-module. A proper submodule $M \subsetneq U$ is called a* maximal submodule *if whenever*

$$M \subseteq N \subseteq U,$$

then either $N = M$ or $N = U$.

Lemma 5.1.5. *Let W be a proper submodule of an R-module U. Assume that U/W is finitely generated. Then there exists a maximal submodule M of U such that*

$$W \subseteq M \subsetneq U.$$

Proof. Consider the set
$$\mathcal{S} := \{N \leq U \mid W \subseteq N \subsetneq U\},$$

ordered by inclusion. Since $W \in \mathcal{S}$, the set is non-empty. Let $\{N_\lambda\}_{\lambda\in\Lambda}$ be a totally ordered subset of $\mathcal{S}$, and set
$$N := \bigcup_{\lambda\in\Lambda} N_\lambda.$$
Then N is a submodule containing W. We claim that $N \neq U$. Indeed, write

$$U/W = R\overline{u}_1 + \cdots + R\overline{u}_n.$$

If $N = U$, then each u_i lies in some N_{λ_i}. Because the family is totally ordered, there exists μ such that
$$N_{\lambda_1}, \ldots, N_{\lambda_n} \subseteq N_\mu.$$
Hence $u_1, \ldots, u_n \in N_\mu$, so $U/W \subseteq N_\mu/W$, which implies $N_\mu = U$, contradicting $N_\mu \in \mathcal{S}$. Therefore N is an upper bound for the chain in $\mathcal{S}$.

By Zorn's lemma,$\mathcal{S}$ has a maximal element M, and this is precisely a maximal submodule of U containing W. $\square$

Corollary 5.1.6. *Every proper left ideal of R is contained in a maximal left ideal.*

Proof. Apply the lemma to the left regular module ${}_RR$ and to the proper submodule $W = I$. ☐

Definition 5.1.7. *Let U be an R-module. The* radical *of U is the intersection of all maximal submodules of U, and is denoted by* $\mathrm{Rad}(U)$*. If U has no maximal submodules, we set* $\mathrm{Rad}(U) = U$*.*

Theorem 5.1.8 (Nakayama's lemma)**.** *Let U be an R-module and let $V \subseteq U$ be a submodule such that U/V is finitely generated. Then*

$$V + \mathrm{Rad}(U) = U \qquad \Longleftrightarrow \qquad V = U.$$

Proof. If $V = U$, then the equality is obvious.

Conversely, assume $V \neq U$. Since U/V is finitely generated, the previous lemma yields a maximal submodule M of U with

$$V \subseteq M \subsetneq U.$$

By definition,

$$\mathrm{Rad}(U) \subseteq M.$$

Therefore

$$V + \mathrm{Rad}(U) \subseteq M \subsetneq U,$$

so $V + \mathrm{Rad}(U) \neq U$. This proves the contrapositive. ☐

Corollary 5.1.9. *If U is a finitely generated R-module, then the quotient map*

$$\pi : U \to U/\mathrm{Rad}(U)$$

is an essential epimorphism.

Proof. Let $W \leq U$ be a submodule such that $\pi(W) = U/\mathrm{Rad}(U)$. Then

$$W + \mathrm{Rad}(U) = U.$$

Since U/W is a quotient of the finitely generated module U, it is finitely generated. Nakayama's lemma therefore gives $W = U$, so π is essential. ☐

Lemma 5.1.10. *Let M and N be finitely generated R-modules, and let*

$$f : M \to N$$

be an R-homomorphism.

1. *We have*
$$f(\mathrm{Rad}(M)) \subseteq \mathrm{Rad}(N).$$

2. *If f is surjective and* $\ker(f) \subseteq \mathrm{Rad}(M)$*, then*
$$\mathrm{Rad}(N) = f(\mathrm{Rad}(M)).$$

Proof. **(1).** Let $x \in \mathrm{Rad}(M)$. Suppose that $f(x) \notin \mathrm{Rad}(N)$. Then there exists a maximal submodule L of N such that
$$f(x) \notin L.$$

Let
$$\rho : N \to N/L$$

be the quotient map. Since N/L is simple and $\rho(f(x)) \neq 0$, the morphism $\rho \circ f : M \to N/L$ is nonzero, hence surjective. Therefore

$$M/\ker(\rho \circ f) \cong N/L$$

is simple, so $\ker(\rho \circ f)$ is a maximal submodule of M. But

$$x \notin \ker(\rho \circ f),$$

contradicting $x \in \mathrm{Rad}(M)$. Hence $f(x) \in \mathrm{Rad}(N)$.

(2). By (1),
$$f(\mathrm{Rad}(M)) \subseteq \mathrm{Rad}(N).$$

For the reverse inclusion, let $x \in M$ with $f(x) \in \mathrm{Rad}(N)$. If $x \notin \mathrm{Rad}(M)$, choose a maximal submodule K of M with
$$x \notin K.$$

Since $\ker(f) \subseteq \mathrm{Rad}(M) \subseteq K$, the map f induces a surjection

$$\overline{f} : M/K \to N/f(K).$$

Because M/K is simple and $x \notin K$, the quotient $N/f(K)$ is nonzero and hence simple. Thus $f(K)$ is a maximal submodule of N. But $f(x) \notin f(K)$, so $f(x) \notin \mathrm{Rad}(N)$, contradiction. Therefore
$$f^{-1}(\mathrm{Rad}(N)) = \mathrm{Rad}(M),$$

and applying f gives the desired equality

$$\mathrm{Rad}(N) = f(\mathrm{Rad}(M)).$$

∎

Lemma 5.1.11. *Let M and N be finitely generated R-modules, and let*

$$f : M \to N$$

be an R-homomorphism. Then f is an essential epimorphism if and only if the induced morphism

$$\overline{f} : M/\operatorname{Rad}(M) \to N/\operatorname{Rad}(N)$$

is an essential epimorphism.

Proof. First assume that f is an essential epimorphism. In particular,f is surjective. By the previous lemma,

$$f(\operatorname{Rad}(M)) \subseteq \operatorname{Rad}(N),$$

so $\overline{f}$ is well defined and surjective. Let

$$X/\operatorname{Rad}(M) \subseteq M/\operatorname{Rad}(M)$$

be a submodule such that $\overline{f}(X/\operatorname{Rad}(M)) = N/\operatorname{Rad}(N)$. Then

$$f(X) + \operatorname{Rad}(N) = N.$$

Since $N/f(X)$ is a quotient of the finitely generated module N, Nakayama's lemma gives $f(X) = N$. As f is essential, we conclude that $X = M$. Hence $\overline{f}$ is essential.

Conversely, assume that $\overline{f}$ is an essential epimorphism. Since $\overline{f}$ is surjective, we have

$$f(M) + \operatorname{Rad}(N) = N.$$

By Nakayama's lemma applied to the submodule $f(M) \subseteq N$, it follows that $f(M) = N$, so f is surjective. Now let $X \leq M$ satisfy $f(X) = N$. Then

$$\overline{f}\big((X + \operatorname{Rad}(M))/\operatorname{Rad}(M)\big) = N/\operatorname{Rad}(N).$$

Since $\overline{f}$ is essential, we obtain

$$X + \operatorname{Rad}(M) = M.$$

Again Nakayama's lemma yields $X = M$. Hence f is an essential epimorphism. $\square$

5.2 Composition Series and Jordan–Hölder Theory

Definition 5.2.1. *A finite chain of submodules*

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$

is called a composition series *of U if each factor*

$$U_i/U_{i+1}$$

is simple. The integer n is called the composition length *of U.*

Lemma 5.2.2. *A nonzero R-module U has a composition series if and only if U is both Noetherian and Artinian.*

Proof. Suppose first that
$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$
is a composition series. We argue by induction on n. If $n = 1$, then U is simple, hence both Noetherian and Artinian. If $n > 1$, then U_1 has a composition series of length $n-1$, so by induction U_1 is Noetherian and Artinian. The quotient U/U_1 is simple, hence also Noetherian and Artinian. By the permanence of both properties in short exact sequences,U is Noetherian and Artinian.

Conversely, assume that U is Noetherian and Artinian. Since $U \neq 0$, choose a maximal proper submodule U_1 of U. Then U/U_1 is simple. Because U_1 is a submodule of both a Noetherian and an Artinian module, it is again Noetherian and Artinian. Repeating the argument inductively and using the Artinian condition to rule out an infinite descending chain, we obtain a composition series. □

Lemma 5.2.3. *Suppose that*

$$U = S_1 + \cdots + S_n$$

where each S_i is a simple submodule of U. If $V \leq U$, then there exists a subset $I \subseteq \{1, \ldots, n\}$ such that
$$U = V \oplus \bigoplus_{i \in I} S_i.$$
In particular, every submodule of a finite direct sum of simple modules is a direct summand.

Proof. Choose a subset $I \subseteq \{1, \ldots, n\}$ maximal with the property that

$$V \oplus \bigoplus_{i \in I} S_i$$

is a direct sum, and set
$$W := V \oplus \bigoplus_{i \in I} S_i.$$
We claim that $W = U$. If not, then there exists $j \notin I$ such that

$$S_j \nsubseteq W.$$

Now $W \cap S_j$ is a submodule of the simple module S_j, so either

$$W \cap S_j = 0 \qquad \text{or} \qquad W \cap S_j = S_j.$$

The second case is impossible because it would imply $S_j \subseteq W$. Hence $W \cap S_j = 0$, and therefore

$$W \oplus S_j$$

is a direct sum, contradicting the maximality of I. Thus $W = U$. □

Theorem 5.2.4. *Let U be an R-module with a composition series. The following are equivalent.*

1. *U is semisimple.*
2. *U is a sum of simple submodules.*
3. *Every submodule of U is a direct summand of U.*

Proof. **(1) $\Longrightarrow$(2).** This is immediate from the definition of a semisimple module.

(2) $\Longrightarrow$(3). Since U has finite composition length, a finite number of simple summands already generate U. Thus

$$U = S_1 + \cdots + S_n$$

for simple submodules S_i. The previous lemma shows that every submodule of U is a direct summand.

(3) $\Longrightarrow$(1). Let

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$

be a composition series. Since U_1 is a direct summand of U, there exists a submodule S_1 such that

$$U = U_1 \oplus S_1.$$

Because U/U_1 is simple, the submodule S_1 is simple. Repeating the argument inside U_1, we obtain simple submodules $S_2, \ldots, S_n$ with

$$U_i = U_{i+1} \oplus S_{i+1}$$

for all i. Hence

$$U = S_1 \oplus \cdots \oplus S_n,$$

so U is semisimple. □

Corollary 5.2.5. *Let U be an R-module of finite composition length.*

1. *The sum of all simple submodules of U is semisimple, hence is the unique largest semisimple submodule of U.*
2. *If S is a simple submodule of U, then the sum of all submodules of U isomorphic*

to S is a submodule of U isomorphic to a direct sum of copies of S. It is the unique largest submodule of U with this property.

Proof. For (1), let Σ be the sum of all simple submodules of U. Since U has finite composition length, only finitely many simple submodules are needed to generate Σ. By the theorem,Σ is semisimple. Any semisimple submodule is a sum of simple submodules and is therefore contained in Σ.

For (2), let Σ_S be the sum of all submodules of U isomorphic to S. Again finite composition length implies that Σ_S is generated by finitely many such submodules, so Σ_S is semisimple. Every simple submodule of Σ_S is isomorphic to S, hence

$$\Sigma_S \cong S^{\oplus m}$$

for some $m \geq 0$. The maximality statement is immediate from the construction. □

Theorem 5.2.6 (Zassenhaus lemma)**.** *Let $A' \subseteq A$ and $B' \subseteq B$ be submodules of an R-module M. Then*

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{B' + (A \cap B)}{B' + (A' \cap B)}.$$

Proof. By the second isomorphism theorem,

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{A \cap B}{(A \cap B') + (A' \cap B)}.$$

Interchanging the roles of A and B gives

$$\frac{B' + (A \cap B)}{B' + (A' \cap B)} \cong \frac{A \cap B}{(A \cap B') + (A' \cap B)}.$$

Hence the two quotients are naturally isomorphic. □

Theorem 5.2.7 (Schreier refinement theorem)**.** *Any two finite submodule series of a module admit equivalent refinements.*

Proof. Starting from two finite submodule series, form the usual Schreier refinements by intersecting the terms of one series with the terms of the other. The Zassenhaus lemma shows that the successive factors in these two refinements are pairwise isomorphic up to reordering. Hence the refinements are equivalent. □

Theorem 5.2.8 (Jordan–Hölder theorem)**.** *Let U be an R-module with a composition series. Then any two composition series of U have the same length, and their composition factors are the same up to permutation and isomorphism.*

Proof. A composition series is, by definition, a submodule series whose successive factors are simple. Therefore it admits no proper refinement. Applying the Schreier refinement theorem to two composition series of U, we obtain equivalent refinements of both. Since no further refinement is possible, the two original series were already equivalent. This means precisely that they have the same length and the same multiset of composition factors up to isomorphism. $\square$

5.3 Socles and Projective Covers

Definition 5.3.1. *The unique largest semisimple submodule of an R-module U is called the* socle *of U and is denoted by*

$$\mathrm{Soc}(U).$$

Lemma 5.3.2. *Let*

$$U = S_1^{\oplus a_1} \oplus \cdots \oplus S_r^{\oplus a_r}$$

be a semisimple R-module, where the S_i are pairwise nonisomorphic simple modules. Then, for each i, the summand $S_i^{\oplus a_i}$ is uniquely determined: it is the unique largest submodule of U that is isomorphic to a direct sum of copies of S_i.

Proof. Let $T \leq U$ be a submodule which is a direct sum of copies of S_i. Write

$$U_i := S_i^{\oplus a_i} \qquad \text{and} \qquad U' := \bigoplus_{j \neq i} S_j^{\oplus a_j}.$$

Consider the projection

$$p : U = U_i \oplus U' \to U'.$$

If $p(T) \neq 0$, then $p(T)$ contains a simple submodule isomorphic to a simple submodule of U', hence to some S_j with $j \neq i$. This contradicts the fact that every simple submodule of T is isomorphic to S_i. Therefore $p(T) = 0$, so $T \subseteq U_i$. Thus U_i contains every submodule of U which is a direct sum of copies of S_i, and therefore U_i is the unique largest such submodule. $\square$

Lemma 5.3.3. *Let U be an R-module.*

1. *Suppose $M_1, \ldots, M_n$ are maximal submodules of U. Then there exists a subset $I \subseteq \{1, \ldots, n\}$ such that*

$$U/(M_1 \cap \cdots \cap M_n) \cong \bigoplus_{i \in I} U/M_i.$$

   In particular,$U/(M_1 \cap \cdots \cap M_n)$ is semisimple.

2. *If U is finitely generated and Artinian, then $U/\,\mathrm{Rad}(U)$ is semisimple, and $\mathrm{Rad}(U)$ is the unique smallest submodule of U with semisimple quotient.*

Proof. **(1).** Choose a subset $I \subseteq \{1, \dots, n\}$ maximal with the property that the diagonal map

$$\delta_I : U \to \bigoplus_{i \in I} U/M_i, \qquad u \mapsto (u + M_i)_{i \in I}$$

is surjective. Then

$$\ker(\delta_I) = \bigcap_{i \in I} M_i.$$

If some $j \notin I$ satisfied

$$\bigcap_{i \in I} M_i \nsubseteq M_j,$$

then the induced map

$$U \to \left(\bigoplus_{i \in I} U/M_i \right) \oplus U/M_j$$

would still be surjective, contradicting the maximality of I. Hence

$$\bigcap_{i \in I} M_i \subseteq M_j \qquad \text{for all } j \notin I,$$

and therefore

$$\bigcap_{i \in I} M_i = M_1 \cap \dots \cap M_n.$$

Applying the first isomorphism theorem gives the required decomposition.

(2). Because U is finitely generated and Artinian, every proper submodule is contained in a maximal submodule, and the intersection of all maximal submodules stabilizes to a finite intersection. Thus

$$\mathrm{Rad}(U) = M_1 \cap \dots \cap M_n$$

for suitable maximal submodules M_i. By part (1),

$$U/\,\mathrm{Rad}(U)$$

is semisimple.

Now let $V \leq U$ be such that U/V is semisimple. Then every maximal submodule containing V is the kernel of a simple quotient of U/V, so V is contained in every maximal submodule of U. Hence

$$V \subseteq \mathrm{Rad}(U).$$

Therefore $\mathrm{Rad}(U)$ is the unique smallest submodule with semisimple quotient. □

Lemma 5.3.4. *For all R-modules U and V,*

$$\mathrm{Rad}(U \oplus V) = \mathrm{Rad}(U) \oplus \mathrm{Rad}(V) \qquad \text{and} \qquad \mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Proof. The projection maps show that every maximal submodule of $U \oplus V$ is of the form

$$M \oplus V \qquad \text{or} \qquad U \oplus N,$$

where M and N are maximal submodules of U and V, respectively. Intersecting all such maximal submodules yields

$$\mathrm{Rad}(U \oplus V) = \mathrm{Rad}(U) \oplus \mathrm{Rad}(V).$$

For the socle, every simple submodule of $U \oplus V$ projects nontrivially onto at least one of the two factors; by simplicity it must embed into one factor. Therefore every simple submodule of $U \oplus V$ lies in

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V),$$

and the reverse inclusion is clear. □

Proposition 5.3.5. *Suppose that*

$$f : P \to U$$

is a projective cover of an R-module U, and that

$$g : Q \to U$$

is an epimorphism with Q projective.

1. *There exists a decomposition*

$$Q = Q_1 \oplus Q_2$$

such that, with respect to this decomposition,

$$g = (g_1, 0),$$

and there is an isomorphism $\gamma : Q_1 \to P$ satisfying

$$f \circ \gamma = g_1.$$

2. *If a projective cover of U exists, then it is unique up to isomorphism.*

Proof. Since Q is projective and $f : P \to U$ is surjective, there exists

$$\alpha : Q \to P$$

such that

$$f \circ \alpha = g.$$

Since P is projective and $g : Q \to U$ is surjective, there exists

$$\beta : P \to Q$$

such that

$$g \circ \beta = f.$$

Consequently,

$$f = f \circ (\alpha\beta).$$

Because f is an essential epimorphism, the endomorphism $\alpha\beta$ of P must be surjective. Since P is projective,$\alpha\beta$ splits. Let

$$P = \ker(\alpha\beta) \oplus P'.$$

If $\ker(\alpha\beta) \neq 0$, then P' is a proper submodule of P and

$$f(P') = f(\alpha\beta(P')) = f(P) = U,$$

contradicting the essentiality of f. Hence $\ker(\alpha\beta) = 0$, so $\alpha\beta$ is an automorphism.

Replacing β by $\beta \circ (\alpha\beta)^{-1}$, we may assume that

$$\alpha\beta = \mathrm{id}_P .$$

Thus β is a split monomorphism and

$$Q = \beta(P) \oplus \ker(\alpha).$$

Set

$$Q_1 := \beta(P), \qquad Q_2 := \ker(\alpha),$$

and let $\gamma : Q_1 \to P$ be the inverse of the isomorphism $\beta : P \to Q_1$. Then

$$g|_{Q_1} = f \circ \gamma.$$

Moreover,

$$g(Q_2) = f(\alpha(Q_2)) = 0,$$

so $g = (g_1, 0)$ with $g_1 = f \circ \gamma$. This proves (1).

For (2), if

$$f : P \to U \qquad \text{and} \qquad f' : P' \to U$$

are projective covers, apply (1) twice: once with $Q = P'$ and once with $Q = P$. We deduce that P is isomorphic to a direct summand of P' and P' is isomorphic to a direct summand of P. The decompositions obtained in (1) force the complementary summands to map trivially to U, hence to vanish by essentiality. Therefore

$$P \cong P',$$

so the projective cover is unique up to isomorphism. $\square$

Chapter 6

Lecture 6

6.1 Projective covers, radicals, and finiteness conditions

Corollary 6.1.1. *For finitely generated projective modules P and Q,*

$$P \cong Q \quad \iff \quad P/\operatorname{Rad}(P) \cong Q/\operatorname{Rad}(Q).$$

Proof. Let

$$T := P/\operatorname{Rad}(P).$$

By Nakayama lemma, the quotient map $P \to T$ is an essential epimorphism, so P is the projective cover of T. Similarly $Q \to Q/\operatorname{Rad}(Q)$ is a projective cover. Then apply uniqueness of projective covers from the previous lecture. $\square$

Definition 6.1.2. *A unital ring R is called left Noetherian if ${}_RR$ is Noetherian as a left R-module.*

Remark 6.1.3. *Similar definitions are given for right Noetherian and for left/right Artinian rings.*

Remark 6.1.4. *one-sided Noetherian / Artinian conditions generally do not imply the opposite-sided versions*

Example 6.1.5. *There exists a ring which is left Noetherian but not right Noetherian. Let*

$$R = \left\{ \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \middle| \, a, b \in \mathbb{Q}, \; n \in \mathbb{Z} \right\}.$$

Then R is left Noetherian, but not right Noetherian.

Proof. Let

$$e_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then, as left R-modules,

$${}_RR = Re_{11} \oplus Re_{22}.$$

We first show that both Re_{11} and Re_{22} are Noetherian as left R-modules.

For Re_{11}, we have

$$Re_{11} = \left\{ \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} \,\middle|\, q \in \mathbb{Q} \right\}.$$

This is a simple left R-module. Indeed, if

$$0 \neq \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} \in Re_{11},$$

then for any $x \in \mathbb{Q}$,

$$\begin{pmatrix} xq^{-1} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}.$$

Hence every nonzero submodule of Re_{11} is all of Re_{11}, so Re_{11} is simple, in particular Noetherian.

Now consider

$$Re_{22} = \left\{ \begin{pmatrix} 0 & q \\ 0 & n \end{pmatrix} \,\middle|\, q \in \mathbb{Q},\ n \in \mathbb{Z} \right\}.$$

Define

$$\pi : Re_{22} \longrightarrow \mathbb{Z}, \qquad \pi\left(\begin{pmatrix} 0 & q \\ 0 & n \end{pmatrix} \right) = n.$$

We regard $\mathbb{Z}$ as a left R-module via

$$\begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \cdot m = nm.$$

Then π is a surjective R-homomorphism. Its kernel is

$$\ker \pi = \left\{ \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \,\middle|\, q \in \mathbb{Q} \right\}.$$

This is again a simple left R-module, because

$$\begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & aq \\ 0 & 0 \end{pmatrix},$$

so the action is through multiplication by elements of $\mathbb{Q}$. Moreover, ${}_R\mathbb{Z}$ is Noetherian, since its submodules are exactly the ideals of $\mathbb{Z}$, and $\mathbb{Z}$ is Noetherian.

Therefore Re_{22} is an extension of two Noetherian left R-modules, hence is Noetherian. It follows that

$${}_R R = Re_{11} \oplus Re_{22}$$

is a Noetherian left R-module. Thus R is left Noetherian.

Next we show that R is not right Noetherian.

For each integer $k \geq 0$, define

$$I_k = \left\{ \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \,\middle|\, q \in 2^{-k}\mathbb{Z} \right\}.$$

We claim that each I_k is a right ideal of R. Indeed, if

$$\begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \in I_k, \qquad \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \in R,$$

then

$$\begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} = \begin{pmatrix} 0 & qn \\ 0 & 0 \end{pmatrix}.$$

Since $q \in 2^{-k}\mathbb{Z}$ and $n \in \mathbb{Z}$, we have $qn \in 2^{-k}\mathbb{Z}$, so the product again lies in I_k.

Moreover,

$$I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \cdots$$

is a strictly ascending chain of right ideals, because

$$2^{-(k+1)} \in 2^{-(k+1)}\mathbb{Z} \quad \text{but} \quad 2^{-(k+1)} \notin 2^{-k}\mathbb{Z}.$$

Hence R does not satisfy the ascending chain condition on right ideals.

Therefore R is not right Noetherian. □

Lemma 6.1.6. *Let R be a left Artinian ring. Then its Jacobson radical $J(R)$ is nilpotent.*

Proof. Set

$$J := J(R).$$

Since R is left Artinian, the descending chain of left ideals

$$J \supseteq J^2 \supseteq J^3 \supseteq \cdots$$

stabilizes. Hence there exists some $n \geq 1$ such that

$$J^n = J^{n+1}.$$

Let

$$N := J^n.$$

We claim that $N = 0$.

Suppose, for contradiction, that $N \neq 0$. Then

$$N^2 = J^{2n} = J^n = N,$$

since the chain has stabilized.

Now consider the set

$$\Sigma = \{\, I \leq {}_R R \mid NI \neq 0 \,\}$$

of left ideals of R. This set is nonempty because

$$N^2 = N \neq 0,$$

so $N \in \Sigma$.

Since R is left Artinian, the set of left ideals satisfies the descending chain condition, so Σ contains a minimal element. Let $I_0 \in \Sigma$ be such a minimal left ideal.

Because $NI_0 \neq 0$, there exists some $a \in I_0$ such that

$$Na \neq 0.$$

Then Na is a left ideal of R contained in I_0, and moreover

$$N(Na) = N^2 a = Na \neq 0.$$

Thus $Na \in \Sigma$. By the minimality of I_0, we must have

$$Na = I_0.$$

Since $a \in I_0 = Na$, there exists $b \in N$ such that

$$a = ba.$$

Therefore

$$(1 - b)a = 0.$$

But $b \in N \subseteq J(R)$, and for every element of the Jacobson radical, $1 - b$ has a left inverse. Hence there exists $c \in R$ such that

$$c(1 - b) = 1.$$

Multiplying the equation $(1 - b)a = 0$ on the left by c, we get

$$a = c(1 - b)a = 0,$$

contradicting the choice of a with $Na \neq 0$.

This contradiction shows that $N = 0$. Hence

$$J^n = 0,$$

so $J(R)$ is nilpotent. □

Theorem 6.1.7 (Levitzki)**.** *Every left Artinian ring is left Noetherian.*

Proof. Let R be a left Artinian ring, and let $J = J(R)$ be its Jacobson radical.

Since R is left Artinian, the quotient ring R/J is also left Artinian. Moreover, $J(R/J) = 0$, hence R/J is semisimple. Therefore every left R/J-module is semisimple.

Also, since R is left Artinian, J is nilpotent. Thus there exists $n \geq 1$ such that

$$J^n = 0.$$

Consider the finite chain of left ideals

$$0 = J^n \subseteq J^{n-1} \subseteq \cdots \subseteq J \subseteq R.$$

For each $i = 0, 1, \ldots, n-1$, the factor module J^i/J^{i+1} is naturally a left R/J-module, because

$$J \cdot J^i \subseteq J^{i+1}.$$

Hence J^i/J^{i+1} is semisimple.

On the other hand, J^i/J^{i+1} is Artinian, being a quotient of the Artinian left R-module ${}_RR$. A semisimple Artinian module must be a finite direct sum of simple modules, so it is Noetherian. Therefore each factor

$$J^i/J^{i+1}$$

is Noetherian.

Now we argue by descending induction on i. Since $J^n = 0$ is Noetherian, and for each $i = 0, 1, \ldots, n-1$ we have a short exact sequence

$$0 \longrightarrow J^{i+1} \longrightarrow J^i \longrightarrow J^i/J^{i+1} \longrightarrow 0,$$

it follows that if J^{i+1} is Noetherian, then so is J^i. Hence

$$J^{n-1},\ J^{n-2},\ \ldots,\ J,\ R$$

are all Noetherian as left R-modules.

In particular, ${}_RR$ is Noetherian. Therefore R is a left Noetherian ring. □

Theorem 6.1.8. *If R is left Noetherian/Artinian, then every finitely generated left R-module is Noetherian/Artinian.*

Proof. Prove for Noetherian case: If M is generated by m elements, then $M = Rm_1 + \ldots + Rm_n$, each Rm_i is a homomorphic image of ${}_RR$. Hence Rm_i is Noetherian. Finite sums of Noetherian modules are Noetherian, so M is Noetherian. □

6.2 Artin-Wedderburn

Theorem 6.2.1. *Let*

$$ {}_RV = V_1 \oplus \cdots \oplus V_n $$

be a direct sum of R-modules, and suppose all V_i are isomorphic to each other. Then

$$ \mathrm{End}_R({}_RV) \cong M_n\big(\mathrm{End}_R(V_i)\big). $$

Proof. For each i, let

$$ \iota_i : V_i \hookrightarrow V \qquad \text{and} \qquad \pi_i : V \to V_i $$

be the canonical inclusion and projection. Fix isomorphisms

$$ \theta_i : V_1 \xrightarrow{\sim} V_i \qquad (i = 1, \dots, n). $$

For $\sigma \in \mathrm{End}_R(V)$, define

$$ \varphi_{ij}(\sigma) := \theta_i^{-1}\pi_i\,\sigma\,\iota_j\theta_j \in \mathrm{End}_R(V_1). $$

Thus we obtain a map

$$ \Phi : \mathrm{End}_R(V) \longrightarrow M_n(E_1), \qquad \sigma \longmapsto \big(\varphi_{ij}(\sigma)\big), $$

where

$$ E_1 := \mathrm{End}_R(V_1). $$

We first check that Φ is a ring homomorphism.

Additivity is immediate. For multiplicativity, let $\sigma, \beta \in \mathrm{End}_R(V)$. Since

$$ 1_V = \sum_{j=1}^{n} \iota_j\pi_j, $$

we have

$$ \begin{aligned} \varphi_{ik}(\beta\sigma) &= \theta_i^{-1}\pi_i\,\beta\sigma\,\iota_k\theta_k \\ &= \theta_i^{-1}\pi_i\,\beta\left(\sum_{j=1}^{n}\iota_j\pi_j\right)\sigma\,\iota_k\theta_k \\ &= \sum_{j=1}^{n}\theta_i^{-1}\pi_i\,\beta\,\iota_j\pi_j\,\sigma\,\iota_k\theta_k \\ &= \sum_{j=1}^{n}\big(\theta_i^{-1}\pi_i\,\beta\,\iota_j\theta_j\big)\big(\theta_j^{-1}\pi_j\,\sigma\,\iota_k\theta_k\big) \\ &= \sum_{j=1}^{n}\varphi_{ij}(\beta)\varphi_{jk}(\sigma). \end{aligned} $$

Hence

$$\Phi(\beta\sigma) = \Phi(\beta)\Phi(\sigma).$$

So Φ is a ring homomorphism.

Next we show that Φ is injective. Suppose $\Phi(\sigma) = 0$. Then

$$\varphi_{ij}(\sigma) = 0 \qquad \text{for all } i, j,$$

so

$$\theta_i^{-1}\pi_i\, \sigma\, \iota_j\theta_j = 0 \qquad \text{for all } i, j.$$

Since each θ_i and θ_j is an isomorphism, this implies

$$\pi_i\, \sigma\, \iota_j = 0 \qquad \text{for all } i, j.$$

Therefore $\pi_i(\sigma(v)) = 0$ for every $v \in V_j$ and every i, j. Hence

$$\sigma(V_j) = 0 \qquad \text{for all } j.$$

Because

$$V = V_1 \oplus \cdots \oplus V_n,$$

it follows that $\sigma = 0$. Thus Φ is injective.

Finally we prove surjectivity. Let

$$(a_{ij}) \in M_n(E_1).$$

Define

$$\sigma := \sum_{i,j=1}^{n} \iota_i\theta_i\, a_{ij}\, \theta_j^{-1}\pi_j \in \mathrm{End}_R(V).$$

Then for each i, j,

$$\begin{aligned}
\varphi_{ij}(\sigma) &= \theta_i^{-1}\pi_i \left(\sum_{r,s=1}^{n} \iota_r\theta_r\, a_{rs}\, \theta_s^{-1}\pi_s \right) \iota_j\theta_j \\
&= \sum_{r,s=1}^{n} \theta_i^{-1}(\pi_i\iota_r)\theta_r\, a_{rs}\, \theta_s^{-1}(\pi_s\iota_j)\theta_j.
\end{aligned}$$

Now

$$\pi_i\iota_r = \begin{cases} 1_{V_i}, & i = r, \\ 0, & i \neq r, \end{cases} \qquad \pi_s\iota_j = \begin{cases} 1_{V_j}, & s = j, \\ 0, & s \neq j, \end{cases}$$

so all terms vanish except the one with $r = i$ and $s = j$. Hence

$$\varphi_{ij}(\sigma) = \theta_i^{-1}\theta_i\, a_{ij}\, \theta_j^{-1}\theta_j = a_{ij}.$$

Therefore

$$\Phi(\sigma) = (a_{ij}),$$

and Φ is surjective.

Thus Φ is a ring isomorphism, and we conclude that

$$\mathrm{End}_R(V) \cong M_n\big(\mathrm{End}_R(V_1)\big).$$

□

Definition 6.2.2. *A ring R is called semisimple if ${}_RR$ is semisimple as a left R-module.*

Lemma 6.2.3. *For a unital ring R, the following are equivalent: (1) R is semisimple. (2) Every left R-module is semisimple.*

Proof. If ${}_RR$ is semisimple, then every free module is semisimple, and every module is a quotient of a free module. Use closure properties of semisimple modules to deduce that every left R-module is semisimple. The converse is immediate by applying the statement to ${}_RR$ itself. □

Lemma 6.2.4. *For an R-module ${}_RM$, there is a standard identification*

$$\mathrm{Hom}_R({}_RR_R, {}_RM) \cong {}_RM, \qquad f \longmapsto f(1).$$

Proof. Define

$$\Phi : \mathrm{Hom}_R({}_RR_R, {}_RM) \longrightarrow {}_RM, \qquad \Phi(f) = f(1).$$

We first show that Φ is bijective.

For any $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$ and any $r \in R$, since f is left R-linear, we have

$$f(r) = f(r \cdot 1) = r\, f(1).$$

Thus f is completely determined by the element $f(1) \in M$.

Conversely, for each $m \in M$, define

$$\Psi(m) : {}_RR_R \longrightarrow {}_RM, \qquad \Psi(m)(r) = rm \quad (r \in R).$$

Then $\Psi(m)$ is left R-linear, because for all $a, r \in R$,

$$\Psi(m)(ar) = (ar)m = a(rm) = a\,\Psi(m)(r).$$

Hence $\Psi(m) \in \mathrm{Hom}_R({}_RR_R, {}_RM)$.

Now for $m \in M$,

$$(\Phi \circ \Psi)(m) = \Phi(\Psi(m)) = \Psi(m)(1) = 1m = m.$$

And for $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$ and $r \in R$,

$$(\Psi \circ \Phi)(f)(r) = \Psi(f(1))(r) = r\,f(1) = f(r).$$

So $\Psi \circ \Phi = \mathrm{id}$ and $\Phi \circ \Psi = \mathrm{id}$, hence Φ is a bijection.

Finally, we check that this identification is R-linear. Recall that $\mathrm{Hom}_R({}_RR_R, {}_RM)$ carries the left R-module structure

$$(a \cdot f)(r) = f(ra) \qquad (a, r \in R).$$

Then for $a \in R$ and $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$,

$$\Phi(a \cdot f) = (a \cdot f)(1) = f(1a) = f(a) = a\,f(1) = a\,\Phi(f).$$

Therefore Φ is an isomorphism of left R-modules:

$$\mathrm{Hom}_R({}_RR_R, {}_RM) \cong {}_RM.$$

□

Corollary 6.2.5. *For any ring R, there is a ring isomorphism*

$$\mathrm{End}_R({}_RR) \cong R^{\mathrm{op}}, \qquad f \longmapsto f(1).$$

Proof. Consider the map

$$\Phi : \mathrm{End}_R({}_RR) \longrightarrow R, \qquad \Phi(f) = f(1).$$

By the preceding lemma with $M = R$, this map is a bijection of sets.

We now check the ring structure. Let $f_1, f_2 \in \mathrm{End}_R({}_RR)$. Then

$$\Phi(f_1 \circ f_2) = (f_1 \circ f_2)(1) = f_1\big(f_2(1)\big).$$

Since f_1 is left R-linear, for any $r \in R$ we have

$$f_1(r) = f_1(r \cdot 1) = r\,f_1(1).$$

Hence

$$f_1\big(f_2(1)\big) = f_2(1)\,f_1(1).$$

Therefore

$$\Phi(f_1 \circ f_2) = \Phi(f_2)\Phi(f_1),$$

where the product on the right is the multiplication in R. This means exactly that

$$\Phi(f_1 \circ f_2) = \Phi(f_1) \cdot_{\mathrm{op}} \Phi(f_2)$$

when the codomain is viewed as R^{op}.

Thus Φ is a ring isomorphism

$$\mathrm{End}_R({}_R R) \cong R^{\mathrm{op}}.$$

$\square$

Lemma 6.2.6. *For any ring R and any $n \geq 1$, there is a ring isomorphism*

$$M_n(R)^{\mathrm{op}} \cong M_n\big(R^{\mathrm{op}}\big), \qquad A \longmapsto A^{\top}.$$

Proof. Define

$$\Psi : M_n(R)^{\mathrm{op}} \longrightarrow M_n\big(R^{\mathrm{op}}\big), \qquad \Psi(A) = A^{\top}.$$

This map is clearly additive and bijective, since transpose is its own inverse.

It remains to check multiplicativity. Let

$$A = (a_{ij}), \qquad B = (b_{ij}) \in M_n(R).$$

Recall that multiplication in $M_n(R)^{\mathrm{op}}$ is reversed, so

$$A \cdot_{\mathrm{op}} B = BA$$

with the usual matrix product on the right-hand side.

Now fix i, j. Then

$$\big(\Psi(A \cdot_{\mathrm{op}} B)\big)_{ij} = \big((BA)^{\top}\big)_{ij} = (BA)_{ji} = \sum_{k=1}^{n} b_{jk} a_{ki}.$$

On the other hand, in the ring R^{op} the product is reversed, so

$$\big(\Psi(A)\Psi(B)\big)_{ij} = \sum_{k=1}^{n} (A^{\top})_{ik} \cdot_{R^{\mathrm{op}}} (B^{\top})_{kj} = \sum_{k=1}^{n} a_{ki} \cdot_{R^{\mathrm{op}}} b_{jk} = \sum_{k=1}^{n} b_{jk} a_{ki}.$$

Thus

$$\Psi(A \cdot_{\mathrm{op}} B) = \Psi(A)\Psi(B).$$

So Ψ is a ring homomorphism, hence a ring isomorphism.

Therefore

$$M_n(R)^{\mathrm{op}} \cong M_n\big(R^{\mathrm{op}}\big).$$

$\square$

Theorem 6.2.7. *(Artin-Wedderburn) Let A be a semisimple Artinian ring. Then*

$$A \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r),$$

where each D_i is a division ring. More concretely, if

$$ {}_A A \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r} $$

with pairwise nonisomorphic simple A-modules S_i, then

$$ D_i = \mathrm{End}_A(S_i)^{op}. $$

Proof. Decompose $\mathrm{End}_A(A)$ using the decomposition of the regular module. Use

$$ \mathrm{Hom}_A(S_i, S_j) = 0 \quad (i \neq j) $$

and Schur's lemma to identify the diagonal blocks with division rings. Then use the previous matrix-ring result to rewrite

$$ \mathrm{End}_A\big(S_i^{\oplus n_i}\big) $$

as

$$ M_{n_i}\big(\mathrm{End}_A(S_i)\big) = M_{n_i}(D_i^{op}). $$

Therefore

$$ A \cong \mathrm{End}_A(A)^{op} \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r). $$

□

Definition 6.2.8. *Let R be a commutative ring with 1, and let A be an associative ring with 1. An R-algebra is an R-module A whose multiplication is compatible with the R-action. Equivalently, there is a ring homomorphism*

$$ R \to A $$

whose image lies in the center $Z(A)$.

Remark 6.2.9. *If $R = k$ is a field and A is finite-dimensional over k, then A is called a finite-dimensional k-algebra. Such an algebra is Noetherian and Artinian as a ring/module object.*

Example 6.2.10. *Let R be a commutative ring with 1. The following are R-algebras:*

1. $R[x_1, \dots, x_n]$.
2. $M_n(R)$.
3. *Let G be a finite group. The group algebra*

$$ RG = \left\{ \sum_{g \in G} a_g g \;\middle|\; a_g \in R \right\} $$

is an R-algebra, with multiplication defined by

$$\left(\sum_{g\in G} a_g g\right)\left(\sum_{h\in G} b_h h\right) = \sum_{r\in G}\left(\sum_{\substack{g,h\in G\\ gh=r}} a_g b_h\right) r.$$

Lemma 6.2.11. *Let A be a k-algebra, where k is a field, and let S be a simple A-module. Then $\mathrm{End}_A(S)$ is a division ring, and it is naturally a k-algebra.*

Proof. We first show that $\mathrm{End}_A(S)$ is a division ring.

Let

$$0 \neq f \in \mathrm{End}_A(S).$$

Since f is A-linear, both $\ker(f)$ and $\mathrm{im}(f)$ are A-submodules of S. Because S is simple and $f \neq 0$, we have $\mathrm{im}(f) \neq 0$, hence

$$\mathrm{im}(f) = S.$$

Also $\ker(f) \neq S$ since $f \neq 0$, so by simplicity,

$$\ker(f) = 0.$$

Thus f is bijective. Since the inverse of a bijective A-linear map is again A-linear, $f^{-1} \in \mathrm{End}_A(S)$. Therefore every nonzero element of $\mathrm{End}_A(S)$ is invertible, so $\mathrm{End}_A(S)$ is a division ring.

Next we show that $\mathrm{End}_A(S)$ is naturally a k-algebra.

Because A is a k-algebra, S becomes a k-vector space via

$$\lambda \cdot s := (\lambda 1_A)s \qquad (\lambda \in k,\ \ s \in S).$$

Now let $f \in \mathrm{End}_A(S)$. Then for $\lambda \in k$ and $s \in S$,

$$f(\lambda s) = f\big((\lambda 1_A)s\big) = (\lambda 1_A)f(s) = \lambda f(s).$$

Hence every A-linear endomorphism of S is automatically k-linear. So

$$\mathrm{End}_A(S) \subseteq \mathrm{End}_k(S)$$

as a k-subspace. Equivalently, there is a k-algebra homomorphism

$$k \longrightarrow \mathrm{End}_A(S), \qquad \lambda \longmapsto \lambda\, \mathrm{id}_S,$$

whose image lies in the center of $\mathrm{End}_A(S)$. Therefore $\mathrm{End}_A(S)$ is naturally a k-algebra. $\square$

Lemma 6.2.12. *If A is finite-dimensional over k, then $\mathrm{End}_A(S)$ is finite-dimensional over k.*

Proof. Choose $0 \neq s \in S$. Since S is simple, the cyclic submodule As is nonzero, hence

$$As = S.$$

Consider the map

$$\varphi : A \longrightarrow S, \qquad a \longmapsto as.$$

This is a surjective k-linear map. Therefore

$$\dim_k S \leq \dim_k A < \infty.$$

So S is finite-dimensional over k.

Since every A-endomorphism of S is k-linear, we have

$$\mathrm{End}_A(S) \subseteq \mathrm{End}_k(S)$$

as k-vector spaces. Hence

$$\dim_k \mathrm{End}_A(S) \leq \dim_k \mathrm{End}_k(S) = (\dim_k S)^2 < \infty.$$

Thus $\mathrm{End}_A(S)$ is finite-dimensional over k. □

Lemma 6.2.13. *If A is finite-dimensional over k and k is algebraically closed, then*

$$\mathrm{End}_A(S) = k.$$

Proof. By the previous lemma, S is finite-dimensional over k, and $\mathrm{End}_A(S)$ is a finite-dimensional k-algebra.

Let

$$f \in \mathrm{End}_A(S).$$

Since S is a finite-dimensional k-vector space and k is algebraically closed, the k-linear operator $f : S \to S$ has an eigenvalue $\lambda \in k$. Hence

$$f - \lambda \,\mathrm{id}_S$$

has a nonzero kernel, so it is not injective.

But $f - \lambda \,\mathrm{id}_S$ is still an element of $\mathrm{End}_A(S)$. Since $\mathrm{End}_A(S)$ is a division ring by the first lemma, every nonzero element is invertible, hence injective. Therefore

$$f - \lambda \,\mathrm{id}_S = 0.$$

So

$$f = \lambda \,\mathrm{id}_S\,.$$

This shows that every A-endomorphism of S is scalar. Under the natural embedding

$$k \hookrightarrow \operatorname{End}_A(S), \qquad \lambda \mapsto \lambda \operatorname{id}_S,$$

we therefore obtain

$$\operatorname{End}_A(S) = k.$$

$\square$

Theorem 6.2.14. *If k is algebraically closed and A is a finite-dimensional semisimple k-algebra, then*

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

Corollary 6.2.15. *Let A be a finite-dimensional semisimple k-algebra, where k is a field.*
In any decomposition

$${}_AA \cong S_1^{n_1} \oplus \cdots \oplus S_r^{n_r},$$

where the S_i's are pairwise non-isomorphic simple A-modules, we have that $S_1, \ldots, S_r$ form a complete set of representatives of the isomorphism classes of simple A-modules.
If moreover $k = \overline{k}$, then $n_i = \dim_k S_i$ and

$$\dim_k A = n_1^2 + \cdots + n_r^2.$$

Proof. Let S be a simple module. Then

$$S \cong A/M$$

where M is a maximal submodule.

By semisimplicity,

$${}_AA = M \oplus S.$$

Hence all simple submodules appear in $\{S_1, \ldots, S_r\}$.

Since $k = \overline{k}$, we have

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

Moreover,

$$M_{n_i}(k) = \underbrace{\left\{ \begin{pmatrix} * & \cdots & 0 \\ \vdots & \ddots & \vdots \\ * & \cdots & 0 \end{pmatrix} \right\} \oplus \cdots \oplus \left\{ \begin{pmatrix} 0 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & * \end{pmatrix} \right\}}_{\text{each of them is a simple } M_{n_i}(k)\text{-module.}}$$

$\square$

6.3 k-algebras, indecomposables, and idempotents

Definition 6.3.1. *A nonzero R-module M is called indecomposable if every decomposition*

$$M = M_1 \oplus M_2$$

forces $M_1 = 0$ or $M_2 = 0$.

Theorem 6.3.2 (Krull-Schmidt)**.** *Let M be an R-module with a composition series. Suppose*

$$M \cong V_1 \oplus \cdots \oplus V_m \cong U_1 \oplus \cdots \oplus U_n$$

where all V_i and U_j are indecomposable. Then

$$m = n,$$

and after reordering the summands one has

$$V_i \cong U_i \quad \textit{for all } i.$$

Proof. Replacing the given isomorphisms by identifications, we may assume

$$M = V_1 \oplus \cdots \oplus V_m = U_1 \oplus \cdots \oplus U_n.$$

If $M = 0$, the statement is trivial. So assume $M \neq 0$.

Since M has a composition series, it has finite length. Hence each direct summand V_i and U_j also has finite length. Moreover, for every indecomposable module X of finite length, Fitting's lemma implies that every endomorphism of X is either invertible or nilpotent; therefore $\mathrm{End}_R(X)$ is a local ring.

We prove the theorem by induction on the length $\ell(M)$ of M. If $\ell(M) = 1$, then M is simple, hence indecomposable, so necessarily $m = n = 1$ and $V_1 \cong U_1$.

Now assume $\ell(M) > 1$ and that the statement is known for all modules of smaller length. Set

$$A := V_1, \qquad B := V_2 \oplus \cdots \oplus V_m,$$

so that

$$M = A \oplus B.$$

Let

$$\iota : A \to M, \qquad \pi : M \to A$$

be the canonical inclusion and projection. For the decomposition

$$M = U_1 \oplus \cdots \oplus U_n,$$

let

$$\kappa_j : U_j \to M, \qquad p_j : M \to U_j$$

be the canonical inclusion and projection for $1 \le j \le n$.

Since

$$1_M = \sum_{j=1}^{n} \kappa_j p_j,$$

we get

$$1_A = \pi\iota = \pi\Big(\sum_{j=1}^{n} \kappa_j p_j\Big)\iota = \sum_{j=1}^{n} (\pi\kappa_j)(p_j\iota).$$

For each j, put

$$u_j := (\pi\kappa_j)(p_j\iota) \in \mathrm{End}_R(A).$$

If all u_j were noninvertible, then, since $\mathrm{End}_R(A)$ is local, their sum would also be noninvertible. But their sum is 1_A, a contradiction. Hence u_j is a unit for some j. After reordering the U_j, we may assume that u_1 is a unit.

Set

$$a := \pi\kappa_1 : U_1 \to A, \qquad b := p_1\iota : A \to U_1.$$

Then

$$ab = u_1$$

is an automorphism of A. Consider

$$e := b(ab)^{-1}a \in \mathrm{End}_R(U_1).$$

Then

$$e^2 = b(ab)^{-1}a\, b(ab)^{-1}a = b(ab)^{-1}(ab)(ab)^{-1}a = b(ab)^{-1}a = e,$$

so e is an idempotent. Also,

$$eb = b(ab)^{-1}ab = b \neq 0,$$

hence $e \neq 0$. Since $\mathrm{End}_R(U_1)$ is local, its only idempotents are 0 and 1, therefore $e = 1_{U_1}$.

Thus

$$b(ab)^{-1}a = 1_{U_1},$$

so a is injective. On the other hand, ab is surjective, hence a is surjective as well. Therefore a is an isomorphism, and we obtain

$$A \cong U_1.$$

Now define

$$f := \kappa_1 a^{-1} : A \to M.$$

Then

$$\pi f = \pi\kappa_1 a^{-1} = aa^{-1} = 1_A.$$

Hence

$$M = f(A) \oplus B.$$

Indeed, for any $x \in M$,

$$x = f(\pi(x)) + \big(x - f(\pi(x))\big),$$

and

$$\pi\big(x - f(\pi(x))\big) = \pi(x) - \pi f(\pi(x)) = 0,$$

so $x - f(\pi(x)) \in \ker\pi = B$. Also, if $f(a) \in B = \ker\pi$, then

$$a = \pi f(a) = 0,$$

so the sum is direct. Since $f(A) = \kappa_1(U_1) = U_1$, we have

$$M = U_1 \oplus B.$$

Therefore

$$B \cong M/U_1 \cong U_2 \oplus \cdots \oplus U_n.$$

But also

$$B = V_2 \oplus \cdots \oplus V_m.$$

Now $\ell(B) < \ell(M)$, so by the induction hypothesis,

$$m - 1 = n - 1,$$

and after reordering $U_2, \dots, U_n$, we have

$$V_i \cong U_i \qquad (i = 2, \dots, m).$$

Together with $V_1 \cong U_1$, this gives

$$m = n, \qquad V_i \cong U_i \quad \text{for all } i$$

after a suitable reordering.

This completes the proof. □

Definition 6.3.3. *An idempotent of a ring R is a nonzero element*

$$e \in R$$

such that

$$e^2 = e.$$

Remark 6.3.4. *If e is an idempotent, then the corner ring*

$$eRe$$

is a ring with identity element e.

Definition 6.3.5. *Two idempotents* $e_1, e_2 \in R$ *are called orthogonal if*

$$e_1 e_2 = e_2 e_1 = 0.$$

More generally, finitely many idempotents are orthogonal if they are pairwise orthogonal.

Lemma 6.3.6. *If* $e_1, \ldots, e_n$ *are orthogonal idempotents, then*

$$e_1 + \cdots + e_n$$

is again an idempotent.

Proof. Let

$$e := e_1 + \cdots + e_n.$$

Since each e_i is an idempotent, we have $e_i^2 = e_i$ for all i. Since the *idempotents* are orthogonal, for $i \neq j$ we have

$$e_i e_j = 0.$$

Therefore

$$e^2 = (e_1 + \cdots + e_n)^2 = \sum_{i=1}^{n} e_i^2 + \sum_{i \neq j} e_i e_j = \sum_{i=1}^{n} e_i + \sum_{i \neq j} 0 = e_1 + \cdots + e_n = e.$$

Hence e is an idempotent. □

Definition 6.3.7. *An idempotent decomposition of an idempotent* e *is an expression*

$$e = e_1 + \cdots + e_n$$

where the e_i *are orthogonal idempotents.*

Remark 6.3.8. *If* $e = e_1 + e_2$ *is an idempotent decomposition, then* e_1 *and* e_2 *are in* eRe*.*

Definition 6.3.9. *An idempotent is called primitive if it cannot be written as a sum of two nonzero orthogonal idempotents.*

Definition 6.3.10. *A primitive orthogonal decomposition is an idempotent decomposition in which all summands are primitive.*

Example 6.3.11. *If* e *is an idempotent in a unital ring, then*

$$1 = e + (1 - e)$$

is an idempotent decomposition of 1.

Example 6.3.12. *Let* e *be an idempotent,* $Re = \{ae | a \in R\}$ *is a left module of* R.

Theorem 6.3.13. *Let* e *be an idempotent of* R.
If

$$e = e_1 + \cdots + e_n$$

is an idempotent decomposition, then

$$Re = Re_1 \oplus \cdots \oplus Re_n.$$

Conversely, if Re *is a direct sum of left ideals*

$$Re = I_1 \oplus \cdots \oplus I_n,$$

then there exists an idempotent decomposition

$$e = e_1 + \cdots + e_n$$

with $e_i \in I_i$ *and* $I_i = Re_i$.
In this way, the idempotent decompositions of e *are in one-to-one correspondence with the direct sum decompositions of the left* R*-module* Re.

Proof. Assume first that

$$e = e_1 + \cdots + e_n$$

is an idempotent decomposition. Since

$$e_i e = e e_i = e_i,$$

we have

$$Re_i \subseteq Re.$$

Indeed, if $x \in Re_i$, then $xe = x$.

Now let $a \in R$. Then

$$ae = ae_1 + \cdots + ae_n \in Re_1 + \cdots + Re_n,$$

so

$$Re \subseteq Re_1 + \cdots + Re_n.$$

The reverse inclusion is clear, hence

$$Re = Re_1 + \cdots + Re_n.$$

To show that the sum is direct, suppose

$$a_1e_1 + \cdots + a_ne_n = 0.$$

Multiplying on the right by e_i, we get

$$a_ie_i = 0$$

for each i, since $e_je_i = 0$ for $j \neq i$. Therefore the expression is unique, and

$$Re = Re_1 \oplus \cdots \oplus Re_n.$$

Conversely, suppose

$$Re = I_1 \oplus \cdots \oplus I_n.$$

Since $e \in Re$, we may write

$$e = e_1 + \cdots + e_n, \qquad e_i \in I_i.$$

For any $a_i \in I_i \subseteq Re$, we have

$$a_i = a_ie = a_ie_1 + \cdots + a_ie_n.$$

Since $e_j \in I_j$, we have

$$Re_j \subseteq I_j,$$

hence

$$a_ie_j \in I_j.$$

But $a_i \in I_i$, and the decomposition

$$Re = I_1 \oplus \cdots \oplus I_n$$

is direct. Therefore

$$a_i = a_ie_i, \qquad a_ie_j = 0 \quad (j \neq i).$$

It follows that

$$I_i \subseteq Re_i.$$

On the other hand, since $e_i \in I_i$ and I_i is a left ideal, we have

$$Re_i \subseteq I_i.$$

Thus

$$I_i = Re_i.$$

Now take $a_i = e_i$. Then

$$e_i = e_i^2, \qquad e_ie_j = 0 \quad (j \neq i).$$

Hence
$$e = e_1 + \cdots + e_n$$
is an idempotent decomposition. $\square$

Corollary 6.3.14. *Let e be an idempotent of R. Then the following are equivalent:*

(1) e is primitive.

(2) Re is indecomposable as a left R-module.

(3) The only idempotents in eRe are 0 and e.

Proof. **(1) $\iff$ (2).** This follows immediately from the previous theorem, which gives a bijection between idempotent decompositions of e and direct sum decompositions of the left ideal Re.

(1) $\implies$ (3). Suppose that $e' \in eRe$ is an idempotent with
$$e' \neq 0, \qquad e' \neq e.$$
Since $e' \in eRe$, we have
$$ee' = e' = e'e.$$
Hence
$$e = e' + (e - e').$$
We claim that this is an idempotent decomposition of e.

First,
$$(e-e')^2 = e^2 - ee' - e'e + (e')^2 = e - e' - e' + e' = e - e'.$$
So $e - e'$ is an idempotent.

Next,
$$e'(e-e') = e'e - (e')^2 = e' - e' = 0,$$
and similarly
$$(e-e')e' = ee' - (e')^2 = e' - e' = 0.$$
Thus e' and $e - e'$ are orthogonal idempotents. Since both are nonzero, this contradicts the primitiveness of e.

Therefore no such e' exists, and the only idempotents in eRe are 0 and e.

(3) $\implies$ (1). Suppose that e is not primitive. Then there exist nonzero orthogonal idempotents e_1, e_2 such that
$$e = e_1 + e_2.$$
Since
$$ee_1e = e_1, \qquad ee_2e = e_2,$$

we have

$$e_1, e_2 \in eRe.$$

But e_1 is a nonzero idempotent different from e, contradicting (3). Hence e must be primitive. $\square$

Chapter 7

Homework

7.1 Homework Week 1

Exercise 7.1.1. *Let R be a ring, and let M be an R-module. Suppose that S is a ring and*

$$\theta : S \to R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S-module.

Proof. This action is well-defined because $\forall s \in S$,$\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_S x = \theta(1_S)x = 1_R x = x$

Hence M is an S-module. $\square$

Exercise 7.1.2. *Let R be a ring, and let M and N be two right R-modules. Let $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ be the set of group homomorphisms from M to N(as abelian groups). The action of R on $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is given as follows: for any $a \in R$ and $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, define*

$$(\varphi a)(x) = \varphi(xa), \qquad \forall x \in M.$$

Prove that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module.

Proof. This action is well defined: for every $r \in R$ and every $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, the map

$$\varphi r \colon M \to N, \qquad (\varphi r)(x) = \varphi(xr),$$

is again a group homomorphism.

We now verify the module axioms. Let $r_1, r_2 \in R$, let $\varphi, \psi \in \operatorname{Hom}_{\mathbb{Z}}(M, N)$, and let $x \in M$. Then

$$\begin{aligned}
((\varphi+\psi)r_1)(x) &= (\varphi+\psi)(xr_1) \\
&= \varphi(xr_1) + \psi(xr_1) \\
&= (\varphi r_1)(x) + (\psi r_1)(x), \\
(\varphi(r_1+r_2))(x) &= \varphi(x(r_1+r_2)) \\
&= \varphi(xr_1 + xr_2) \\
&= \varphi(xr_1) + \varphi(xr_2) \\
&= (\varphi r_1)(x) + (\varphi r_2)(x), \\
((\varphi r_1)r_2)(x) &= (\varphi r_1)(xr_2) \\
&= \varphi((xr_2)r_1) \\
&= \varphi(x(r_2 r_1)) \\
&= (\varphi(r_2 r_1))(x), \\
(\varphi 1_R)(x) &= \varphi(x 1_R) \\
&= \varphi(x).
\end{aligned}$$

Hence $\operatorname{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module. □

Exercise 7.1.3. *Let R be a ring, and let M be an R-module. Denote*

$$\operatorname{Ann}_R M = \{r \in R \mid rm = 0,\ \forall m \in M\}.$$

We call M a faithful R-module if $\operatorname{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\operatorname{Ann}_R M$ on M by

$$\overline{r}\,x = rx,$$

where $x \in M$,$r \in R$, and $\overline{r} = r + \operatorname{Ann}_R M$. Prove that M is a faithful $R/\operatorname{Ann}_R M$-module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\operatorname{Ann}_R M.$$

Then $r - r' \in \operatorname{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}\,x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R-module structure on M.

Now we prove faithfulness. Suppose $\bar{r} \in R/\operatorname{Ann}_R M$ acts trivially on M, that is,

$$\bar{r}\,x = 0 \qquad \forall x \in M.$$

By definition, this means

$$rx = 0 \qquad \forall x \in M.$$

Therefore $r \in \operatorname{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence

$$\operatorname{Ann}_{R/\operatorname{Ann}_R M}(M) = \{\bar{0}\},$$

and M is faithful as an $R/\operatorname{Ann}_R M$-module. $\square$

Exercise 7.1.4. *Let $\varphi : M \to M$ be a homomorphism of R-modules such that*

$$\varphi\varphi = \varphi.$$

Prove that

$$M = \ker\varphi \oplus \operatorname{Im}\varphi.$$

Proof. Let $x \in M$. Then

$$x = \varphi(x) + (x - \varphi(x)),$$

where $\varphi(x) \in \operatorname{Im}\varphi$ and $x - \varphi(x) \in \ker\varphi$. Hence

$$M = \operatorname{Im}\varphi + \ker\varphi.$$

It remains to show that

$$\ker\varphi \cap \operatorname{Im}\varphi = \{0\}.$$

Suppose $x \in \ker\varphi \cap \operatorname{Im}\varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Thus

$$0 = \varphi(x) = \varphi(\varphi(y)) = \varphi(y) = x,$$

so $x = 0$. Hence $\ker\varphi \cap \operatorname{Im}\varphi = \{0\}$. $\square$

Exercise 7.1.5. *Show that any R-module is a homomorphism image of some free R-module.*

Proof. Let M be a left R-module. Consider the free R-module

$$F = \bigoplus_{x \in M} Re_x$$

with basis $\{e_x\}_{x\in M}$ indexed by the underlying set of M. Define an R-module homomorphism

$$\pi : F \to M$$

by

$$\pi(e_x) = x \qquad (x \in M),$$

and extend R-linearly. Thus

$$\pi\Big(\sum_{i=1}^{n} r_i e_{x_i}\Big) = \sum_{i=1}^{n} r_i x_i.$$

This is well-defined and R-linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F. Equivalently,

$$M \cong F/\ker \pi.$$

$\square$

Exercise 7.1.6. *Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$-module?*

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$-module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n\frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$-module. $\square$

7.2 Homework Week 2

Exercise 7.2.1. *Let R be a commutative ring, and suppose that φ is an endomorphism of the free R-module R^m. Prove that if φ is surjective, then φ is injective. Consider that: if φ is injective, is φ surjective?*

Proof. Let $\{e_1, \ldots, e_m\}$ be the standard basis of R^m, and let $A \in M_m(R)$ be the matrix of φ with respect to this basis. Since φ is surjective, for each j there exists $v_j \in R^m$ with $\varphi(v_j) = e_j$. Write $v_j = \sum_{i=1}^m b_{ij} e_i$ and let $B = (b_{ij}) \in M_m(R)$. Then $AB = I_m$, so

$$\det(A)\det(B) = \det(AB) = 1.$$

Hence $\det(A)$ is a unit in R. It follows that A is invertible (with $A^{-1} = \det(A)^{-1}\operatorname{adj}(A)$), so φ is an isomorphism. In particular φ is injective.

Counterexample for the converse. The converse is false in general. Let $R = \mathbb{Z}$ and $m = 1$, so $R^m = \mathbb{Z}$. Define $\varphi \colon \mathbb{Z} \to \mathbb{Z}$ by $\varphi(n) = 2n$. Then φ is injective but not surjective; for example, $1 \notin \operatorname{Im}\varphi$. $\square$

Exercise 7.2.2. *Let $\varphi : M \to N$ be a homomorphism of R-modules. Prove that:*

1. *φ is a monomorphism if and only if for any R-module T and any two R-module homomorphisms $\xi, \eta : T \to M$, $\varphi\xi = \varphi\eta$ implies $\xi = \eta$;*

2. *φ is an epimorphism if and only if for any R-module S and any two R-module homomorphisms $\rho, \theta : N \to S$,$\rho\varphi = \theta\varphi$ implies $\rho = \theta$.*

Proof. 1. ($\Rightarrow$) Suppose φ is injective and $\varphi\xi = \varphi\eta$ for $\xi, \eta : T \to M$. Then for every $t \in T$,

$$\varphi(\xi(t)) = \varphi(\eta(t)),$$

so $\xi(t) - \eta(t) \in \ker \varphi = \{0\}$. Hence $\xi(t) = \eta(t)$ for all $t \in T$, and therefore $\xi = \eta$.

($\Leftarrow$) We show $\ker \varphi = \{0\}$. Let $T = \ker \varphi$, let $\xi : T \hookrightarrow M$ be the inclusion, and let $\eta = 0 : T \to M$ be the zero map. Then $\varphi\xi(t) = \varphi(t) = 0 = \varphi\eta(t)$ for all $t \in T$, so $\varphi\xi = \varphi\eta$. By hypothesis $\xi = \eta$, which means $t = \xi(t) = \eta(t) = 0$ for all $t \in T$. Hence $\ker \varphi = \{0\}$, so φ is injective.

2. ($\Rightarrow$) Suppose φ is surjective and $\rho\varphi = \theta\varphi$. For any $n \in N$, pick $m \in M$ with $\varphi(m) = n$. Then

$$\rho(n) = \rho(\varphi(m)) = \theta(\varphi(m)) = \theta(n),$$

so $\rho = \theta$.

($\Leftarrow$) We show φ is surjective. Let $S = N/\operatorname{Im}\varphi$, let $\rho : N \to S$ be the canonical projection, and let $\theta = 0 : N \to S$ be the zero map. For any $m \in M$, we have

$$\rho(\varphi(m)) = \overline{\varphi(m)} = \bar{0} = \theta(\varphi(m)),$$

so $\rho\varphi = \theta\varphi$. By hypothesis $\rho = \theta$, i.e. the projection $N \to N/\operatorname{Im}\varphi$ is the zero map. This means $N = \operatorname{Im}\varphi$, so φ is surjective.

$\square$

Exercise 7.2.3. *Let M and M_i $(1 \le i \le n)$ be R-modules. Show that $M \simeq \bigoplus_{i=1}^n M_i$ if and only if for every $i \in \{1, \dots, n\}$, there exists an $\varphi_i \in \operatorname{End}_R(M)$ such that $\operatorname{Im}\varphi_i \simeq M_i$,$\varphi_i\varphi_j = 0$ for $i \neq j$,$\varphi_i^2 = \varphi_i$, and $\varphi_1 + \varphi_2 + \cdots + \varphi_n = \mathrm{id}_M$.*

Proof. ($\to$) Suppose $f : M \xrightarrow{\sim} \bigoplus_{i=1}^n M_i$. Let $\iota_i : M_i \hookrightarrow \bigoplus_{j=1}^n M_j$ be the canonical inclusion and $\pi_i : \bigoplus_{j=1}^n M_j \twoheadrightarrow M_i$ the canonical projection. Define

$$\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f \in \operatorname{End}_R(M).$$

Then:

- $\varphi_i^2 = f^{-1}\iota_i\pi_i f f^{-1}\iota_i\pi_i f = f^{-1}\iota_i(\pi_i\iota_i)\pi_i f = f^{-1}\iota_i\pi_i f = \varphi_i$, since $\pi_i\iota_i = \mathrm{id}_{M_i}$.
- For $i \neq j$:$\varphi_i\varphi_j = f^{-1}\iota_i(\pi_i\iota_j)\pi_j f = 0$, since $\pi_i\iota_j = 0$ when $i \neq j$.
- $\sum_{i=1}^n \varphi_i = f^{-1}\left(\sum_{i=1}^n \iota_i\pi_i\right) f = f^{-1} \circ \mathrm{id} \circ f = \mathrm{id}_M$, since $\sum_i \iota_i\pi_i = \mathrm{id}_{\bigoplus M_j}$.
- $\operatorname{Im}\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f(M) \simeq M_i$.

($\Leftarrow$) Suppose such φ_i exist. Define $g : M \to \bigoplus_{i=1}^n \operatorname{Im}\varphi_i$ by $g(x) = (\varphi_1(x), \dots, \varphi_n(x))$.

Surjective: for any $(y_1, \ldots, y_n) \in \bigoplus \operatorname{Im} \varphi_i$, let $x = y_1 + \cdots + y_n \in M$. Then

$$\varphi_j(x) = \sum_{i=1}^{n} \varphi_j(y_i) = \sum_{i=1}^{n} \varphi_j(\varphi_i(x_i)) = \varphi_j(\varphi_j(x_j)) = \varphi_j^2(x_j) = \varphi_j(x_j) = y_j,$$

where we used $\varphi_j \varphi_i = 0$ for $j \neq i$ and $\varphi_j^2 = \varphi_j$. Hence $g(x) = (y_1, \ldots, y_n)$.

Injective: if $g(x) = 0$, then $\varphi_i(x) = 0$ for all i, so $x = \mathrm{id}_M(x) = \sum_i \varphi_i(x) = 0$.

Since $\operatorname{Im} \varphi_i \simeq M_i$, we conclude $M \simeq \bigoplus_{i=1}^n M_i$. □

Exercise 7.2.4.

1. *Prove $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m,n)\mathbb{Z}$, where $m, n \in \mathbb{Z}$.*
2. *Let A be an abelian group, prove $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.*
3. *Let A, B be two finitely generated abelian groups. What is $A \otimes_{\mathbb{Z}} B$?*

Proof.

1. We use the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$. Tensoring with $\mathbb{Z}/m\mathbb{Z}$ over $\mathbb{Z}$ gives the right-exact sequence

$$\mathbb{Z}/m\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

   Hence $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z})$. Now $n(\mathbb{Z}/m\mathbb{Z}) = (n\mathbb{Z} + m\mathbb{Z})/m\mathbb{Z}$. Since $n\mathbb{Z} + m\mathbb{Z} = (m,n)\mathbb{Z}$, we get

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \frac{\mathbb{Z}/m\mathbb{Z}}{(m,n)\mathbb{Z}/m\mathbb{Z}} \simeq \mathbb{Z}/(m,n)\mathbb{Z}.$$

2. Similarly, tensor the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$ with A:

$$A \xrightarrow{\cdot n} A \to A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

   By right-exactness,$A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.

3. By the fundamental theorem of finitely generated abelian groups,

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^{s} \mathbb{Z}/d_i\mathbb{Z}, \qquad B \simeq \mathbb{Z}^t \oplus \bigoplus_{j=1}^{u} \mathbb{Z}/e_j\mathbb{Z}.$$

   Since tensor product distributes over direct sums, using (1), (2), and $\mathbb{Z} \otimes_{\mathbb{Z}} M \simeq M$, we get

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rt} \oplus \bigoplus_{j=1}^{u} (\mathbb{Z}/e_j\mathbb{Z})^r \oplus \bigoplus_{i=1}^{s} (\mathbb{Z}/d_i\mathbb{Z})^t \oplus \bigoplus_{i=1}^{s} \bigoplus_{j=1}^{u} \mathbb{Z}/(d_i, e_j)\mathbb{Z}.$$

□

Exercise 7.2.5. *Let $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ be the unique monomorphism. Prove that*

$$\mathrm{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

Proof. The unique monomorphism $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ is given by $\varphi(\bar{1}) = \bar{2}$. By the previous exercise,$\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$.

It suffices to check on the generator $\bar{1} \otimes \bar{1}$ of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$:

$$(\mathrm{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \bar{1} \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = \bar{1} \otimes 2 \cdot \bar{1} = 2 \cdot (\bar{1} \otimes \bar{1}) = (2 \cdot \bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0,$$

where we used $2 \cdot \bar{1} = \bar{0}$ in $\mathbb{Z}/2\mathbb{Z}$. Since the generator maps to 0, the map $\mathrm{id} \otimes \varphi$ is the zero homomorphism.

This also shows that tensoring a monomorphism need not yield a monomorphism, i.e. the functor $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} -$ is not left-exact. □

Exercise 7.2.6. *Let V_1 and V_2 be vector spaces over a field F, with $\dim_F V_1 = n$ and $\dim_F V_2 = m$. Let $\{\epsilon_1, \dots, \epsilon_n\}$ and $\{\eta_1, \dots, \eta_m\}$ be bases of V_1 and V_2, respectively. Suppose that $\mathscr{A} : V_1 \to V_1$ and $\mathscr{B} : V_2 \to V_2$ are F-linear maps, represented by the matrices $A = (a_{ij})_{n \times n}$ and $B = (b_{ij})_{m \times m}$ with respect to these bases. Prove that*

$$\{\epsilon_i \otimes \eta_j \mid 1 \le i \le n,\ 1 \le j \le m\}$$

is an F-basis of $V_1 \otimes_F V_2$. Determine the matrix representing the F-linear map $\mathscr{A} \otimes \mathscr{B}$ under this basis.

Proof. **Step 1.** The set $\{\epsilon_i \otimes \eta_j\}$ is a basis.

Spanning. Every element of $V_1 \otimes_F V_2$ is a finite sum of simple tensors $v \otimes w$. Writing $v = \sum_i a_i \epsilon_i$ and $w = \sum_j b_j \eta_j$, by bilinearity of the tensor product we get

$$v \otimes w = \sum_{i,j} a_i b_j\, (\epsilon_i \otimes \eta_j).$$

Hence $\{\epsilon_i \otimes \eta_j\}$ spans $V_1 \otimes_F V_2$.

Linear independence. Suppose $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$. For each pair (p, q), let $f_p \in V_1^*$ and $g_q \in V_2^*$ be the dual basis functionals:$f_p(\epsilon_i) = \delta_{pi}$ and $g_q(\eta_j) = \delta_{qj}$. The bilinear map $(v, w) \mapsto f_p(v) g_q(w)$ from $V_1 \times V_2$ to F induces (by the universal property of the tensor product) an F-linear map $\Phi_{pq} : V_1 \otimes_F V_2 \to F$ with $\Phi_{pq}(\epsilon_i \otimes \eta_j) = \delta_{pi}\delta_{qj}$. Applying Φ_{pq} to the relation $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$ yields $c_{pq} = 0$. Since (p, q) was arbitrary, all $c_{ij} = 0$.

Therefore $\{\epsilon_i \otimes \eta_j : 1 \le i \le n,\ 1 \le j \le m\}$ is an F-basis of $V_1 \otimes_F V_2$, and $\dim_F(V_1 \otimes_F V_2) = nm$.

Step 2: matrix of $\mathscr{A} \otimes \mathscr{B}$.

Order the basis lexicographically:$\epsilon_1 \otimes \eta_1,\ \epsilon_1 \otimes \eta_2,\ \dots,\ \epsilon_1 \otimes \eta_m,\ \epsilon_2 \otimes \eta_1,\ \dots,\ \epsilon_n \otimes \eta_m$. Since

$\mathscr{A}(\epsilon_i) = \sum_{k=1}^{n} a_{ki}\epsilon_k$ and $\mathscr{B}(\eta_j) = \sum_{l=1}^{m} b_{lj}\eta_l$, we have

$$(\mathscr{A} \otimes \mathscr{B})(\epsilon_i \otimes \eta_j) = \mathscr{A}(\epsilon_i) \otimes \mathscr{B}(\eta_j) = \sum_{k=1}^{n}\sum_{l=1}^{m} a_{ki}b_{lj}\,(\epsilon_k \otimes \eta_l).$$

The entry in row (k, l) and column (i, j) of the representing matrix is $a_{ki}b_{lj}$. Written as an $nm \times nm$ block matrix, this is the **Kronecker product**

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}.$$

□

7.3 Homework Week 3

Exercise 7.3.1. *Let M, N, $(M_i)_{i\in I}$, and $(N_j)_{j\in J}$ be R-modules. Prove the following isomorphisms of abelian groups:*

1.

$$\operatorname{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \cong \prod_{i\in I} \operatorname{Hom}_R(M_i, N).$$

2.

$$\operatorname{Hom}_R\left(M,\, \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \operatorname{Hom}_R(M, N_j).$$

Proof. We prove the two statements separately.

(1) Define

$$\Phi : \operatorname{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \longrightarrow \prod_{i\in I} \operatorname{Hom}_R(M_i, N)$$

by

$$\Phi(f) = (f \circ \iota_i)_{i\in I},$$

where

$$\iota_i : M_i \to \bigoplus_{i\in I} M_i$$

is the canonical inclusion.

We show that Φ is an isomorphism.

First,Φ is a group homomorphism, since composition is compatible with addition:

$$\Phi(f + g) = \big((f + g) \circ \iota_i\big)_{i\in I} = \big(f \circ \iota_i + g \circ \iota_i\big)_{i\in I} = \Phi(f) + \Phi(g).$$

Now let $(f_i)_{i\in I} \in \prod_{i\in I} \operatorname{Hom}_R(M_i, N)$. Define

$$f : \bigoplus_{i\in I} M_i \to N$$

by

$$f\big((m_i)_{i\in I}\big) = \sum_{i\in I} f_i(m_i).$$

This is well-defined, because an element of $\bigoplus_{i\in I} M_i$ has only finitely many nonzero components, so the above sum is finite. It is immediate that f is R-linear, and

$$f \circ \iota_i = f_i \qquad \text{for all } i \in I.$$

Hence $\Phi(f) = (f_i)_{i\in I}$, so Φ is surjective.

To prove injectivity, suppose $\Phi(f) = \Phi(g)$. Then

$$f \circ \iota_i = g \circ \iota_i \qquad \text{for all } i \in I.$$

Take any $x = (m_i)_{i\in I} \in \bigoplus_{i\in I} M_i$. Since x has finite support, we may write

$$x = \sum_{i\in I} \iota_i(m_i),$$

with only finitely many nonzero terms. Therefore,

$$f(x) = \sum_{i\in I} f(\iota_i(m_i)) = \sum_{i\in I} g(\iota_i(m_i)) = g(x).$$

So $f = g$, and Φ is injective.

Thus Φ is an isomorphism:

$$\operatorname{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \cong \prod_{i\in I} \operatorname{Hom}_R(M_i, N).$$

(2) Let

$$\pi_j : \prod_{j\in J} N_j \to N_j$$

be the canonical projection. Define

$$\Psi : \operatorname{Hom}_R\left(M, \prod_{j\in J} N_j\right) \longrightarrow \prod_{j\in J} \operatorname{Hom}_R(M, N_j)$$

by

$$\Psi(f) = (\pi_j \circ f)_{j\in J}.$$

Again,Ψ is a group homomorphism.

We show that Ψ is an isomorphism. Given a family

$$(g_j)_{j\in J} \in \prod_{j\in J} \mathrm{Hom}_R(M, N_j),$$

define

$$g : M \to \prod_{j\in J} N_j$$

by

$$g(m) = (g_j(m))_{j\in J}.$$

Since each g_j is R-linear,g is also R-linear. Moreover,

$$\pi_j \circ g = g_j \qquad \text{for all } j \in J,$$

so

$$\Psi(g) = (g_j)_{j\in J}.$$

Hence Ψ is surjective.

For injectivity, suppose $\Psi(f) = \Psi(g)$. Then

$$\pi_j \circ f = \pi_j \circ g \qquad \text{for all } j \in J.$$

Thus for every $m \in M$ and every $j \in J$,

$$\pi_j(f(m)) = \pi_j(g(m)).$$

Since two elements of $\prod_{j\in J} N_j$ are equal if and only if all their components are equal, it follows that

$$f(m) = g(m) \qquad \text{for all } m \in M.$$

Hence $f = g$, and Ψ is injective.

Therefore

$$\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \mathrm{Hom}_R(M, N_j).$$

This completes the proof. □

Exercise 7.3.2 (Five Lemma)**.** *Suppose that the following diagram of homomorphisms of R-modules*

$$\begin{tikzcd}
M_1 \arrow[r] \arrow[d, "\varphi_1"'] & M_2 \arrow[r] \arrow[d, "\varphi_2"'] & M_3 \arrow[r] \arrow[d, "\varphi_3"'] & M_4 \arrow[r] \arrow[d, "\varphi_4"'] & M_5 \arrow[d, "\varphi_5"'] \\
N_1 \arrow[r] & N_2 \arrow[r] & N_3 \arrow[r] & N_4 \arrow[r] & N_5
\end{tikzcd}$$

is commutative and has exact rows. Prove that:

1. *If φ_1 is surjective, and φ_2 and φ_4 are injective, then φ_3 is injective.*

2. *If φ_5 is injective, and φ_2 and φ_4 are surjective, then φ_3 is surjective.*

Proof. We prove the two assertions separately.

(1) Assume that φ_1 is surjective and that φ_2, φ_4 are injective. We show that φ_3 is injective.

Let $x \in M_3$ and suppose that
$$\varphi_3(x) = 0.$$
We must prove that $x = 0$.

Let $\alpha_3 : M_3 \to M_4$ and $\beta_3 : N_3 \to N_4$ denote the third horizontal maps. By commutativity,
$$\varphi_4(\alpha_3(x)) = \beta_3(\varphi_3(x)) = \beta_3(0) = 0.$$
Since φ_4 is injective, it follows that $\alpha_3(x) = 0$. Exactness of the top row at M_3 yields
$$x = \alpha_2(y)$$
for some $y \in M_2$, where $\alpha_2 : M_2 \to M_3$ is the second horizontal map.

Again by commutativity,
$$0 = \varphi_3(x) = \varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)),$$
where $\beta_2 : N_2 \to N_3$ is the second horizontal map. Hence $\varphi_2(y) \in \ker(\beta_2)$. By exactness of the bottom row at N_2,
$$\ker(\beta_2) = \mathrm{Im}(\beta_1),$$
so there exists $z \in N_1$ such that
$$\beta_1(z) = \varphi_2(y),$$
where $\beta_1 : N_1 \to N_2$ is the first horizontal map.

Since φ_1 is surjective, there exists $w \in M_1$ such that
$$\varphi_1(w) = z.$$
Using commutativity once more,
$$\varphi_2(\alpha_1(w)) = \beta_1(\varphi_1(w)) = \beta_1(z) = \varphi_2(y),$$
where $\alpha_1 : M_1 \to M_2$ is the first horizontal map. Because φ_2 is injective, we get
$$\alpha_1(w) = y.$$
Therefore
$$x = \alpha_2(y) = \alpha_2(\alpha_1(w)) = 0,$$
since the composition of two consecutive maps in an exact sequence is zero. Thus $\ker(\varphi_3) = 0$, so φ_3 is injective.

(2) Assume that φ_5 is injective and that φ_2, φ_4 are surjective. We show that φ_3 is surjective.

Let $n \in N_3$. We must find $m \in M_3$ such that

$$\varphi_3(m) = n.$$

Let $\beta_3 : N_3 \to N_4$ and $\alpha_3 : M_3 \to M_4$ denote the third horizontal maps. Since φ_4 is surjective, there exists $u \in M_4$ such that

$$\varphi_4(u) = \beta_3(n).$$

Let $\alpha_4 : M_4 \to M_5$ and $\beta_4 : N_4 \to N_5$ be the next horizontal maps. Then

$$\varphi_5(\alpha_4(u)) = \beta_4(\varphi_4(u)) = \beta_4(\beta_3(n)) = 0.$$

Because φ_5 is injective, it follows that $\alpha_4(u) = 0$. Exactness of the top row at M_4 shows that there exists $m \in M_3$ such that

$$\alpha_3(m) = u.$$

Consider

$$n - \varphi_3(m) \in N_3.$$

By commutativity,

$$\beta_3\big(n - \varphi_3(m)\big) = \beta_3(n) - \beta_3(\varphi_3(m)) = \beta_3(n) - \varphi_4(\alpha_3(m)) = \beta_3(n) - \varphi_4(u) = 0.$$

Hence $n - \varphi_3(m) \in \ker(\beta_3)$. By exactness of the bottom row at N_3,

$$\ker(\beta_3) = \mathrm{Im}(\beta_2),$$

so there exists $v \in N_2$ such that

$$\beta_2(v) = n - \varphi_3(m),$$

where $\beta_2 : N_2 \to N_3$ is the second horizontal map.

Since φ_2 is surjective, there exists $y \in M_2$ such that

$$\varphi_2(y) = v.$$

Then

$$\varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)) = \beta_2(v) = n - \varphi_3(m),$$

where $\alpha_2 : M_2 \to M_3$ is the second horizontal map. Therefore

$$n = \varphi_3(m) + \varphi_3(\alpha_2(y)) = \varphi_3\big(m + \alpha_2(y)\big).$$

Thus n lies in the image of φ_3, proving that φ_3 is surjective. □

Exercise 7.3.3 (Snake Lemma)**.** *Suppose that the following diagram of homomorphisms of*

R-modules

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\varphi'"'] & M \arrow[r, "\beta"] \arrow[d, "\varphi"'] & M'' \arrow[r] \arrow[d, "\varphi''"'] & 0 \\
0 \arrow[r] & N' \arrow[r, "\mu"] & N \arrow[r, "\nu"] & N'' \arrow[r] & 0
\end{tikzcd}$$

is commutative and has exact rows. Prove that

$$0 \longrightarrow \ker\varphi' \xrightarrow{\alpha'} \ker\varphi \xrightarrow{\beta'} \ker\varphi'' \xrightarrow{\delta} \operatorname{coker}\varphi' \xrightarrow{\mu'} \operatorname{coker}\varphi \xrightarrow{\nu'} \operatorname{coker}\varphi'' \longrightarrow 0$$

is a long exact sequence, where $\alpha', \beta', \mu', \nu'$ are the homomorphisms induced by α, β, μ, ν, respectively.

Proof. Since the rows are exact,α and μ are injective, while β and ν are surjective.

We first define the connecting homomorphism

$$\delta : \ker\varphi'' \to \operatorname{coker}\varphi'.$$

Let $x \in \ker\varphi'' \subseteq M''$. Since $\beta : M \to M''$ is surjective, choose $y \in M$ such that

$$\beta(y) = x.$$

By commutativity,

$$\nu(\varphi(y)) = \varphi''(\beta(y)) = \varphi''(x) = 0.$$

Hence $\varphi(y) \in \ker\nu = \operatorname{Im}\mu$. Therefore there exists $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Define

$$\delta(x) = \overline{z} = z + \operatorname{Im}\varphi' \in \operatorname{coker}\varphi'.$$

We must check that this is well-defined. First, suppose $y_1, y_2 \in M$ both satisfy

$$\beta(y_1) = \beta(y_2) = x.$$

Then $y_1 - y_2 \in \ker\beta = \operatorname{Im}\alpha$, so there exists $m' \in M'$ such that

$$y_1 - y_2 = \alpha(m').$$

If $\mu(z_i) = \varphi(y_i)$ for $i = 1, 2$, then

$$\mu(z_1 - z_2) = \varphi(y_1) - \varphi(y_2) = \varphi(\alpha(m')) = \mu(\varphi'(m')),$$

where we used commutativity. Since μ is injective, it follows that

$$z_1 - z_2 = \varphi'(m') \in \operatorname{Im}\varphi'.$$

Thus $\overline{z_1} = \overline{z_2}$ in $\operatorname{coker} \varphi'$, so $\delta(x)$ is independent of the choice of y.

Next, if $z, z' \in N'$ both satisfy

$$\mu(z) = \mu(z') = \varphi(y),$$

then $\mu(z - z') = 0$, and since μ is injective,$z = z'$. Hence $\delta(x)$ is also independent of the choice of z.

It is straightforward to verify that δ is a homomorphism.

Now we prove exactness.

Exactness at $\ker \varphi'$. Since α induces α', we have

$$\alpha'(\ker \varphi') \subseteq \ker \varphi.$$

Because α is injective, so is α'. Hence

$$\ker \alpha' = 0.$$

Therefore the sequence is exact at $\ker \varphi'$.

Exactness at $\ker \varphi$. We show that

$$\operatorname{Im} \alpha' = \ker \beta'.$$

Take $x' \in \ker \varphi'$. Then

$$\beta'(\alpha'(x')) = \beta(\alpha(x')) = 0,$$

so $\operatorname{Im} \alpha' \subseteq \ker \beta'$.

Conversely, let $y \in \ker \varphi$ satisfy $\beta'(y) = 0$. Then $\beta(y) = 0$, hence by exactness of the top row there exists $x' \in M'$ such that

$$\alpha(x') = y.$$

Since $\varphi(y) = 0$, we get

$$0 = \varphi(\alpha(x')) = \mu(\varphi'(x')).$$

As μ is injective,$\varphi'(x') = 0$, so $x' \in \ker \varphi'$. Thus

$$y = \alpha'(x'),$$

proving $\ker \beta' \subseteq \operatorname{Im} \alpha'$.

Exactness at $\ker \varphi''$. We show that

$$\operatorname{Im} \beta' = \ker \delta.$$

First let $y \in \ker \varphi$. Then $\beta(y) \in \ker \varphi''$, since

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = 0.$$

Thus $\beta'(y) \in \ker \varphi''$. To compute $\delta(\beta'(y))$, choose the lift y itself. Then $\varphi(y) = 0$, so we may take $z = 0$, and hence

$$\delta(\beta'(y)) = \overline{0} = 0.$$

Therefore $\operatorname{Im} \beta' \subseteq \ker \delta$.

Conversely, let $x \in \ker \varphi''$ and suppose $\delta(x) = 0$. Choose $y \in M$ with

$$\beta(y) = x,$$

and choose $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Since $\delta(x) = \overline{z} = 0$ in $\operatorname{coker} \varphi'$, there exists $x' \in M'$ such that

$$z = \varphi'(x').$$

Hence

$$\varphi(y - \alpha(x')) = \varphi(y) - \varphi(\alpha(x')) = \mu(z) - \mu(\varphi'(x')) = 0.$$

So $y - \alpha(x') \in \ker \varphi$, and

$$\beta'(y - \alpha(x')) = \beta(y - \alpha(x')) = \beta(y) = x,$$

because $\beta\alpha = 0$. Thus $x \in \operatorname{Im} \beta'$, proving

$$\ker \delta \subseteq \operatorname{Im} \beta'.$$

Exactness at $\operatorname{coker} \varphi'$. We show that

$$\operatorname{Im} \delta = \ker \mu'.$$

Let $x \in \ker \varphi''$, and write

$$\delta(x) = \overline{z} \in \operatorname{coker} \varphi'$$

as above, with $\mu(z) = \varphi(y)$. Then

$$\mu'(\delta(x)) = \overline{\mu(z)} = \overline{\varphi(y)} = 0$$

in $\operatorname{coker} \varphi$. Hence $\operatorname{Im} \delta \subseteq \ker \mu'$.

Conversely, let $\overline{z} \in \operatorname{coker} \varphi'$ satisfy

$$\mu'(\overline{z}) = 0.$$

This means that $\mu(z) \in \operatorname{Im} \varphi$, so there exists $y \in M$ such that

$$\varphi(y) = \mu(z).$$

Applying ν, we get
$$\varphi''(\beta(y)) = \nu(\varphi(y)) = \nu(\mu(z)) = 0.$$
Thus $\beta(y) \in \ker \varphi''$. By construction of δ,
$$\delta(\beta(y)) = \overline{z}.$$
So $\overline{z} \in \operatorname{Im} \delta$, proving
$$\ker \mu' \subseteq \operatorname{Im} \delta.$$

Exactness at $\operatorname{coker} \varphi$**.** We show that
$$\operatorname{Im} \mu' = \ker \nu'.$$

For $\overline{z} \in \operatorname{coker} \varphi'$, we have
$$\nu'(\mu'(\overline{z})) = \overline{\nu(\mu(z))} = \overline{0} = 0,$$
so $\operatorname{Im} \mu' \subseteq \ker \nu'$.

Conversely, let $\overline{n} \in \operatorname{coker} \varphi$ satisfy
$$\nu'(\overline{n}) = 0.$$
Then $\overline{\nu(n)} = 0$ in $\operatorname{coker} \varphi''$, so there exists $m'' \in M''$ such that
$$\varphi''(m'') = \nu(n).$$
Since β is surjective, choose $m \in M$ such that
$$\beta(m) = m''.$$
Then
$$\nu(n - \varphi(m)) = \nu(n) - \varphi''(\beta(m)) = \nu(n) - \varphi''(m'') = 0.$$
Hence
$$n - \varphi(m) \in \ker \nu = \operatorname{Im} \mu,$$
so there exists $z \in N'$ such that
$$\mu(z) = n - \varphi(m).$$
Therefore in $\operatorname{coker} \varphi$,
$$\overline{n} = \overline{\mu(z)} = \mu'(\overline{z}),$$
showing that $\overline{n} \in \operatorname{Im} \mu'$. Thus
$$\ker \nu' \subseteq \operatorname{Im} \mu'.$$

Exactness at $\operatorname{coker} \varphi''$ **and termination.** Since $\nu : N \to N''$ is surjective, the induced map

$$\nu' : \operatorname{coker} \varphi \to \operatorname{coker} \varphi''$$

is surjective as well: for any $\overline{n''} \in \operatorname{coker} \varphi''$, choose $n \in N$ such that

$$\nu(n) = n'',$$

then

$$\nu'(\overline{n}) = \overline{n''}.$$

Hence

$$\operatorname{Im} \nu' = \operatorname{coker} \varphi'',$$

which is equivalent to exactness at $\operatorname{coker} \varphi''$ and the final 0.

Collecting all the above, we obtain the exact sequence

$$0 \longrightarrow \ker \varphi' \xrightarrow{\alpha'} \ker \varphi \xrightarrow{\beta'} \ker \varphi'' \xrightarrow{\delta} \operatorname{coker} \varphi' \xrightarrow{\mu'} \operatorname{coker} \varphi \xrightarrow{\nu'} \operatorname{coker} \varphi'' \longrightarrow 0.$$

□

7.4 Homework Week 4

Exercise 7.4.1. *Let J be an R-module. Show that the following are equivalent.*

1. *J is an injective module.*
2. *For any R-module monomorphism $\varphi \colon J \to M$, there exists an R-module homomorphism $\psi \colon M \to J$ such that $\psi\varphi = \mathrm{id}_J$.*
3. *If J is a submodule of some R-module M, then J is a direct summand of M.*

Proof. We prove (1) ⇒ (2) ⇒ (3) ⇒ (1).

(1) ⇒ (2)**.** Assume that J is injective, and let $\varphi \colon J \to M$ be a monomorphism. Apply the defining property of injectivity to the monomorphism φ and the map $\mathrm{id}_J \colon J \to J$:

$$\begin{tikzcd} J \arrow[r, hook, "\varphi"] \arrow[d, "\mathrm{id}_J"'] & M \arrow[dl, dashed, "\exists\,\psi"] \\ J & \end{tikzcd}$$

Then there exists an R-module homomorphism $\psi \colon M \to J$ such that

$$\psi \circ \varphi = \mathrm{id}_J\,.$$

(2) ⇒ (3)**.** Assume (2). If $J \subseteq M$, let $\iota \colon J \hookrightarrow M$ be the inclusion map. By (2), there exists $\psi \colon M \to J$ such that

$$\psi \circ \iota = \mathrm{id}_J\,.$$

We claim that
$$M = J \oplus \ker \psi.$$
Indeed, for any $m \in M$,
$$m = \iota(\psi(m)) + \big(m - \iota(\psi(m))\big),$$
and
$$\psi\big(m - \iota(\psi(m))\big) = \psi(m) - \psi(\iota(\psi(m))) = \psi(m) - \psi(m) = 0,$$
so $m - \iota(\psi(m)) \in \ker \psi$. Hence $M = J + \ker \psi$. If $x \in J \cap \ker \psi$, then
$$x = (\psi \circ \iota)(x) = \psi(x) = 0,$$
so $J \cap \ker \psi = 0$. Therefore $M = J \oplus \ker \psi$, and J is a direct summand of M.

(3) $\Rightarrow$ (1). Assume (3). To prove that J is injective, let
$$i \colon A \hookrightarrow B$$
be a monomorphism of R-modules, and let
$$f \colon A \to J$$
be an R-module homomorphism. We must show that f extends to a map $g \colon B \to J$.

Consider the quotient module
$$P = (B \oplus J)/N, \qquad N = \{(i(a), -f(a)) \mid a \in A\}.$$
Define maps
$$\eta \colon J \to P, \qquad \eta(j) = [(0, j)],$$
and
$$\beta \colon B \to P, \qquad \beta(b) = [(b, 0)].$$
We first show that η is injective. If $\eta(j) = 0$, then $(0, j) \in N$, so for some $a \in A$,
$$(0, j) = (i(a), -f(a)).$$
Hence $i(a) = 0$. Since i is injective,$a = 0$, and therefore $j = -f(0) = 0$. Thus η is a monomorphism, so we may identify J with the submodule $\eta(J) \subseteq P$.

By (3), this submodule is a direct summand of P. Therefore there exists an R-module homomorphism
$$s \colon P \to J$$
such that
$$s \circ \eta = \mathrm{id}_J\,.$$
Now define
$$g = s \circ \beta \colon B \to J.$$

For any $a \in A$, we have in P:

$$[(i(a), 0)] = [(0, f(a))],$$

because

$$(i(a), 0) - (0, f(a)) = (i(a), -f(a)) \in N.$$

Hence

$$g(i(a)) = s([(i(a), 0)]) = s([(0, f(a))]) = (s \circ \eta)(f(a)) = f(a).$$

So $g \circ i = f$, and f extends to B. Therefore J is injective. $\square$

Exercise 7.4.2. *Suppose that $(M_i)_{i \in I}$ are R-modules. Prove that*

$$\bigoplus_{i \in I} M_i$$

is a flat R-module if and only if each M_i is a flat R-module.

Proof. Set

$$M = \bigoplus_{i \in I} M_i.$$

First assume that each M_i is flat. Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of R-modules. Since each M_i is flat, for every $i \in I$ the sequence

$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$

is exact. Taking direct sums over $i \in I$, we obtain an exact sequence

$$0 \longrightarrow \bigoplus_{i \in I}(M_i \otimes_R A) \longrightarrow \bigoplus_{i \in I}(M_i \otimes_R B) \longrightarrow \bigoplus_{i \in I}(M_i \otimes_R C) \longrightarrow 0.$$

Using the canonical isomorphisms

$$\left(\bigoplus_{i \in I} M_i\right) \otimes_R X \cong \bigoplus_{i \in I}(M_i \otimes_R X) \qquad (X = A, B, C),$$

we get

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0.$$

Hence M is flat.

Conversely, assume that M is flat. Fix $i \in I$, and write

$$M = M_i \oplus N_i, \qquad N_i = \bigoplus_{j \neq i} M_j.$$

Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of R-modules. Since M is flat, the sequence

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0$$

is exact. Using

$$M \otimes_R X \cong (M_i \otimes_R X) \oplus (N_i \otimes_R X) \qquad (X = A, B, C),$$

this becomes

$$0 \longrightarrow (M_i \otimes_R A) \oplus (N_i \otimes_R A) \longrightarrow (M_i \otimes_R B) \oplus (N_i \otimes_R B) \longrightarrow (M_i \otimes_R C) \oplus (N_i \otimes_R C) \longrightarrow 0.$$

Since kernels and images of direct sum maps decompose componentwise, it follows that

$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$

is exact. Therefore M_i is flat. As i was arbitrary, each M_i is flat. $\square$

Exercise 7.4.3. *Prove that the following three conditions on an R-module V are equivalent.*

1. (Ascending chain condition.) *For every infinite chain of R-submodules of V,*

$$V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots,$$

   there is a number m such that

$$V_m = V_{m+1} = \cdots .$$

2. *Any non-empty set of submodules of V contains a maximal element with respect to inclusion.*

3. *Every R-submodule of V is finitely generated.*

Proof. We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2)**.** Assume that (2) fails. Then there exists a non-empty set $\mathcal{S}$ of submodules of V having no maximal element. Choose $V_1 \in \mathcal{S}$. Since V_1 is not maximal in $\mathcal{S}$, there exists $V_2 \in \mathcal{S}$ such that

$$V_1 \subsetneq V_2.$$

Inductively, having chosen $V_n \in \mathcal{S}$, since V_n is not maximal there exists $V_{n+1} \in \mathcal{S}$ with

$$V_n \subsetneq V_{n+1}.$$

Thus we obtain a strictly ascending chain

$$V_1 \subsetneq V_2 \subsetneq \cdots,$$

which contradicts (1). Hence every non-empty set of submodules contains a maximal element.

(2) $\Rightarrow$ (3). Let U be a submodule of V. Consider the set

$$\mathcal{T} = \{N \leq U \mid N \text{ is finitely generated}\}.$$

This set is non-empty, since $0 \in \mathcal{T}$. By (2),$\mathcal{T}$ has a maximal element, say N.

We claim that $N = U$. If not, choose $u \in U \setminus N$. Then

$$N + Ru$$

is a finitely generated submodule of U properly containing N, contradicting the maximality of N in $\mathcal{T}$. Therefore $U = N$, so U is finitely generated. Hence every submodule of V is finitely generated.

(3) $\Rightarrow$ (1). Let

$$V_1 \subseteq V_2 \subseteq \cdots$$

be an ascending chain of submodules of V. Set

$$U = \bigcup_{n \geq 1} V_n.$$

Since the chain is ascending,U is a submodule of V. By (3),U is finitely generated, say

$$U = Ru_1 + \cdots + Ru_r.$$

For each j, choose n_j such that $u_j \in V_{n_j}$, and let

$$m = \max\{n_1, \ldots, n_r\}.$$

Then $u_1, \ldots, u_r \in V_m$, so $U \subseteq V_m$. Since always $V_m \subseteq U$, we get $U = V_m$. Therefore for all $n \geq m$,

$$V_n = U = V_m,$$

so the chain stabilizes. Hence (1) holds. □

Exercise 7.4.4. *Let W be an R-submodule of V. Prove that V is Noetherian if and only if both W and V/W are Noetherian.*

Proof. We use the characterization that an R-module is Noetherian if and only if every submodule is finitely generated.

Assume first that V is Noetherian. Then every submodule of V is finitely generated.

Since every submodule of W is also a submodule of V, it follows that every submodule of W is finitely generated. Hence W is Noetherian.

Now let $\overline{U}$ be a submodule of V/W, and let

$$\pi \colon V \to V/W$$

be the quotient map. Then $U := \pi^{-1}(\overline{U})$ is a submodule of V, so U is finitely generated, say

$$U = Rv_1 + \cdots + Rv_n.$$

Applying π, we get

$$\overline{U} = R(v_1 + W) + \cdots + R(v_n + W).$$

Thus every submodule of V/W is finitely generated, and therefore V/W is Noetherian.

Conversely, assume that both W and V/W are Noetherian. Let U be any submodule of V. We show that U is finitely generated.

First,

$$U \cap W$$

is a submodule of W, so it is finitely generated. Choose generators

$$U \cap W = Rw_1 + \cdots + Rw_m.$$

Next, since

$$(U + W)/W$$

is a submodule of V/W, it is finitely generated as well. Choose elements $u_1, \ldots, u_n \in U$ such that

$$(U + W)/W = R(u_1 + W) + \cdots + R(u_n + W).$$

Now let $u \in U$. Since $u + W \in (U + W)/W$, there exist $a_1, \ldots, a_n \in R$ such that

$$u + W = \sum_{i=1}^{n} a_i(u_i + W).$$

Therefore

$$u - \sum_{i=1}^{n} a_i u_i \in W.$$

Because also $u, u_i \in U$, we have

$$u - \sum_{i=1}^{n} a_i u_i \in U \cap W.$$

So there exist $b_1, \ldots, b_m \in R$ such that

$$u - \sum_{i=1}^{n} a_i u_i = \sum_{j=1}^{m} b_j w_j.$$

Hence

$$u = \sum_{i=1}^{n} a_i u_i + \sum_{j=1}^{m} b_j w_j.$$

This shows that U is generated by

$$u_1, \ldots, u_n, w_1, \ldots, w_m.$$

Thus every submodule of V is finitely generated, so V is Noetherian. $\square$

Exercise 7.4.5. *Let V be a Noetherian R-module and let $f\colon V \to V$ be an epimorphism. Show that f is an R-isomorphism.*

Proof. Since f is an epimorphism, it is surjective. It therefore suffices to prove that f is injective.

For each $n \geq 1$, set

$$K_n = \ker(f^n).$$

Then

$$K_1 \subseteq K_2 \subseteq K_3 \subseteq \cdots,$$

because if $f^n(x) = 0$, then

$$f^{n+1}(x) = f(f^n(x)) = 0.$$

Since V is Noetherian, this ascending chain stabilizes. Hence there exists $m \geq 1$ such that

$$K_m = K_{m+1}.$$

We claim that $K_m = 0$. Let $x \in K_m$. Because f is surjective,f^m is also surjective, there exists $y \in V$ such that

$$f^m(y) = x.$$

Now

$$f^{2m}(y) = f^m(f^m(y)) = f^m(x) = 0,$$

so $y \in K_{2m} = K_m$. Therefore

$$x = f^m(y) = 0.$$

Thus $K_m = 0$.

Since

$$\ker(f) = K_1 \subseteq K_m = 0,$$

we obtain $\ker(f) = 0$. So f is injective. Being both surjective and injective,f is an isomorphism. $\square$

7.5 Homework Week 5

Exercise 7.5.1. *Are the following statements correct?*

(1) Any submodule of a semisimple module is also semisimple.

(2) Any semisimple Noetherian module is Artinian.

Proof. **(1)** True. Let V be a semisimple module, so

$$V = \bigoplus_{i \in I} S_i$$

where each S_i is simple. Let $W \leq V$ be a submodule. We show that W is a sum of simple submodules, hence semisimple.

Consider the set $\mathcal{S}$ of all subsets $J \subseteq I$ such that the sum

$$W + \bigoplus_{j \in J} S_j$$

is direct, i.e.

$$W \cap \bigoplus_{j \in J} S_j = 0.$$

By Zorn's lemma, there exists a maximal such subset J_0. Set

$$U := W \oplus \bigoplus_{j \in J_0} S_j.$$

For any $i \in I \setminus J_0$, the maximality of J_0 forces $U \cap S_i \neq 0$. Since S_i is simple, we have $S_i \subseteq U$. Therefore $S_i \subseteq U$ for every $i \in I$, so $V = U$. Hence

$$V = W \oplus \bigoplus_{j \in J_0} S_j,$$

and

$$W \cong V \Big/ \bigoplus_{j \in J_0} S_j \cong \bigoplus_{i \in I \setminus J_0} S_i,$$

which is semisimple.

(2) True. Let V be a semisimple Noetherian module. Since V is semisimple,

$$V = \bigoplus_{i \in I} S_i$$

Since V is Noetherian, it is finitely generated. Let $v_1, \ldots, v_n$ generate V. For each k, because $V = \bigoplus_{i \in I} S_i$ is a direct sum, the element v_k has only finitely many nonzero components, so there exists a finite subset $I_k \subseteq I$ such that

$$v_k \in \bigoplus_{i \in I_k} S_i.$$

Hence

$$V = \langle v_1, \ldots, v_n \rangle \subseteq \bigoplus_{i \in I_1 \cup \cdots \cup I_n} S_i.$$

Since the reverse inclusion is obvious, we get

$$V = \bigoplus_{i \in I_1 \cup \cdots \cup I_n} S_i,$$

and therefore V is a finite direct sum of simple modules. Each S_i is simple, hence both Noetherian and Artinian. A finite direct sum of Artinian modules is Artinian, so V is Artinian. $\square$

Exercise 7.5.2. *Let V be an R-module with a composition series. Prove that the following conditions are equivalent.*

(1) V is semisimple.

(2) Every irreducible submodule of V is a direct summand of V.

(3) Every maximal submodule of V is a direct summand of V.

Proof. We prove

$$(1) \Longrightarrow (2) \Longrightarrow (3) \Longrightarrow (1).$$

(1) $\Longrightarrow$ (2). If V is semisimple, then every submodule of V is a direct summand. In particular, every irreducible submodule of V is a direct summand.

(2) $\Longrightarrow$ (3). Let M be a maximal submodule of V. Then V/M is simple.

Since M is a maximal submodule of V, the quotient V/M is simple. Now let S be a submodule of V such that $S \not\subseteq M$. Then $S + M \neq M$, so

$$(S + M)/M \neq 0.$$

By the second isomorphism theorem,

$$S/(S \cap M) \cong (S + M)/M.$$

Since $(S + M)/M$ is a nonzero submodule of the simple module V/M, it follows that $(S + M)/M$ is simple. Hence $S/(S \cap M)$ is simple.

Choose $S \subseteq V$ minimal subject to $S \not\subseteq M$. Then $S \cap M$ is a maximal submodule of S, and by minimality of S, the quotient $S/(S \cap M)$ is simple. But $S \cap M$ is a proper submodule of S, so by minimality of S, it must be 0. Thus S is simple.

Therefore S is an irreducible submodule of V with $S \not\subseteq M$. By (2), S is a direct summand of V, say

$$V = S \oplus T.$$

Since $S \not\subseteq M$ and S is simple, we have $S \cap M = 0$. Also M is maximal and $S \not\subseteq M$, so

$$M + S = V.$$

Hence

$$V = M \oplus S,$$

and therefore M is a direct summand of V.

(3) $\Longrightarrow$ (1). Let

$$V = V_0 \supsetneq V_1 \supsetneq \cdots \supsetneq V_n = 0$$

be a composition series of V. We argue by induction on n, the composition length of V.

If $n = 1$, then V is simple, hence semisimple.

Assume $n > 1$, and suppose the statement holds for modules of composition length $< n$. Since V/V_1 is simple, V_1 is a maximal submodule of V. By (3), V_1 is a direct summand of V; thus

$$V = V_1 \oplus S$$

for some simple submodule $S \cong V/V_1$.

Now V_1 has composition length $n - 1$. We claim that every maximal submodule of V_1 is a direct summand of V_1. Indeed, let N be a maximal submodule of V_1. Then

$$V/N \cong (V_1/N) \oplus S,$$

and since V_1/N is simple, $N \oplus S$ is a maximal submodule of V. By (3), $N \oplus S$ is a direct summand of V, say

$$V = (N \oplus S) \oplus T.$$

Because

$$V = V_1 \oplus S,$$

intersecting with V_1 gives

$$V_1 = N \oplus T.$$

So N is a direct summand of V_1.

Hence V_1 satisfies condition (3), and by the induction hypothesis V_1 is semisimple. Therefore

$$V = V_1 \oplus S$$

is semisimple as well.

This proves (3) $\Longrightarrow$ (1), and therefore the three conditions are equivalent. □

Exercise 7.5.3. *Prove that*

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Proof. We first show

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V) \subseteq \mathrm{Soc}(U \oplus V).$$

Indeed, $\mathrm{Soc}(U)$ and $\mathrm{Soc}(V)$ are semisimple, so $\mathrm{Soc}(U) \oplus \mathrm{Soc}(V)$ is a semisimple submodule of $U \oplus V$. Hence it is contained in $\mathrm{Soc}(U \oplus V)$.

For the reverse inclusion, let S be a simple submodule of $U \oplus V$, and let

$$\pi_U : U \oplus V \to U, \qquad \pi_V : U \oplus V \to V$$

be the canonical projections. Consider the restrictions

$$\pi_U|_S : S \to U, \qquad \pi_V|_S : S \to V.$$

Since S is simple, the image of any homomorphism defined on S is either 0 or simple. Therefore $\pi_U(S)$ is either 0 or a simple submodule of U, and similarly $\pi_V(S)$ is either 0 or a simple submodule of V. Hence

$$\pi_U(S) \subseteq \mathrm{Soc}(U), \qquad \pi_V(S) \subseteq \mathrm{Soc}(V).$$

Now for any $s \in S$, we can write

$$s = (\pi_U(s), \pi_V(s)).$$

Since $\pi_U(s) \in \mathrm{Soc}(U)$ and $\pi_V(s) \in \mathrm{Soc}(V)$, it follows that

$$s \in \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Thus

$$S \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Since $\mathrm{Soc}(U \oplus V)$ is the sum of all simple submodules of $U \oplus V$, we obtain

$$\mathrm{Soc}(U \oplus V) \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Combining the two inclusions, we conclude that

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

□

Exercise 7.5.4. *Let R be an arbitrary ring. Then every two-sided ideal of the matrix ring $M_n(R)$ can be expressed in the form $M_n(I)$, where I is a two-sided ideal of R. In particular, R is simple if and only if $M_n(R)$ is simple.*

Hint. For an ideal L of $M_n(R)$, let I be the set of $(1,1)$-entries of L. Show that I is an ideal of R and $L = M_n(I)$.

Proof. Let E_{ij} denote the matrix with 1 in position (i,j) and 0 elsewhere. For a two-sided ideal L of $M_n(R)$, define

$$I := \{a \in R \mid aE_{11} \in L\}.$$

We first note that for any $A = (a_{kl}) \in L$, we have

$$E_{1k} \cdot A \cdot E_{l1} = a_{kl}\, E_{11} \in L,$$

so every entry of every matrix in L belongs to I. In particular,

$$I = \{a_{kl} \mid A = (a_{kl}) \in L,\ 1 \le k, l \le n\}.$$

I is a two-sided ideal of R. If $a, b \in I$, then $aE_{11}, bE_{11} \in L$, so $(a-b)E_{11} = aE_{11} - bE_{11} \in L$, hence $a - b \in I$. For $r \in R$, we have $(rE_{11}) \cdot (aE_{11}) = raE_{11} \in L$ and $(aE_{11}) \cdot (rE_{11}) = arE_{11} \in L$, so $ra, ar \in I$. Thus I is a two-sided ideal of R.

$L = M_n(I)$. Since every entry of every matrix in L lies in I, we have $L \subseteq M_n(I)$. Conversely, let $aE_{ij} \in M_n(I)$ with $a \in I$. Then $aE_{11} \in L$, and

$$aE_{ij} = E_{i1} \cdot (aE_{11}) \cdot E_{1j} \in L.$$

Since every matrix in $M_n(I)$ is a sum of such elements aE_{ij}, we get $M_n(I) \subseteq L$. Therefore $L = M_n(I)$.

R is simple $\iff$ $M_n(R)$ is simple. The map $I \mapsto M_n(I)$ is a bijection from the set of two-sided ideals of R to the set of two-sided ideals of $M_n(R)$. Hence R has no nontrivial two-sided ideals if and only if $M_n(R)$ has no nontrivial two-sided ideals. $\square$

Exercise 7.5.5. *Let W be a submodule of a semisimple module V. Show that W and V/W are also semisimple.*

Proof. Since V is semisimple, write $V = \bigoplus_{i \in I} S_i$ with each S_i simple.

W is semisimple. By Zorn's lemma, choose a maximal subset $J \subseteq I$ such that $W \cap \bigoplus_{j \in J} S_j = 0$. Set $U = W \oplus \bigoplus_{j \in J} S_j$. For any $i \in I \setminus J$, the maximality of J gives $U \cap S_i \neq 0$, and since S_i is simple,$S_i \subseteq U$. Therefore $V \subseteq U$, so $V = U = W \oplus \bigoplus_{j \in J} S_j$. Thus

$$W \cong V \Big/ \bigoplus_{j \in J} S_j \cong \bigoplus_{i \in I \setminus J} S_i,$$

which is a direct sum of simple modules, hence semisimple.

V/W is semisimple. From the decomposition $V = W \oplus \bigoplus_{j \in J} S_j$, we obtain

$$V/W \cong \bigoplus_{j \in J} S_j,$$

which is a direct sum of simple modules, hence semisimple. $\square$

7.6 Homework Week 6

Exercise 7.6.1. *Let D be a division ring and n a positive integer.*

(a) Show that the natural left $M_n(D)$-module, consisting of column vectors of length n with entries in D, is a simple module.

(b) Show that $M_n(D)$ is semisimple and has up to isomorphism only one simple module.

(c) Show that every ring of the form

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

is semisimple, where D_i are division rings.

(d) *Show that $M_n(D)$ is a simple ring, namely, one in which the only 2-sided ideals are the zero ideal and the whole ring.*

Proof. Let $R = M_n(D)$ and let $V = D^n$ be the natural left R-module of column vectors.

(a) We show that V is simple. Let $0 \neq W \leq V$, and choose

$$v = (v_1, \dots, v_n)^t \in W, \qquad v \neq 0.$$

Then $v_j \neq 0$ for some j. For each $i \in \{1, \dots, n\}$, let $A_i \in R$ be the matrix whose (i, j)-entry is v_j^{-1} and whose other entries are 0. Then

$$A_i v = e_i,$$

where e_i is the i-th standard basis vector in D^n. Since W is an R-submodule and $v \in W$, we get $e_i \in W$ for all i. Hence

$$W = D^n = V.$$

Therefore V is a simple left R-module.

(b) Let $E_{11}, \dots, E_{nn}$ be the standard diagonal matrix units. Then

$$1 = E_{11} + \cdots + E_{nn},$$

and hence

$$R = RE_{11} \oplus \cdots \oplus RE_{nn}.$$

Indeed, RE_{ii} is precisely the set of matrices whose only possibly nonzero column is the i-th column, so every matrix is uniquely the sum of its columns.

For each i, define

$$\phi_i : RE_{ii} \longrightarrow V$$

by sending a matrix in RE_{ii} to its i-th column. This is an isomorphism of left R-modules. By part (a), V is simple, so each RE_{ii} is simple. Thus ${}_RR$ is a direct sum of simple submodules, and therefore R is semisimple.

Now let S be a simple left R-module. Pick $0 \neq s \in S$. Then the map

$$\psi : R \longrightarrow S, \qquad A \mapsto As$$

is a nonzero R-homomorphism, hence surjective since S is simple. Because R is semisimple, every quotient of R is semisimple. Since S is simple, it must be isomorphic to one of the simple direct summands RE_{ii}. Therefore

$$S \cong RE_{ii} \cong V$$

for some i. Hence, up to isomorphism, R has only one simple left module.

(c) Let
$$A = M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r).$$
As a left A-module over itself,
$${}_AA = {}_AM_{n_1}(D_1) \oplus \cdots \oplus {}_AM_{n_r}(D_r).$$
Each summand is semisimple by part (b), and a finite direct sum of semisimple modules is semisimple. Hence A is a semisimple ring.

(d) Let I be a nonzero two-sided ideal of R. Choose $0 \neq A = (a_{pq}) \in I$, and pick indices i, j such that $a_{ij} \neq 0$. For arbitrary r, s, let E_{uv} denote the standard matrix units. Then
$$E_{ri}AE_{js} = a_{ij}E_{rs} \in I.$$
Since $a_{ij} \in D^\times$, we also have
$$(a_{ij}^{-1}E_{rr})(a_{ij}E_{rs}) = E_{rs} \in I.$$
Thus every matrix unit E_{rs} belongs to I. Therefore
$$1 = E_{11} + \cdots + E_{nn} \in I,$$
so $I = R$. Hence the only two-sided ideals of R are 0 and R, i.e. R is a simple ring.

□

Exercise 7.6.2. *Let U be an A-module for a semisimple finite-dimensional algebra A (over a field). Show that if $\mathrm{End}_A(U)$ is a division ring then U is simple.*

Proof. Since A is a semisimple finite-dimensional algebra over a field, it is a semisimple Artinian ring. Hence every left A-module is semisimple. In particular, U is a semisimple A-module.

We claim that U must be simple. Suppose not. Then U is not simple, so because U is semisimple, there exist nonzero submodules $U_1, U_2 \leq U$ such that
$$U = U_1 \oplus U_2.$$
Let
$$\pi_1 : U \to U_1$$
be the projection onto the first summand. Since U_1 is an A-submodule, π_1 is an A-module endomorphism of U, so
$$\pi_1 \in \mathrm{End}_A(U).$$
Moreover,
$$\pi_1^2 = \pi_1,$$
so π_1 is an idempotent.

Now $\pi_1 \neq 0$ because $U_1 \neq 0$, and $\pi_1 \neq 1_U$ because $U_2 \neq 0$. Thus π_1 is a nontrivial idempotent in $\mathrm{End}_A(U)$.

But a division ring has no idempotents other than 0 and 1: indeed, if $e^2 = e$ and $e \neq 0$, then multiplying by e^{-1} gives $e = 1$. This contradicts the assumption that $\mathrm{End}_A(U)$ is a division ring.

Therefore U must be simple. □

Exercise 7.6.3. *Let A be a ring. Then A is a semisimple Artinian ring if and only if every A-module is projective.*

Proof. Here all modules are left A-modules.

$$(\Rightarrow)$$

Assume that A is a semisimple Artinian ring. Then every A-module is semisimple.

Let M be any A-module. To show that M is projective, consider an arbitrary short exact sequence

$$0 \longrightarrow K \longrightarrow N \xrightarrow{\pi} M \longrightarrow 0.$$

Since N is semisimple, the submodule K is a direct summand of N. Thus there exists a submodule $N' \leq N$ such that

$$N = K \oplus N'.$$

Now $\pi|_{N'} : N' \to M$ is surjective, and

$$\ker(\pi|_{N'}) = N' \cap K = 0.$$

Hence $\pi|_{N'}$ is an isomorphism. Therefore the above short exact sequence splits. Since every short exact sequence ending in M splits, M is projective.

Since M was arbitrary, every A-module is projective.

$$(\Leftarrow)$$

Assume that every A-module is projective. Let M be any A-module and let $N \leq M$ be a submodule. Then the quotient M/N is projective, so the short exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

splits. Hence N is a direct summand of M.

Therefore every submodule of every A-module is a direct summand. It follows that every A-module is semisimple. In particular, the left regular module ${}_AA$ is semisimple.

Now A is cyclic as a left A-module, generated by 1. Since ${}_AA$ is semisimple, it is a sum of simple submodules:

$$A = \sum_{\lambda \in \Lambda} S_\lambda.$$

Because $1 \in A$, we may write

$$1 = s_1 + \cdots + s_n \qquad (s_i \in S_{\lambda_i}).$$

Then for any $a \in A$,

$$a = a \cdot 1 = as_1 + \cdots + as_n \in S_{\lambda_1} + \cdots + S_{\lambda_n}.$$

Thus

$$A = S_{\lambda_1} + \cdots + S_{\lambda_n}.$$

Since A is semisimple, this finite sum is in fact a direct sum (after deleting repetitions if necessary):

$$A \cong S_1 \oplus \cdots \oplus S_m$$

with each S_i simple.

Hence ${}_AA$ is a finite direct sum of simple modules, so it is Artinian. Therefore A is a semisimple Artinian ring.

This proves that A is a semisimple Artinian ring if and only if every A-module is projective. $\square$

Exercise 7.6.4. *Let k be a field. Suppose the group algebra kG is semisimple, and the simple kG-modules are $S_1, \dots, S_r$ with degrees $d_i = \dim_k S_i$. Show that*

$$\sum_{i=1}^{r} d_i^2 \geq |G|$$

with equality if and only if $\mathrm{End}_{kG}(S_i) = k$ *for all i.*

Proof. Let

$$A = kG, \qquad E_i = \mathrm{End}_A(S_i) \quad (1 \leq i \leq r).$$

Since A is a semisimple finite-dimensional k-algebra, by the Wedderburn–Artin theorem there exist positive integers $n_1, \dots, n_r$ such that

$$A \cong \bigoplus_{i=1}^{r} M_{n_i}(E_i^{\mathrm{op}}),$$

where $S_1, \dots, S_r$ form a complete set of representatives of the isomorphism classes of simple A-modules.

For each i, the simple module S_i is naturally a right E_i-vector space of dimension n_i. Hence

$$d_i = \dim_k S_i = n_i \dim_k E_i.$$

Taking k-dimensions in the Wedderburn decomposition gives

$$|G| = \dim_k(kG) = \dim_k A = \sum_{i=1}^{r} \dim_k M_{n_i}(E_i^{\mathrm{op}}) = \sum_{i=1}^{r} n_i^2 \dim_k E_i.$$

Using $d_i = n_i \dim_k E_i$, we obtain

$$|G| = \sum_{i=1}^{r} n_i^2 \dim_k E_i = \sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i}.$$

Now each E_i is a division k-algebra, so $\dim_k E_i \geq 1$. Therefore

$$|G| = \sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i} \leq \sum_{i=1}^{r} d_i^2.$$

Thus

$$\sum_{i=1}^{r} d_i^2 \geq |G|.$$

It remains to determine when equality holds. Since each $d_i > 0$ and each $\dim_k E_i \geq 1$, we have

$$\sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i} = \sum_{i=1}^{r} d_i^2$$

if and only if

$$\dim_k E_i = 1 \qquad \text{for all } i.$$

But $E_i = \mathrm{End}_A(S_i)$ is a k-algebra, so $\dim_k E_i = 1$ is equivalent to

$$E_i = k.$$

Hence equality holds if and only if

$$\mathrm{End}_{kG}(S_i) = k \qquad \text{for all } i.$$

This proves the claim. □

Exercise 7.6.5. *Suppose that we have R-module homomorphisms*

$$U \xrightarrow{f} V \xrightarrow{g} W.$$

Show that: if gf is an essential epimorphism and f is an epimorphism, then both f and g are essential epimorphisms.

Proof. Assume that

$$U \xrightarrow{f} V \xrightarrow{g} W$$

are R-module homomorphisms such that gf is an essential epimorphism and f is an epimorphism.

We must show that both f and g are essential epimorphisms.

Step 1: f is an essential epimorphism.

Since f is already assumed to be an epimorphism, it remains to show that it is essential.

Let $U' \leq U$ be a submodule such that

$$f(U') = V.$$

Then applying g, we get

$$(gf)(U') = g(f(U')) = g(V) = W,$$

because gf is an epimorphism. Thus U' maps onto W under gf.

But gf is an *essential* epimorphism, so no proper submodule of U can map surjectively onto W. Hence we must have

$$U' = U.$$

Therefore f is an essential epimorphism.

Step 2: g is an essential epimorphism.

First, g is an epimorphism: indeed, since gf is surjective and f is surjective, for any $w \in W$ there exists $u \in U$ such that

$$g(f(u)) = w.$$

Thus $w \in g(V)$, so $g(V) = W$.

Now let $V' \leq V$ be a submodule such that

$$g(V') = W.$$

We claim that $V' = V$.

Consider the inverse image

$$f^{-1}(V') = \{u \in U \mid f(u) \in V'\}.$$

This is a submodule of U, and

$$(gf)\big(f^{-1}(V')\big) = g(V') = W.$$

Since gf is an essential epimorphism, it follows that

$$f^{-1}(V') = U.$$

Applying f to both sides gives

$$V = f(U) = f\big(f^{-1}(V')\big) \subseteq V'.$$

Hence $V' = V$.

Therefore no proper submodule of V maps surjectively onto W under g, so g is an essential epimorphism.

We conclude that both f and g are essential epimorphisms. □